

An FPRAS for Counting Common Bases of Two Matroids

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Abstract

We design the first polynomial-time algorithms for approximately counting and uniformly sampling common bases of two matroids given by their independence oracles. Moreover, our algorithms generalize far beyond this to Hadamard products of two probability measures on the Boolean cube satisfying a simple nonnegative curvature condition. These algorithmic primitives have myriad applications in statistical physics, polyhedral combinatorics, the study of quantum many-body systems, and beyond.

Our approach has two key ingredients.

- We relax the intersection by imposing an overlap penalty on the product measure formed by the two input measures. We prove, via an integrated Bochner-type method, that this “soft intersection” satisfies a Poincaré Inequality uniformly over all external fields.
- We solve a dual maximum entropy convex program to compute external fields under which the hard constraint is satisfied with high probability under the soft intersection measure. We bound this success probability directly using the uniform Poincaré Inequality and smallness of the gradient norm.

AI Disclosure GPT-5.6 Sol Ultra and GPT-6 Astra Ultra were heavily used to develop the ideas in this paper. A more complete discussion is included in [Section 1.5](#).

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1 Introduction

In this paper, we design the first polynomial-time algorithms that solve the following fundamental problems in full generality: Let $\mathcal{M}, \mathcal{N}$ be two matroids with the same rank r on a common ground set $\mathcal{U}$ of size n , both given via an independence oracle, and write $\mathcal{B}(\mathcal{M})$ (resp. $\mathcal{B}(\mathcal{N})$) for the collection of bases of $\mathcal{M}$ (resp. $\mathcal{N}$); for background on matroids, see [Section 3.2](#).

Approximate Counting: Estimate the number $|\mathcal{B}(\mathcal{M}) \cap \mathcal{B}(\mathcal{N})|$ of common bases up to $(1 \pm \varepsilon)$ -multiplicative error, for any desired target accuracy $0 < \varepsilon < 1$. Note that exact counting is #P-hard and hence, unlikely to admit an efficient algorithm [[Val79](#)].

Approximate Sampling: Sample a common basis from the uniform distribution over $\mathcal{B}(\mathcal{M}) \cap \mathcal{B}(\mathcal{N})$ up to $\pm\varepsilon$ -additive total variation error, for any $0 < \varepsilon < 1$.

These are fundamental algorithmic problems in theoretical computer science that have abundant applications in statistical physics, high-dimensional statistics, and combinatorics; see [Appendix C](#) for a more comprehensive discussion.

Counting and sampling common bases of two matroids has received significant attention in the literature over the past several decades. For the moment, we highlight two landmark results that previously solved important special cases of the problem. Jerrum–Sinclair–Vigoda designed the first polynomial-time algorithms for counting and sampling (edge weighted) perfect matchings in bipartite graphs [[JSV04](#)]. As a result, they also obtained the first FPRAS for estimating the permanent of any nonnegative matrix. Anari–Liu–Oveis Gharan–Vinzant gave the first polynomial-time algorithms for counting and sampling bases of a single matroid [[Ana+19](#)]. Both results are based on the Markov chain Monte Carlo (MCMC) method.

Our algorithms are also based on MCMC and hence, are randomized. We note that randomization is necessary when the matroids can only be accessed via their independence oracles. Indeed, even for a single matroid $\mathcal{M}$, no efficient deterministic algorithm can approximate $|\mathcal{B}(\mathcal{M})|$ to within a simply exponential multiplicative factor in the oracle model [[ABF94](#)]. This lower bound is nearly matched by the deterministic $2^{O(r)}$ -approximation algorithm of Anari–Oveis Gharan–Vinzant [[AOV18](#)], which also applies to common bases of two matroids.

Definition 1.1 (FPRAS). Let $Z \geq 0$ be a quantity determined by an input instance. A *fully polynomial-time randomized approximation scheme* (FPRAS) for Z is a randomized algorithm that, given the instance and an accuracy parameter $0 < \varepsilon \leq 1$, returns $\hat{Z} \geq 0$ satisfying

$$\Pr \left[(1 - \varepsilon) \cdot Z \leq \hat{Z} \leq (1 + \varepsilon) \cdot Z \right] \geq \frac{3}{4},$$

in time polynomial in the size of the instance and $1/\varepsilon$.¹ In the oracle model, each oracle call has unit cost.

Theorem 1.2 (Counting Common Bases). *There is an FPRAS that, given independence oracles for two matroids $\mathcal{M}, \mathcal{N}$ on a common ground set $\mathcal{U}$, estimates the number $|\mathcal{B}(\mathcal{M}) \cap \mathcal{B}(\mathcal{N})|$ of common bases.*

Remark 1.3. The running time of our algorithm is $\tilde{O}(n^{13}/\varepsilon^2)$, which certainly isn't optimal. Since our goal is to present our algorithms and proofs in the simplest way possible, we made no attempt to optimize the running time.

All problems we consider in this paper will be self-reducible in the sense of [[JVV86](#)]. Hence, approximate counting and sampling are equivalent under polynomial-time reductions and we will move fluidly between them. In particular, [Theorem 1.2](#) also implies a polynomial-time algorithm for approximately sampling a uniformly random common basis.

As a special case, [Theorem 1.2](#) gives an FPRAS for counting *strong orientations* and *undirected Euler tours*, resolving the approximate counting questions in [[Jer24](#), Section 3.5] and [[GŠ10](#)], respectively; see [Appendices C.1](#) and [C.2](#). A variant of [Theorem 1.2](#) for weighted common bases (proved below) applies to *network reliability*, where the goal is to estimate the probability of directed root-to-all reachability, or of an unpaired vertex-disjoint linkage between two terminal sets, under independent edge failures; see [Appendices C.3](#) and [C.4](#). The former settles the general directed reachability question in [[GJ19](#), Section 7]; the latter extends the two-terminal result of [[FFG26](#)] to multiple disjoint paths. Finally, we mention that [Theorem 1.2](#) also yields a new FPRAS for perfect matchings in bipartite graphs (and the permanent of nonnegative matrices more generally).

¹By the standard median trick, returning the median of $O(\log(1/\delta))$ independent calls boosts the success probability to $1 - \delta$ for any $0 < \delta < 1$.

1.1 Subset Convolution of Log-Concave Measures

[Theorem 1.2](#) is an important special case of “intersections” of multi-affine log-concave measures, which we now describe. Let $\mu : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ be a nonnegative weight function, which need not be normalized to sum to 1, and define its multi-affine generating polynomial with variables $\{x_u : u \in \mathcal{U}\}$ by

$$g_\mu(\mathbf{x}) := \sum_{S \subseteq \mathcal{U}} \mu(S) \cdot \mathbf{x}^S, \quad (1)$$

where $\mathbf{x}^S := \prod_{u \in S} x_u$. We write $[\mathbf{x}^S] g_\mu$ for the coefficient $\mu(S)$ of $\mathbf{x}^S$ in the polynomial g_μ .

Definition 1.4 (Subset Convolution; [Bjö+07]). Let $\mu, \nu : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ be two nonnegative weight functions. We define their *subset convolution* by the quantity

$$(\mu * \nu)(T) := \sum_{S \subseteq T} \mu(S) \cdot \nu(T \setminus S) = [\mathbf{x}^T] (g_\mu(\mathbf{x}) \cdot g_\nu(\mathbf{x})), \quad \forall T \subseteq \mathcal{U}.$$

Definition 1.5 (Log-Concave Polynomial; [Gur09; AOV18; BH20]). A nonzero polynomial g with nonnegative coefficients is *log-concave* on $\mathbb{R}_{\geq 0}^{\mathcal{U}}$ if

$$\nabla^2 \log g(\mathbf{x}) \preceq 0, \quad \forall \mathbf{x} \in \mathbb{R}_{>0}^{\mathcal{U}}.$$

We will design an FPRAS for the subset convolution (and a natural generalization of it) of two nonnegative weight functions μ, ν assuming g_μ, g_ν are log-concave on $\mathbb{R}_{\geq 0}^{\mathcal{U}}$. To bound the running time, let us also define the *coefficient range* $\mathcal{R}_\mu$ for a weight function μ as

$$\mathcal{R}_\mu := \frac{\max_{S \in \text{supp}(\mu)} \mu(S)}{\min_{S \in \text{supp}(\mu)} \mu(S)} \geq 1. \quad (2)$$

To guarantee that our algorithms run in polynomial-time, we will only require that $\mathcal{R}_\mu, \mathcal{R}_\nu \leq \exp(\text{poly}(n))$, which is easily satisfied in most applications of interest. The algorithms below take these parameters as input; upper bounds supplied by the user may be used instead, with the same replacement in the running-time bounds.

Theorem 1.6 (FPRAS for Log-Concave Subset Convolution). Let $\mu, \nu : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ be two nonnegative weight functions given by evaluation oracles,² and suppose g_μ, g_ν are log-concave. Assume initial sets $S_0, T_0 \subseteq \mathcal{U}$ satisfying $\mu(S_0) > 0, \nu(T_0) > 0$, upper bounds on $\mathcal{R}_\mu, \mathcal{R}_\nu$, and an error parameter $0 < \varepsilon \leq 1$ are given. Then there is an FPRAS for any specified coefficient

$$Z_\alpha := [\mathbf{x}^\alpha] (g_\mu(\mathbf{x}) \cdot g_\nu(\mathbf{x})), \quad \alpha \in \{0, 1, 2\}^{\mathcal{U}}$$

that uses at most

$$\tilde{O} \left(\frac{n^{11} \cdot (n + \log(\mathcal{R}_\mu \cdot \mathcal{R}_\nu))^2}{\varepsilon^2} \right)$$

arithmetic operations and evaluation oracle calls.

Proof of Theorem 1.2. This is an immediate corollary of [Theorem 1.6](#) in the case $\alpha = \mathbf{1}$ by setting $\mu(S) = \mathbf{1}[S \in \mathcal{B}(\mathcal{M})]$ and $\nu(T) = \mathbf{1}[\mathcal{U} \setminus T \in \mathcal{B}(\mathcal{N})]$, both of which can be evaluated given independence oracles for $\mathcal{M}, \mathcal{N}$. Log-concavity of g_μ, g_ν was established in [AOV18; BH20], and we can find initializing sets S_0, T_0 by greedy algorithms. $\square$

We highlight three further applications of the more general [Theorem 1.6](#) that aren’t captured directly by [Theorem 1.2](#). Each admits an FPRAS; precise definitions and reductions appear in [Appendix C](#).

- **Mixed discriminants.** Approximate the mixed discriminant of positive semidefinite matrices, a generalization of the permanent [Cel+17]; see [Appendix C.5](#). This resolves the mixed-discriminant question in [Gur07, Section 4, Problem 2], disproving the conjectured nonexistence of an FPRAS.

²This means that there is a black-box that returns the weight $\mu(S)$ (resp. $\nu(S)$) on input $S \subseteq \mathcal{U}$. The number of calls to this black-box is counted in the overall running time of our algorithms.

- **The six-vertex model.** Estimate its partition function on loopless 4-regular multigraphs when the nonnegative weights satisfy $a \leq b + c$, $b \leq a + c$, and $c \leq a + b$; see [Appendix C.6](#). For positive weights, this completes the general-graph approximation classification left open by [\[CLL19, Section 6\]](#).
- **Quantum Max-Cut and the EPR model.** Estimate the optimal value of Quantum Max-Cut on bipartite graphs and of the EPR model on arbitrary graphs, with nonnegative edge weights; see [Appendix C.7](#). This resolves the bipartite complexity question in [\[Gha23, Open question 3.4\]](#) and proves the EPR-in-BPP conjecture of [\[MS26, Conjecture 1\]](#).

1.2 Counting and Sampling for Hadamard Products via Curvature

The key input into our approximate counting and sampling algorithms is a Poincaré Inequality for the Hadamard product of two probability measures μ, ν assumed to satisfy suitable “curvature” conditions. For the Hadamard-product dynamics and Poincaré inequality below, let $\mu, \nu : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ have a common nonempty support Ω . For weights with different supports, first restrict both to their support intersection and set them to zero outside it.

Definition 1.7 (Hadamard Product). Let $\rho = \mu \odot \nu$ denote the probability measure supported on Ω defined by $(\mu \odot \nu)(S) \propto \mu(S) \cdot \nu(S)$ for all $S \subseteq \mathcal{U}$.

Let us now introduce a natural local Markov chain that samples from $\rho = \mu \odot \nu$. Write $S + u := S \cup \{u\}$ and $S - u := S \setminus \{u\}$. For each $u \in \mathcal{U}$, define the Laplacian $\mathcal{L}_u$ via the following operator on functions $f : 2^{\mathcal{U}} \rightarrow \mathbb{R}$, evaluated at $S \in \Omega$:

$$(\mathcal{L}_u f)(S) = r_u(S) \cdot (f(S + u) - f(S - u)) \quad \text{where } r_u(S) := \begin{cases} -\mu(S + u)/\mu(S - u), & \text{if } u \notin S \\ \nu(S - u)/\nu(S + u), & \text{if } u \in S \end{cases}$$

gives the relative birth and death rates. The operator $\mathcal{L}_u$ encodes the updates at coordinate u . The full Laplacian is defined as

$$\mathcal{L} := \sum_{u \in \mathcal{U}} \mathcal{L}_u. \quad (3)$$

It acts on $S \in \Omega$ with infeasible additions and deletions omitted.

Fact 1.8. $\mathcal{L}_u$ is reversible with respect to ρ for all $u \in \mathcal{U}$, i.e. $\langle f, \mathcal{L}_u g \rangle_\rho = \langle \mathcal{L}_u f, g \rangle_\rho$ for all functions $f, g : 2^{\mathcal{U}} \rightarrow \mathbb{R}$. Moreover, $\mathcal{L}_u$ is positive semidefinite, i.e. $\langle f, \mathcal{L}_u f \rangle_\rho \geq 0$ for all $f : 2^{\mathcal{U}} \rightarrow \mathbb{R}$.

Remark 1.9 (Discrete-time implementation). The continuous-time chain with generator $-\mathcal{L}$ can be simulated by a simple discrete-time chain. Let $B \geq \max_{S \in \Omega, u \in \mathcal{U}} |r_u(S)|$ be any known bound on the rates. From the current state $S \in \Omega$:

- (1) With probability $1/2$, stay at S .
- (2) Otherwise, choose $u \in \mathcal{U}$ uniformly at random, and
 - if $u \notin S$ and $S + u \in \Omega$, move to $S + u$ with probability $\mu(S + u)/(B\mu(S))$;
 - if $u \in S$ and $S - u \in \Omega$, move to $S - u$ with probability $\nu(S - u)/(B\nu(S))$;
 - otherwise, stay at S .

Each probability in (2) is at most 1 by the choice of B . The resulting transition matrix is $P = I - \mathcal{L}/(2nB)$, which is lazy and reversible with respect to ρ , and its spectral gap is that of $\mathcal{L}$ divided by $2nB$. The Poincaré inequality below therefore gives the mixing bound [Eq. \(12\)](#) with $R = nB$.

Having defined the dynamics, let us now introduce the local “curvature” criterion that will be used to deduce a Poincaré Inequality for $\mathcal{L}$.

Definition 1.10 ((Upward) κ -Spectral Curvature). Fix a parameter $\kappa \in \mathbb{R}$. For a nonnegative weight function $\mu : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ that is not identically zero, and each $S \in \text{supp}(\mu)$, define the $\mathcal{U} \times \mathcal{U}$ (upward) κ -curvature matrix³

$$A_S^{\mu, \kappa}(u, v) := \begin{cases} \frac{\mu(S+v)^2}{\mu(S)^2} - \kappa \cdot \frac{\mu(S+v)}{\mu(S)}, & \text{if } u, v \notin S, u = v \\ \frac{\mu(S+u)}{\mu(S)} \cdot \frac{\mu(S+v)}{\mu(S)} - \frac{\mu(S+u+v)}{\mu(S)}, & \text{if } u, v \notin S, u \neq v \\ 0, & \text{otherwise} \end{cases}$$

³Note that the only nonzero entries A_S^{μ} come from elements in $\mathcal{U} \setminus S$.

We say μ has (upward) κ -spectral curvature if $A_S^{\mu,\kappa} \succeq 0$ for every $S \in \text{supp}(\mu)$.

Remark 1.11. In this paper, we are primarily interested in the $\kappa \geq 0$ regime. However, we note that κ can be any real number; see [Appendix A.7](#) for examples.

Remark 1.12. Unlike the influence/correlation/covariance matrices appearing in the theory of spectral independence [[ALO20](#)] and localization schemes [[CE22](#)], whose entries are #P-hard to compute exactly, the entries of $A_S^{\mu,\kappa}$ are straightforward to compute for any $S \in \text{supp}(\mu)$. Hence, for many applications, verifying spectral curvature can be done easily by hand.

The notion of curvature appearing in [Definition 1.10](#) has previously appeared in the literature in various guises; we discuss these connections in greater depth in [Section 1.3](#). Before we continue, note that we can also define a “downward” analog of κ -spectral curvature by considering the (upward) κ -spectral curvature of the dual μ^* , defined as follows.

Definition 1.13 (Dual). Let $\mu : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ be nonzero. We define its dual $\mu^* : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ by $\mu^*(S) := \mu(\mathcal{U} \setminus S)$ for all $S \subseteq \mathcal{U}$.

We elaborate on the role of the “downward” analog of κ -spectral curvature in [Section 4](#). Our main technical theorem is the following.

Theorem 1.14 (Main Poincaré Inequality). Let $\mu, \nu : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ have a common nonempty support Ω , connected under single-element additions and deletions, and let $\rho = \mu \odot \nu$ be their Hadamard product, which is supported on Ω . If μ has η -spectral curvature and ν^* has κ -spectral curvature, where $\eta, \kappa \in \mathbb{R}$ satisfy $\eta + \kappa > 0$, then ρ obeys the Poincaré Inequality

$$(\eta + \kappa) \cdot \text{Var}_{\rho}[f] \leq \langle f, \mathcal{L}f \rangle_{\rho}, \quad \forall f : \Omega \rightarrow \mathbb{R}.$$

A family $\mathcal{F} \subseteq 2^W$ is downward closed if it contains every subset of each of its members, and upward closed if it contains every superset contained in W of each of its members.

Remark 1.15. By [Lemma A.1](#), a positive spectral-gap bound from [Theorem 1.14](#) forces Ω to be upward or downward closed after removing coordinates that are constant on Ω .

Through [Theorem 1.14](#), we can design an FPRAS for the subset convolution of a pair of downward closed measures that satisfy nonnegative curvature.

Theorem 1.16. Suppose $\mu, \nu : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ have downward closed supports and upward spectral curvatures η, κ , with $\eta + \kappa \geq 0$. Put $\gamma := \max\{\eta, \kappa\}$; if $\gamma > 0$, let $\chi \geq 1$ be a supplied bound, polynomial in n , such that, for each $\omega \in \{\mu, \nu\}$,

$$\omega(S + u) \leq \chi \cdot \gamma \cdot \omega(S), \quad \forall S \in \text{supp } \omega, \forall u \notin S.$$

In all other cases, take $\chi = 1$ without imposing this bound. Given value oracles for μ, ν , upper bounds on $\mathcal{R}_{\mu}, \mathcal{R}_{\nu}$, and an error parameter $0 < \varepsilon < 1$, there is an FPRAS for any specified coefficient

$$Z_{\alpha} := [\mathbf{x}^{\alpha}](g_{\mu}(\mathbf{x}) \cdot g_{\nu}(\mathbf{x})), \quad \alpha \in \{0, 1, 2\}^{\mathcal{U}}$$

that uses at most

$$\tilde{O}\left(\frac{n^{11}(n + \log(\mathcal{R}_{\mu}\mathcal{R}_{\nu}))^2}{\varepsilon^2} + \frac{n^5\chi(n + \log(\mathcal{R}_{\mu}\mathcal{R}_{\nu}))}{\varepsilon^3}\right)$$

arithmetic operations and evaluation oracle calls.

Remark 1.17. [Theorem 1.16](#) can be viewed as a generalization of [Theorem 1.6](#). For log-concave measures, translating $g_{\mu}(\mathbf{x}), g_{\nu}(\mathbf{x})$ to $g_{\mu}(\mathbf{1} + \mathbf{x}), g_{\nu}(\mathbf{1} + \mathbf{x})$ gives downward-closed measures with zero curvature by [Proposition 1.22](#). After reducing to homogeneous polynomials whose degrees sum to n (see [Appendix A.1](#)), these translations preserve the coefficient of $\mathbf{x}^{\mathcal{U}}$. The translated weights, i.e. the coefficients of $g_{\mu}(\mathbf{1} + \mathbf{x}), g_{\nu}(\mathbf{1} + \mathbf{x})$, are respectively

$$\hat{\mu}(S) = \sum_{T \supseteq S} \mu(T), \quad \hat{\nu}(S) = \sum_{T \supseteq S} \nu(T).$$

These sums each admit an FPRAS since μ, ν are log-concave [[Ana+19](#)]. Using sufficiently accurate estimates of the queried weights and controlling the accumulated sampling error gives the reduction, with additional polynomial overhead. We state [Theorem 1.6](#) separately to give a direct algorithm using the original evaluation oracles and a sharper running-time bound.

Remark 1.18 (Optimality of the curvature threshold). The condition $\eta + \kappa \geq 0$ in [Theorem 1.16](#) is optimal among constant lower bounds on the curvature sum. For every fixed $\delta > 0$, there is no FPRAS under the weaker condition $\eta + \kappa > -\delta$, unless $\text{NP} = \text{RP}$. This holds even when $\mu = \nu$ have downward-closed support, polynomial-time value oracles, and $\log(\mathcal{R}_\mu \mathcal{R}_\nu) = O(n \log n)$. The proof is deferred to [Appendix A.3](#).

Additional Applications of [Theorem 1.16](#). As noted in [Remark 1.17](#), [Theorem 1.16](#) generalizes [Theorem 1.6](#). However, we highlight a couple interesting applications of the former that aren't captured by the latter. In particular, we show that each of the following approximate counting problems on simple graphs admits an FPRAS. Both counts include all sizes; precise definitions and proofs appear in [Appendix C.11](#).

- **Rainbow matchings.** Count matchings with no repeated edge color, for an arbitrary edge coloring; see [Corollary C.24](#). This counts all sizes and does not address the rainbow perfect-matching enumeration studied by [\[PS13\]](#).
- **Degree-constrained forests.** Count forests satisfying prescribed upper bounds on the vertex degrees. When every bound is two, this counts unions of vertex-disjoint paths; see [Corollary C.25](#). This adds vertex-degree bounds to the forest-counting result of [\[Ana+19\]](#).

1.3 Connections with Previous Notions

We compare the spectral curvature condition in [Definition 1.10](#) with earlier notions that appeared in the literature.

Trickle-Down Condition for Field Dynamics Previous work of [\[Che+25a\]](#) derived a sufficient condition for the so-called *field dynamics* [\[Che+24\]](#) to satisfy a Poincaré Inequality based on a certain “pairwise dependency matrix”. This matrix turns out to be equivalent to the curvature matrix appearing in [Definition 1.10](#) up to a simple transformation, which we now describe.

Let $\mu : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ be a nonzero weight function with downward closed support. For each $S \in \text{supp}(\mu)$, put

$$D_S := \text{diag} \left(\frac{\mu(S+u)}{\mu(S)} : u \notin S, \mu(S+u) > 0 \right).$$

The matrices below are indexed by the feasible additions $u \notin S$ with $\mu(S+u) > 0$.

Definition 1.19 (Pairwise dependency matrix [\[Che+25a\]](#)). Let $S \in \text{supp}(\mu)$. The *pairwise dependency matrix* of μ at S is the symmetric matrix M_S with entries

$$M_S(u, v) := \begin{cases} \frac{\mu(S+u+v) \cdot \mu(S)}{\mu(S+u) \cdot \mu(S+v)} - 1, & u \neq v, \\ 0, & u = v. \end{cases}$$

A straightforward calculation reveals the identity

$$A_S^{\mu, \kappa} = D_S \cdot (I - M_S - \kappa D_S^{-1}) \cdot D_S.$$

Here we restrict $A_S^{\mu, \kappa}$ to the feasible additions; downward closure makes all other rows and columns zero. Thus the trickle-down criterion for field dynamics $M_S \preceq I + (1 - \delta) D_S^{-1}$ for every S , with $0 < \delta < 1$, in [\[Che+25a, Theorems 1.9 and 1.14\]](#) is exactly upward $(\delta - 1)$ -spectral curvature. The stronger condition $M_S \preceq I$ is the case $\kappa = 0$.

A shorter proof of the spectral gap bound under the aforementioned trickle-down criterion appears in [\[Göb+26\]](#), which is based on an integrated Bochner method. Our proof of [Theorem 1.14](#) is a direct generalization of their argument.

Remark 1.20. For $\delta > 0$, the δ -ultra-log-concavity of [\[CW26, Definition 3.5\]](#) extends this criterion to measures on $\mathbb{N}^{\mathcal{U}}$ with downward-closed support. On the Boolean cube, it is exactly upward $(\delta - 1)$ -spectral curvature; the case $\delta = 1$ corresponds to zero upward curvature.

Log-concave polynomials. Log-concavity of the generating polynomial gives another way to bound spectral curvature.

Definition 1.21 (Fractional log-concavity [Ali+21]). Let $0 < \alpha \leq 1$, and let $p \in \mathbb{R}[x_1, \dots, x_n]$ be a nonzero polynomial with nonnegative coefficients. We say that p is α -fractionally log-concave if $x \mapsto \log p(x_1^\alpha, \dots, x_n^\alpha)$ is concave on $\mathbb{R}_{\geq 0}^n$.

For multiaffine p , this gives a curvature bound for the coefficients of $p(\mathbf{1} + \mathbf{x})$.

Proposition 1.22. Let p be nonzero, multiaffine, and α -fractionally log-concave, with nonnegative coefficients and $0 < \alpha \leq 1$. The coefficient measure $\hat{\mu}$ defined by $g_{\hat{\mu}}(\mathbf{x}) := p(\mathbf{1} + \mathbf{x})$ has downward-closed support and upward $(1 - \alpha^{-1})$ -spectral curvature.

The proof is given in [Appendix A.6](#).

Remark 1.23. Lorentzian polynomials extend the theory of multi-affine homogeneous log-concave polynomials to polynomials with M-convex supports [BH20, Theorems 2.30 and 3.10]. For a nonzero multiaffine Lorentzian polynomial p , [Proposition 1.22](#) with $\alpha = 1$ gives zero upward spectral curvature for the coefficient measure of $p(\mathbf{1} + \mathbf{x})$.

0-spectral curvature is closely related to log-concavity through the trickle-down principle [Opp18] and analogous local-to-global results for polynomials [Ana+19; BH20]. These polynomial results are stated for homogeneous polynomials, meaning that every monomial has the same degree. [Che+25a] gives a direct extension to generating polynomials that need not be homogeneous, for measures with downward-closed support, yielding the following result.

Proposition 1.24 ([Che+25a, Proposition 3.8]). Let $\mu : 2^U \rightarrow \mathbb{R}_{\geq 0}$ have downward closed support. If μ has (upward) 0-spectral curvature, then g_μ is log-concave.

1.4 Prior Work

Trickle-down and field dynamics. Trickle-down arguments grew out of earlier local-to-global ideas [Gar73; KM17; DK17] and establish rapid mixing of “global” random walks by analyzing related “local” random walks, which are usually easier to study [Opp18; KO20; AL20; Ana+19; ALO20]. Refinements and extensions appear in [GM20; ALO21; AO23; GK23; AKV24; LLO25; LO25; GZ26; BLO26]. Related sampling results appear in [CLV23; CLV21; Fen+22; Bla+22; Liu21; Ana+24]. Field dynamics [Che+24] provides a mechanism for reducing sampling from a “hard” distribution to sampling from relatively “easier” conditional distributions with smaller external fields. This approach and related methods [Eld13; EKZ22; CE22; Che+22] have led to rapid-mixing results for the hardcore model (and more general antiferromagnetic two-spin systems) in the tree-uniqueness regime and at the uniqueness threshold [Che+25b; Che+26b]. Our curvature condition builds on top of a criterion that lies between these two lines of work [Che+25a].

Bochner-Type Methods. Bochner-type methods bound the spectral gap of random walks by relating it to the local “curvature” of the underlying space. The terminology originates from Bochner’s formula in Riemannian geometry [Boc46; Lic58], and its application to continuous diffusions in high-dimensional probability is classical [BÉ85]; see also [BGL14]. Adaptations of these techniques to discrete spaces appear in [Bou+06; CP07; Cap08; CDP09; KKO13; DP13; Joh17; JY17; WZ21; Ped25]. A recent flurry of work, starting with [Göb+26], applies a variant of this method and closely related arguments for spin systems on random regular and planar graphs [Göb+26; GZ26], the Sherrington–Kirkpatrick model [Wan26; BLO26], graph colorings [CL26a; Haq26], and “ultra-log-concave” distributions [CW26]. It also gives simpler proofs of known trickle-down results; see [CL26b] and references therein. Our proof of [Theorem 1.14](#) uses this method.

Geometry of polynomials. Properties of generating polynomials have been used to prove inequalities for combinatorial counts and probabilities [BBL09; Gur09; Huh12; HK12; HW17; AHK18; AOV18; Ana+18; BH20].

Gurvits capacity and related entropy maximization programs connect these inequalities to approximate counting and optimization [Gur06; Gur07; NS16; SV17; GL21]. Applications include permanents [LSW00; Gur06], bases and common bases of matroids [AO17; SV17; AOV18], contingency tables and integer flows [Bar09; BLP23], and lattice points in bounded totally unimodular

polytopes [LM26]. These methods often give polynomial-time algorithms with exponential approximation ratios. Recent work has improved deterministic approximation guarantees for the permanent [Ana26; KLM26; DJ26].

For sampling and randomized counting on matroids, earlier algorithms covered special classes [FM92; JS02; AOR16]. Later advances based on log-concavity handled general matroids and improved the running times [Ana+19; CGM21; Ana+21; AD20].

Annealing and adjusting weights. The Markov-chain approach to approximating the permanent was initiated by Broder [Bro86] and developed by Jerrum and Sinclair [JS89]. Jerrum, Sinclair, and Vigoda [JSV04] gave the first FPRAS for the permanent of an arbitrary nonnegative matrix. This algorithm and subsequent FPRASes for the permanent [Bez+08; CVY26; Che+26a] and two-terminal reliability [FFG26] add extra configurations to make sampling easier, and gradually change the input weights (annealing; see [ŠVV09; Guo+26]) while learning weights that keep desired configurations, such as perfect matchings, from becoming rare. Those algorithms learn correction weights along a carefully designed annealing path; our method chooses fields directly by stochastic gradient descent. The resulting fields make the desired pairs frequent. We then estimate the total weight using the standard reduction from sampling to counting [JVV86]. See Section 2.3 for a detailed discussion.

1.5 Acknowledgements

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AI Disclosure For transparency, we detail the progression of this project and the role AI tools played. While the authors did provide many high level directions, we emphasize that AI contributions were critical to the success of this project.

This project began with a much more modest problem, namely establishing rapid mixing of Glauber dynamics for sampling from the distribution $\mu(S) \propto \lambda^{|S|}$ for common independent sets S of all sizes (see Appendix D.1). Although the authors were unaware of any previous work formulating this specific problem, it is a somewhat obvious generalization of the monomer-dimer model on all matchings of a graph, for which Glauber dynamics provably mixes rapidly for all λ [JS89; CLV21; Che+25c; MY26; CCZ26]. Based on this, as well as rapid mixing for a similar distribution over all independent sets of a single matroid, the authors conjectured that no dynamical phase transition occurs, and that the mixing time remains polynomial for every fixed $\lambda \geq 0$. This conjecture was previously posed by the authors, for instance, during “The geometry of polynomials in combinatorics and sampling” workshop at the American Institute of Mathematics in early March 2025. The main motivation for studying this specific problem was two-fold:

- Its resolution would provide a first stepping stone towards an FPRAS for common bases of two matroids, paralleling the chronological development of the first FPRAS for the permanent of nonnegative matrices.
- Independent of whether or not it would eventually lead to an FPRAS for common bases of two matroids, the authors believed this problem was interesting because all previously known techniques for proving rapid mixing appear to fail for this model: There are simple examples of pairs of matroids for which the spectral-independence parameter of the resulting distribution on common independent sets is $\Omega(\text{poly}(n))$.⁴ It is also believed that constructing canonical paths or flows with polynomial congestion would be hard in general.

⁴For example, at $\lambda = 1$, the monomer-dimer model on an even cycle of length $\Theta(n^{1/3})$ with each edge replaced by $\Theta(n^{2/3})$ parallel edges has $\Theta(n^{1/3})$ -spectral-independence, where n is the total number of edges; see [Che+25d] for the parallel-edge construction. This bipartite graph’s matchings are exactly the common independent sets of two partition matroids.

In light of recent exciting work of [Göb+26], the authors suspected that (integrated) Bochner-type methods had a chance of overcoming these obstacles. The authors instructed GPT 5.6 Sol Ultra to take inspiration from [Göb+26] and prove rapid mixing for all $\lambda \geq 0$ using an integrated Bochner-type argument. Initially, GPT 5.6 Sol Ultra worked on this task for 17 hours without success. Fortunately, GPT 6 Astra Ultra was released at this time, and it was able to solve the problem with an additional 3 hours. The result was a fairly short and simple proof, parts of which are reproduced in Section 4, that was tailored to this specific problem. During the process of digesting this proof, the authors independently realized that the argument works much more generally for intersections of probability measures satisfying suitable “curvature” conditions, and that this notion of “curvature” is essentially equivalent to a notion contained in prior work of [Che+25a]. This is how we came to the formulation of Theorem 1.14.

Our progress on sampling common independent sets of matroids led us to ask whether similar ideas could give an FPRAS for common bases. To explore this question, we provided ChatGPT Pro with several separate drafts:

- a revised note on sampling common independent sets;
- a revised note on a common framework for recent progress on Bochner-type methods;
- a draft on a faster FPRAS for the permanent [Che+26a].

ChatGPT first identified an obstruction to obtaining an FPRAS directly from our sampler for common independent sets. We replied that this was understandable and suggested an annealing approach, as in algorithms for the permanent. It agreed, and after a further prompt to “keep trying,” it found a solution.

While studying the proof produced by ChatGPT, we recognized the underlying stochastic gradient descent framework (Section 2.3). With further help from ChatGPT, we generalized the FPRAS for common bases to subset convolution, both for measures with log-concave generating polynomials and for measures with downward closed supports satisfying our nonnegative curvature condition (Theorems 1.6 and 1.16). We also used ChatGPT to search for reductions of other known problems to these subset-convolution problems (see Appendix C).

2 Technical Overview

2.1 Poincaré Inequality via an Integrated Bochner Method

We prove Theorem 1.14 in Section 4 by an integrated Bochner-type method, which bounds the spectral gap of a Laplacian using its square. The following abstract criterion holds for the Laplacian $\mathcal{L}$ of any irreducible reversible Markov chain; see Section 3.3 for the notation and proof.

Proposition 2.1. *Let $\mathcal{L}$ be the Laplacian of an irreducible reversible Markov chain on a finite state space Ω , with stationary distribution π . Then its spectral gap satisfies*

$$\gamma := \inf_{f \text{ nonconstant}} \frac{\langle f, \mathcal{L}f \rangle_\pi}{\text{Var}_\pi[f]} = \inf_{f \text{ nonconstant}} \frac{\langle \mathcal{L}f, \mathcal{L}f \rangle_\pi}{\langle f, \mathcal{L}f \rangle_\pi}.$$

We apply Proposition 2.1 to the Laplacian $\mathcal{L}$ defined in Eq. (3). The common support Ω is connected under single-element additions and deletions, and every feasible transition has positive rate, so the chain with Laplacian $\mathcal{L}$ is irreducible. Thus, to prove $\gamma \geq \eta + \kappa$, it suffices to show

$$\langle \mathcal{L}f, \mathcal{L}f \rangle_\rho \geq (\eta + \kappa) \cdot \langle f, \mathcal{L}f \rangle_\rho \quad \forall f : \Omega \rightarrow \mathbb{R}.$$

We prove this by writing $\langle f, (\mathcal{L}^2 - (\eta + \kappa) \cdot \mathcal{L})f \rangle_\rho$ as a sum-of-squares (Proposition 4.2), taking heavy inspiration from [Göb+26]. The spectral-curvature conditions make the terms involving the curvature matrices nonnegative, and the remaining terms are squared second differences of f with nonnegative weights.

2.2 Reduction to the Homogeneous Log-Concave Case

The following special case of Theorem 1.6 (and Theorem 1.16) is the most interesting case. It is also, in some sense, the most difficult.

Theorem 2.2 (Simplified version of [Theorem 5.1](#)). Let $\mu : \binom{\mathcal{U}}{a} \rightarrow \mathbb{R}_{\geq 0}$ and $\nu : \binom{\mathcal{U}}{b} \rightarrow \mathbb{R}_{\geq 0}$, extended by zero to $2^{\mathcal{U}}$, where $a, b \geq 0$ are integers with $a + b = n := |\mathcal{U}| \geq 1$. Suppose g_μ, g_ν are log-concave. Given value oracles for μ, ν , the coefficient ranges $\mathcal{R}_\mu, \mathcal{R}_\nu$, and one set in the support of each measure, there is an FPRAS for the subset-convolution value

$$Z_\cap := (\mu * \nu)(\mathcal{U}) = [\mathbf{x}^{\mathcal{U}}](g_\mu(\mathbf{x}) \cdot g_\nu(\mathbf{x})) = \sum_{S \subseteq \mathcal{U}} \mu(S) \cdot \nu(\mathcal{U} \setminus S).$$

In this subsection, we describe how we reduce [Theorem 1.16](#) (which generalizes [Theorem 1.6](#) as mentioned in [Remark 1.17](#)) to [Theorem 2.2](#) and [Theorem 1.14](#). A pictorial representation is provided in [Fig. 1](#), with detailed proofs deferred to [Appendix A.1](#). Throughout this subsection, we focus on the case of convolution where $\alpha = 1$; this is essentially without loss of generality, as one can simply pin the coordinates with exponent 0 or 2 for a general $\alpha \in \{0, 1, 2\}^{\mathcal{U}}$. We briefly explain our reductions as follows:

- For [Theorem 1.16](#) with $\max\{\eta, \kappa\} > 0$, we apply an external field to the measure with higher curvature. This external field is of magnitude $O(1/n)$ away from 1, which is enough to make the Poincaré constant in [Theorem 1.14](#) at least $\Omega(1/n)$, but small enough so that the new partition function remains ε -multiplicatively close to the original. We then apply the standard self-reduction [[JVV86](#)].
- For [Theorem 1.6](#), which includes [Theorem 1.16](#) with $\eta = \kappa = 0$, we split by degree into at most $n + 1$ homogeneous pairs and apply [Theorem 2.2](#) to each.

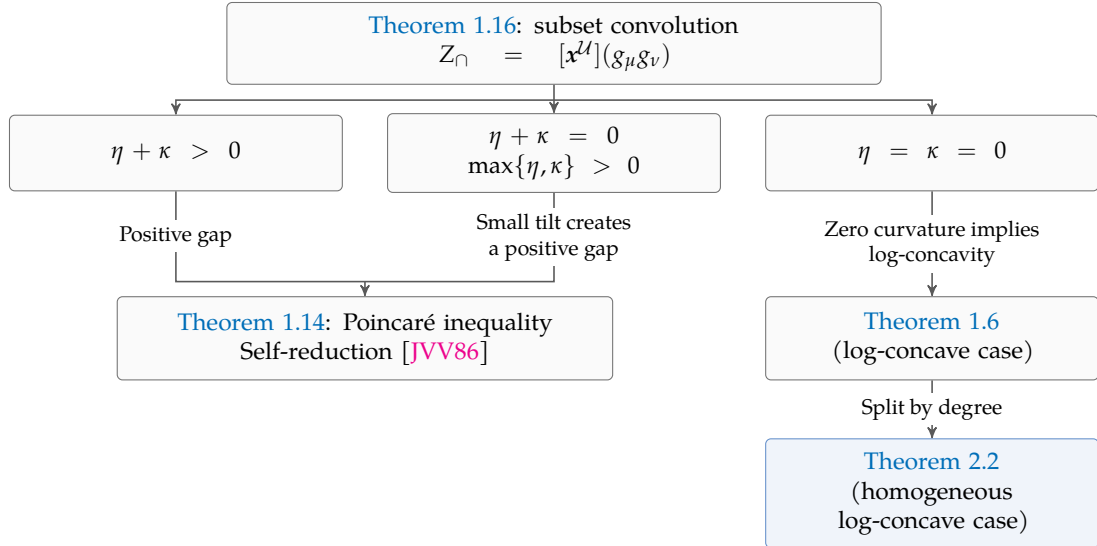


Figure 1: The three curvature cases in the proof of [Theorem 1.16](#). Arrows point toward the result used by each reduction.

2.3 From Poincaré to Counting via Convex Programming and Softening

In this section, we outline the proof of [Theorem 2.2](#). The algorithm and its proof of correctness provide a general framework to turn sampling from a “soft” intersection of two measures on sets into approximate counting for the corresponding “hard” intersection. This method may be of independent interest, and is based on the following general principle: A Poincaré inequality that holds uniformly over external fields turns subset convolution into a dual maximum entropy convex program. Variants of this principle based on stronger assumptions (e.g. log-concavity of the generating polynomial) with significantly weaker approximation guarantees have previously appeared in the literature [[Gur06](#); [AO17](#)]. We discuss this in greater detail in [Remark 2.7](#).

Let $\mu, \nu : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ be nonzero, put $n := |\mathcal{U}|$, and consider

$$Z_\cap := \sum_{S \subseteq \mathcal{U}} \mu(S) \cdot \nu(\mathcal{U} \setminus S).$$

When $Z_\cap > 0$, the direct target measure $\mu \odot \nu^*$, with weights proportional to $\mu(S)\nu(\mathcal{U} \setminus S)$, can be poorly connected under natural local moves: the hard constraint $T = \mathcal{U} \setminus S$ forces changes to the

two sets to be coordinated. To bypass this, it is natural to consider a variant of $\mu \odot \nu^*$ in which the hard constraint $T = \mathcal{U} \setminus S$ is “softened”.

For penalty vectors $\mathbf{t} \in (0, 1]^{\mathcal{U}}$, write $\mathbf{t}^R := \prod_{u \in R} t_u$. For external fields $\mathbf{z} \in \mathbb{R}_{>0}^{\mathcal{U}}$, the *soft intersection* is the distribution on $\text{supp } \mu \times \text{supp } \nu$ given by

$$\pi_{\mathbf{t}, \mathbf{z}}(S, T) \propto \mu(S) \cdot \nu(T) \cdot \mathbf{z}^S \mathbf{z}^T \cdot \mathbf{t}^{S \cap T},$$

that is, the Hadamard product

$$\pi_{\mathbf{t}, \mathbf{z}} = (\mu_{\mathbf{z}} \otimes \nu_{\mathbf{z}}) \odot \psi_{\mathbf{t}}, \quad \mu_{\mathbf{z}}(S) := \mu(S) \cdot \mathbf{z}^S, \quad \nu_{\mathbf{z}}(T) := \nu(T) \cdot \mathbf{z}^T, \quad \psi_{\mathbf{t}}(S, T) := \mathbf{t}^{S \cap T}.$$

Let

$$Z_{\mathbf{t}, \mathbf{z}} := \sum_{S, T \subseteq \mathcal{U}} \mu(S) \cdot \nu(T) \cdot \mathbf{z}^S \mathbf{z}^T \cdot \mathbf{t}^{S \cap T}$$

be its partition function. When all penalties equal a scalar $t \in (0, 1]$, we write a subscript t for the uniform penalty vector $\mathbf{t} = t \cdot \mathbf{1}$.

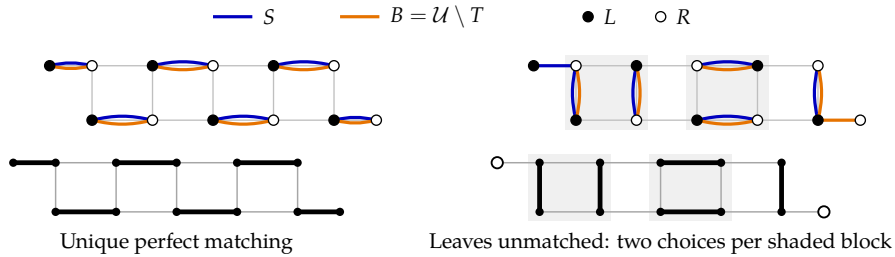
Note that $\pi_{\mathbf{t}, \mathbf{z}}$ penalizes pairs (S, T) where S, T have a large intersection. On the other hand, complementary pairs $T = \mathcal{U} \setminus S$ carry no penalty. Moreover, all such complementary pairs have the common factor $\mathbf{z}^{\mathcal{U}}$, so

$$Z_{\cap} = \frac{Z_{\mathbf{t}, \mathbf{z}}}{\mathbf{z}^{\mathcal{U}}} \cdot \pi_{\mathbf{t}, \mathbf{z}}(T = \mathcal{U} \setminus S), \quad \forall \mathbf{z} \in \mathbb{R}_{>0}^{\mathcal{U}}.$$

Note that in our application, since μ and ν are log-concave measures, so is $\mu \otimes \nu$. Hence, the down-up walk (a.k.a. the basis exchange walk) on $\mu \otimes \nu$ is rapidly mixing [Ana+19]. $\mu \otimes \nu$ corresponds to the case $t = 1$, and we can hope that when t is not too close to 0, local Markov chains for sampling from $\pi_{\mathbf{t}, \mathbf{z}}$ remain rapidly mixing. In the later part of the proof, we will formally prove this by an application of Theorem 1.14 and show that there is an efficiently implementable Markov chain that samples from $\pi_{\mathbf{t}, \mathbf{z}}$ in polynomial time provided $t \gtrsim 1/\text{poly}(n)$. This also allows us to approximate $Z_{\mathbf{t}, \mathbf{z}}$ by standard counting-to-sampling reductions [JVV86].

All that remains is to estimate $\pi_{\mathbf{t}, \mathbf{z}}(T = \mathcal{U} \setminus S)$ with good relative error, since putting it together with an accurate estimate of $Z_{\mathbf{t}, \mathbf{z}}$ yields an accurate estimate of $Z_{\cap}$. The naïve Monte Carlo method can work well as long as $\pi_{\mathbf{t}, \mathbf{z}}(T = \mathcal{U} \setminus S) \gtrsim 1/\text{poly}(n)$. The problem is that choosing t to be as small as $1/\text{poly}(n)$ does not automatically make $\pi_{\mathbf{t}, \mathbf{z}}(T = \mathcal{U} \setminus S)$ large. See Remark 2.3 for an example.

Remark 2.3. An inverse-polynomial t need not give complementary pairs inverse-polynomial probability when $\mathbf{z} = \mathbf{1}$. Consider the following ladder of Jerrum, Sinclair, and Vigoda [JSV04, Figure 2]. Blue S and orange $B = \mathcal{U} \setminus T$ are bases of the partition matroids on filled and hollow vertices, respectively; shared edges are drawn twice above and once below.



The pendant edges force a unique perfect matching M . Leaving both leaves unmatched gives exponentially many near-perfect matchings C , with two choices per shaded block. Completing C as shown gives a distinct soft configuration with $|S \cap T| = |S \setminus B| = 1$, hence only one factor of t . Their total weight makes the perfect matching exponentially unlikely.

Since $Z_{\cap}$ isn't affected by the choice of $\mathbf{z}$ and t , the hope is that we can still make $\pi_{\mathbf{t}, \mathbf{z}}(T = \mathcal{U} \setminus S) \gtrsim 1/\text{poly}(n)$ by finding a good $\mathbf{z} \in \mathbb{R}_{>0}^{\mathcal{U}}$. To find such a $\mathbf{z}$, it is natural to consider the following dual maximum entropy convex program.

Definition 2.4. Fix $t \in (0, 1]$. Write the external field as $\mathbf{z} = \mathbf{e}^{\mathbf{y}}$, where $\mathbf{y} \in \mathbb{R}^{\mathcal{U}}$ gives its logarithmic coordinates, and define

$$\Phi(\mathbf{y}) := \log Z_{\mathbf{t}, \mathbf{z}} - \sum_{u \in \mathcal{U}} y_u, \quad \partial_u \Phi(\mathbf{y}) = \pi_{\mathbf{t}, \mathbf{z}}(u \in S) - \pi_{\mathbf{t}, \mathbf{z}}(u \notin T). \quad (4)$$

We consider the following convex program:

$$\inf_{\mathbf{y} \in \mathbb{R}^{\mathcal{U}}} \Phi(\mathbf{y}). \quad (5)$$

The quantity $\exp(\inf_{\mathbf{y} \in \mathbb{R}^{\mathcal{U}}} \Phi(\mathbf{y})) = \text{Cap}_1(Z_{t,x})$ is the Gurvits capacity of $Z_{t,x}$, viewed as a polynomial in x ; see [Remark 2.7](#) for the definition and its connection to earlier work. The potential Φ is convex because its Hessian is the covariance matrix of $\mathbf{1}_S + \mathbf{1}_T - \mathbf{1}$ for $(S, T) \sim \pi_{t,e^{\mathbf{y}}}$. When $Z_{\cap} > 0$, it holds that

$$\pi_{t,e^{\mathbf{y}}}(T = \mathcal{U} \setminus S) = Z_{\cap} \cdot \exp(-\Phi(\mathbf{y})).$$

Hence, minimizing Φ maximizes the probability $\pi_{t,e^{\mathbf{y}}}(T = \mathcal{U} \setminus S)$.

The next lemma shows that, under suitable Poincaré inequality assumptions, any near stationary point $\mathbf{y}$ (i.e., points with $\|\nabla \Phi(\mathbf{y})\|_{\infty}$ small) make $\pi_{t,e^{\mathbf{y}}}(T = \mathcal{U} \setminus S)$ large when t is sufficiently small. By [Eq. \(4\)](#), the gradient of Φ can be estimated by sampling. Thus, we can find such a near stationary point by using standard stochastic gradient descent algorithms.

Lemma 2.5. *Let $\gamma > 0$. Suppose that for every $t \in (0, 1]^{\mathcal{U}}$ and $\mathbf{z} \in \mathbb{R}_{>0}^{\mathcal{U}}$, there is an irreducible continuous-time Markov chain reversible with respect to $\pi_{t,\mathbf{z}}$, with spectral gap at least γ . Each move updates only one of the two sets S, T . Whenever $u \in S \oplus T$, the total rate of moves removing its unique copy is at most 1. Then for every scalar $t \in (0, 1]$ and every $\mathbf{z} \in \mathbb{R}_{>0}^{\mathcal{U}}$, with $\mathbf{y} = \log \mathbf{z}$,*

$$\pi_{t,\mathbf{z}}(T \neq \mathcal{U} \setminus S) \leq n \left(\|\nabla \Phi(\mathbf{y})\|_{\infty} + \frac{4\sqrt{t}}{\gamma} \right). \quad (6)$$

The proof is deferred to [Section 5.2](#).

To summarize, we choose a small scalar penalty $t \asymp 1/\text{poly}(n)$ and implement stochastic gradient descent on Φ , seeking logarithmic coordinates $\mathbf{y}$ with small $\nabla \Phi(\mathbf{y})$ and setting $\mathbf{z} = e^{\mathbf{y}}$. Note that by [Eq. \(4\)](#), each gradient coordinate can be efficiently estimated as the difference of two empirical frequencies from random samples of $\pi_{t,\mathbf{z}}$. With an inverse-polynomial lower bound on γ , taking $\|\nabla \Phi(\mathbf{y})\|_{\infty} \leq 1/\text{poly}(n)$ sufficiently small makes complementary pairs occur with constant probability, by [Lemma 2.5](#). We can therefore estimate this probability $\pi_{t,\mathbf{z}}(T = \mathcal{U} \setminus S)$ from polynomially many samples as desired.

Application to the Homogeneous Log-Concave Case We apply the previous framework under the hypotheses of [Theorem 2.2](#), with degrees a, b and $a + b = n$. Here $\pi_{t,\mathbf{z}}$ is supported on $\binom{\mathcal{U}}{a} \times \binom{\mathcal{U}}{b}$, and we call it the *bases pair law*; see [Definition 5.4](#). The key input we need to verify is the Poincaré Inequality for $\pi_{t,\mathbf{z}}$ required by [Lemma 2.5](#). We describe here the argument at a high level; the formal argument is provided in [Section 5.1](#).

We first extend the measures to downward-closed set families. Define the weights $\hat{\mu}_{\mathbf{z}}, \hat{\nu}_{\mathbf{z}}$ through their generating polynomials by

$$g_{\hat{\mu}_{\mathbf{z}}}(\mathbf{x}) := g_{\mu}(\mathbf{1} + \mathbf{z}\mathbf{x}), \quad g_{\hat{\nu}_{\mathbf{z}}}(\mathbf{x}) := g_{\nu}(\mathbf{1} + \mathbf{z}\mathbf{x}), \quad \mathbf{z}\mathbf{x} := (z_u x_u)_{u \in \mathcal{U}}.$$

The *independent-set pair law* on $2^{\mathcal{U}} \times 2^{\mathcal{U}}$ ([Definition 5.2](#)) is the normalized Hadamard product

$$\hat{\pi}_{t,\mathbf{z}} := (\hat{\mu}_{\mathbf{z}} \otimes \hat{\nu}_{\mathbf{z}}) \odot \psi_t.$$

Conditioning $\hat{\pi}_{t,\mathbf{z}}$ on $\binom{\mathcal{U}}{a} \times \binom{\mathcal{U}}{b}$ gives exactly $\pi_{t,\mathbf{z}}$, and $\hat{\pi}_{t,\lambda\mathbf{z}}$ converges to $\pi_{t,\mathbf{z}}$ as $\lambda \rightarrow \infty$. The first factor has 0-spectral curvature, and the dual of the second factor has 1-spectral curvature ([Observations 5.8](#) and [5.9](#)). Thus [Theorem 1.14](#) gives a Poincaré bound with constant 1 for $\hat{\pi}_{t,\lambda\mathbf{z}}$, uniformly in λ . Taking $\lambda \rightarrow \infty$ transfers this bound to an exchange chain for $\pi_{t,\mathbf{z}}$ ([Lemma 5.5](#)), yielding an efficient sampler using only the original value oracles.

Connections and Additional Intuition

Remark 2.6 (Partition-matroid interpretation). The penalty ψ_t can be understood as an analogue of the random-cluster model, where there is a soft penalty for non-independent sets. Let $\mathcal{P}$ be the partition matroid on $\mathcal{U} \times \{1, 2\}$ allowing at most one copy of each u , and set $X := (S \times \{1\}) \cup (T \times \{2\})$. For uniform penalties $t_u = t \in (0, 1]$,

$$\psi_{t\mathbf{1}}(S, T) = t^{|S \cap T|} = t^{|X| - \text{rk}_{\mathcal{P}}(X)}.$$

Remark 2.7 (Gurvits capacity). For a nonzero finitely supported measure w on $\mathbb{Z}_{\geq 0}^n$, write $g_w(\mathbf{x}) = \sum_{\mathbf{k}} w(\mathbf{k}) \mathbf{x}^{\mathbf{k}}$. Its *Gurvits capacity* at $\alpha \in \mathbb{R}_{\geq 0}^n$ is

$$\text{Cap}_{\alpha}(g_w) := \inf_{\mathbf{x} \in \mathbb{R}_{>0}^n} \frac{g_w(\mathbf{x})}{\mathbf{x}^{\alpha}}.$$

The Gurvits capacity at $\alpha \in \text{supp } w$ is always an upper bound on $w(\alpha)$. To see this, define the distribution $p_z(\mathbf{k}) = w(\mathbf{k}) z^{\mathbf{k}} / g_w(z)$. If $\alpha \in \text{supp } w$, then

$$\text{Cap}_{\alpha}(g_w) = \frac{w(\alpha)}{\sup_{z>0} p_z(\alpha)} \geq w(\alpha).$$

Intuitively, it gives the best possible approximation of $w(\alpha)$ that we can get by only tuning external fields applied to the base measure.

After taking logarithms, the optimization over external fields above has a dual formulation as a minimum KL-divergence convex program with prescribed mean, or equivalently, a weighted maximum-entropy convex program [SV14]. For $\alpha \in \text{supp } w$, let

$$q^* = \arg \min_{q: \mathbb{E}_q[X] = \alpha} D_{\text{KL}}(q \parallel p_1),$$

where q ranges over probability distributions on $\text{supp } w$, $p_1 = w/g_w(\mathbf{1})$ is the normalized measure, and D_{KL} denotes Kullback–Leibler divergence. Entropy duality and the KL Pythagorean identity [CS04, Theorem 3.2] give

$$\log \frac{\text{Cap}_{\alpha}(g_w)}{w(\alpha)} = D_{\text{KL}}(\delta_{\alpha} \parallel q^*) = -\log q^*(\alpha),$$

where δ_{α} is the point mass at α . In other words, we adjust the original distribution as little as possible in KL divergence to obtain mean α . The approximation ratio is the reciprocal of the probability that this distribution assigns to α itself.

For every homogeneous *real stable* polynomial g of degree n in n variables with nonnegative coefficients,

$$\frac{n!}{n^n} \text{Cap}_{\mathbf{1}}(g) \leq [x_1 \cdots x_n] g \leq \text{Cap}_{\mathbf{1}}(g).$$

The upper bound follows from the definition of capacity. Gurvits [Gur06] proves the lower bound, showing that capacity overestimates the coefficient by at most a factor $n^n/n!$. For the permanent of a nonnegative matrix A , he uses the polynomial $g(\mathbf{x}) = \prod_{i=1}^n (\sum_{j=1}^n A_{ij} x_j)$; for mixed discriminants, he uses $g(\mathbf{x}) = \det(\sum_{i=1}^n x_i A_i)$, where the A_i are positive semidefinite. This approach has since been extended to other settings; see Section 1.4.

In our setting, the target coefficient $[x^{\mathcal{U}}] Z_{t,x} = Z_{\cap}$ is independent of t . For fixed t , the Gurvits capacity is

$$\text{Cap}_{\mathbf{1}}(Z_{t,x}) = \inf_{\mathbf{x} \in \mathbb{R}_{>0}^{\mathcal{U}}} \frac{Z_{t,x}}{\mathbf{x}^{\mathcal{U}}} = \exp \left(\inf_{\mathbf{y} \in \mathbb{R}^{\mathcal{U}}} \Phi(\mathbf{y}) \right).$$

When $Z_{\cap} > 0$, the probability interpretation above gives the approximation ratio

$$\frac{\text{Cap}_{\mathbf{1}}(Z_{t,x})}{Z_{\cap}} = \frac{1}{\sup_{z>0} \pi_{t,z}(T = \mathcal{U} \setminus S)}.$$

Under the hypotheses of Lemma 2.5, this ratio satisfies

$$1 \leq \frac{\text{Cap}_{\mathbf{1}}(Z_{t,x})}{Z_{\cap}} \leq \left(1 - \frac{4n\sqrt{t}}{\gamma} \right)^{-1}, \quad 4n\sqrt{t} < \gamma.$$

Indeed, Φ is bounded below and has a globally bounded Hessian, so it admits points $\mathbf{y}$ with arbitrarily small gradient; applying Lemma 2.5 to the corresponding fields $\mathbf{z} = e^{\mathbf{y}}$ gives the upper bound on the ratio. Thus decreasing t makes the capacity bound nearly tight.

Remark 2.8 (Comparison with permanent algorithms). The permanent algorithm of Jerrum–Sinclair–Vigoda [JSV04] learns correction weights along an annealing schedule to balance the total weight of perfect matchings with that of each near-perfect class, indexed by its pair of unmatched vertices. Our framework chooses external fields directly by stochastic gradient descent. It can serve as a black box for a broad class of counting problems admitting the above formulation, once the sampling and uniform Poincaré hypotheses are verified.

3 Preliminaries

3.1 Notation

For a finite ground set $\mathcal{U}$, write $n := |\mathcal{U}|$ and $\binom{\mathcal{U}}{r} := \{S \subseteq \mathcal{U} : |S| = r\}$. We use $S + u := S \cup \{u\}$, $S - u := S \setminus \{u\}$, and $S \oplus T := (S \setminus T) \cup (T \setminus S)$. The symbol $\mathbf{1}[\cdot]$ denotes an indicator, $\mathbf{1}_S$ the indicator vector of S , and $\mathbf{1}$ the all-ones vector.

Vectors are written in bold. Products $\mathbf{z}\mathbf{x}$, exponentials $e^{\mathbf{y}}$, and logarithms $\log \mathbf{z}$ are taken coordinatewise. We write $\mathbf{z}^S := \prod_{u \in S} z_u$ for $S \subseteq \mathcal{U}$, and $\mathbf{z}^\alpha := \prod_{u \in \mathcal{U}} z_u^{\alpha_u}$ for $\alpha \in \mathbb{Z}_{\geq 0}^{\mathcal{U}}$. For a polynomial p , $[z^\alpha]p$ denotes the coefficient of $\mathbf{z}^\alpha$ in p . For symmetric matrices, $A \succeq B$ means that $A - B$ is positive semidefinite.

A nonnegative set measure $\mu : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ need not be normalized. Its support is $\text{supp } \mu := \{S \subseteq \mathcal{U} : \mu(S) > 0\}$. An *external field* is a vector $\mathbf{z} \in \mathbb{R}_{>0}^{\mathcal{U}}$. Applying it to μ gives the unnormalized measure

$$\mu_{\mathbf{z}}(S) := \mu(S)\mathbf{z}^S.$$

When μ is nonzero, the corresponding probability distribution is $\mu_{\mathbf{z}}/g_{\mu}(\mathbf{z})$. We write $\mathbf{y} = \log \mathbf{z}$ for the logarithmic coordinates of the external field, so $\mathbf{z} = e^{\mathbf{y}}$. The product weights $(\mu \otimes \nu)(S, T) = \mu(S)\nu(T)$ are also unnormalized; $\mu \odot \nu$ denotes the normalized Hadamard product of Definition 1.7.

Logarithms are natural unless a base is specified. $\tilde{O}(\cdot)$ hides polylogarithmic factors.

3.2 Matroids

A *matroid* $\mathcal{M}$ on $\mathcal{U}$ has a family $\mathcal{I}(\mathcal{M}) \subseteq 2^{\mathcal{U}}$ of *independent sets* satisfying:

- $\emptyset \in \mathcal{I}(\mathcal{M})$.
- If $I \in \mathcal{I}(\mathcal{M})$ and $J \subseteq I$, then $J \in \mathcal{I}(\mathcal{M})$.
- If $I, J \in \mathcal{I}(\mathcal{M})$ and $|I| < |J|$, then $I \cup \{e\} \in \mathcal{I}(\mathcal{M})$ for some $e \in J \setminus I$.

A *basis* is a maximal independent set; write $\mathcal{B}(\mathcal{M})$ for the family of bases. The *rank* of $S \subseteq \mathcal{U}$ is

$$\text{rk}_{\mathcal{M}}(S) := \max\{|I| : I \subseteq S, I \in \mathcal{I}(\mathcal{M})\}.$$

All bases have size $\text{rk}_{\mathcal{M}}(\mathcal{U})$. An *independence oracle* tests membership in $\mathcal{I}(\mathcal{M})$. Greedily adding elements of S while preserving independence finds a largest independent subset of S , and hence its rank, with at most $|S|$ oracle queries. The *dual matroid* $\mathcal{M}^*$ is the matroid on $\mathcal{U}$ whose independent sets are the sets $I \subseteq \mathcal{U}$ such that $\mathcal{U} \setminus I$ contains a basis of $\mathcal{M}$. Equivalently, $\mathcal{B}(\mathcal{M}^*) = \{\mathcal{U} \setminus B : B \in \mathcal{B}(\mathcal{M})\}$; see [Oxl11] for further background.

The basis generating polynomial $g_{\mathcal{M}}(\mathbf{x}) := \sum_{B \in \mathcal{B}(\mathcal{M})} \mathbf{x}^B$ is log-concave [AOV18; BH20]. Additional matroid terminology and operations used in the applications are recalled in Appendix B.

3.3 Markov chains and mixing time

Let Ω be a finite state space and π a probability law on Ω with $\pi(x) > 0$ for every $x \in \Omega$. For $f, g : \Omega \rightarrow \mathbb{R}$, define

$$\langle f, g \rangle_{\pi} := \sum_{x \in \Omega} \pi(x) f(x) g(x), \quad \|f\|_{\pi}^2 := \langle f, f \rangle_{\pi}, \quad \mathbf{Var}_{\pi}[f] := \|f - \mathbb{E}_{\pi}[f]\|_{\pi}^2.$$

Let $r(x, y) \geq 0$, $x \neq y$, be transition rates on Ω . These define a continuous-time Markov chain $(X_t)_{t \geq 0}$: assign each directed edge $x \rightarrow y$ an independent Poisson clock of rate $r(x, y)$. When its

clock rings, the chain moves to y if it is currently at x ; otherwise, nothing happens. We assume the rates satisfy the *detailed balance equations*

$$\pi(x) r(x, y) = \pi(y) r(y, x) \quad \text{for all } x \neq y \in \Omega. \quad (7)$$

We then say that the chain is *reversible* with respect to π . All Markov chains used in this paper are finite and reversible. The positive Laplacian and Dirichlet form associated with these rates are

$$(\mathcal{L}_r f)(x) := \sum_{y \neq x} r(x, y) \cdot (f(x) - f(y)), \quad (8)$$

$$\mathcal{E}_r(f, g) := \langle f, \mathcal{L}_r g \rangle_\pi = \frac{1}{2} \sum_{\substack{x, y \in \Omega \\ x \neq y}} \pi(x) r(x, y) \cdot (f(y) - f(x)) \cdot (g(y) - g(x)). \quad (9)$$

The second equality in Eq. (9) follows from detailed balance. The chain has generator $-\mathcal{L}_r$, so its transition matrix $P_t(x, y) := \Pr[X_t = y \mid X_0 = x]$ satisfies

$$P_t = \exp(-t\mathcal{L}_r), \quad t \geq 0.$$

We omit the subscript of $\mathcal{L}_r$ when the rates are clear, and note that $\mathcal{E}_{c,r}(\cdot, \cdot) = c \cdot \mathcal{E}_r(\cdot, \cdot)$ for $c > 0$. We use $\mathcal{L}$ for a general Laplacian and reserve $\mathcal{L}$ for the Laplacian defined in Eq. (3).

Spectral gap and Poincaré inequalities. The chain is *irreducible* if, for every $x, y \in \Omega$, there is a sequence $x = x_0, \dots, x_k = y$ such that $r(x_{j-1}, x_j) > 0$ for $1 \leq j \leq k$. In this case, its *spectral gap* is

$$\gamma(\mathcal{L}_r) := \inf_{f \text{ nonconstant}} \frac{\mathcal{E}_r(f, f)}{\text{Var}_\pi[f]}, \quad (10)$$

the smallest nonzero eigenvalue of $\mathcal{L}_r$; see [LP17, Lemma 13.7]. Thus $\gamma(\mathcal{L}_r) \geq \gamma$ if and only if the following *Poincaré inequality* holds:

$$\gamma \cdot \text{Var}_\pi[f] \leq \mathcal{E}_r(f, f) \quad \text{for every } f : \Omega \rightarrow \mathbb{R}. \quad (11)$$

Mixing time. The continuous-time chain P_t has a lazy discrete-time implementation: choose $R > 0$ with $R \geq \max_x \sum_{y \neq x} r(x, y)$ and set $P := I - \mathcal{L}_r/(2R)$. Applying P at the rings of an independent Poisson clock of rate $2R$ gives the chain P_t . For sampling from π , we run P for a fixed number of steps, bounded as follows. For probability laws ρ, ν on Ω , their *total variation distance* is

$$\|\rho - \nu\|_{\text{TV}} := \frac{1}{2} \sum_{x \in \Omega} |\rho(x) - \nu(x)|.$$

For $0 < \varepsilon < 1$, the *mixing time* of P is

$$t_{\text{mix}}(\varepsilon) := \min \left\{ k \in \mathbb{Z}_{\geq 0} : \max_{x \in \Omega} \|P^k(x, \cdot) - \pi\|_{\text{TV}} \leq \varepsilon \right\}.$$

Write $\pi_{\min} := \min_{x \in \Omega} \pi(x)$. Under Eq. (11) with $\gamma > 0$, the standard spectral mixing bound [LP17, Theorem 12.4] gives

$$t_{\text{mix}}(\varepsilon) \leq \left\lceil \frac{2R}{\gamma} \log \frac{1}{\varepsilon \pi_{\min}} \right\rceil. \quad (12)$$

Running P for this many steps from any starting state gives a sample within total variation distance ε of π . Thus an efficient implementation of each step gives an efficient sampler whenever this mixing-time bound is polynomial.

Proof of Proposition 2.1. Since the chain is reversible, $\mathcal{L}$ is self-adjoint and positive semidefinite with respect to $\langle \cdot, \cdot \rangle_\pi$; since it is irreducible, its kernel consists of the constant functions. Hence $\mathcal{L}$ is positive definite on $\mathbf{1}^\perp$ (the functions with mean zero under π), and $\mathcal{L}^{1/2}$ is a bijection of $\mathbf{1}^\perp$. Both ratios are unchanged when a constant is added to f , so we may take $f \in \mathbf{1}^\perp \setminus \{0\}$, for which $\text{Var}_\pi[f] = \langle f, f \rangle_\pi$. Substituting $g := \mathcal{L}^{1/2}f$ gives

$$\frac{\langle \mathcal{L}f, \mathcal{L}f \rangle_\pi}{\langle f, \mathcal{L}f \rangle_\pi} = \frac{\langle g, \mathcal{L}g \rangle_\pi}{\langle g, g \rangle_\pi},$$

and g ranges over $\mathbf{1}^\perp \setminus \{0\}$ as f does. Hence the two infima coincide. $\square$

3.4 Sampling to counting

We record below the standard reduction from counting to sampling due to Jerrum, Valiant, and Vazirani [JVV86].

Lemma 3.1 ([JVV86]). *Let $m \geq 1$, let $w : \{0, 1\}^m \rightarrow \mathbb{R}_{\geq 0}$ be nonzero, and write $Z := \sum_{\sigma} w(\sigma)$ and $\pi(\sigma) := w(\sigma)/Z$. For every nonempty conditional law obtained by fixing coordinates, suppose a sampler, given any starting state in its support and an accuracy $0 < \tau < 1$, returns a sample in that support whose distribution is within total variation distance τ of the conditional law, using fresh independent randomness. Given an evaluation oracle for w , a state $\sigma_0 \in \text{supp } w$, and $0 < \varepsilon, \delta < 1$, one can compute $\hat{Z}$ satisfying*

$$\Pr \left[(1 - \varepsilon)Z \leq \hat{Z} \leq (1 + \varepsilon)Z \right] \geq 1 - \delta.$$

On every execution, the algorithm uses $Q = O(m^3 \varepsilon^{-2} \log(2m/\delta))$ sampler calls, each with total variation distance at most $\tau = \varepsilon/(16m)$, one evaluation of w , and $O(Q)$ additional arithmetic operations. It supplies a starting state for every call.

3.5 Stochastic Coordinate-Descent

Stochastic coordinate descent finds a point with small gradient using noisy gradient estimates: it averages fresh estimates and steps along any coordinate whose averaged estimate is large. The analysis combines the coordinate-descent estimate of Nutini et al. [Nut+15, Section 3, equation (2)] with concentration bounds for the noise. We state it with two guarantees that are convenient when the gradient oracle is implemented by sampling: every iterate stays in a bounded region, and the cost is bounded on every execution.

Theorem 3.2. *Fix $d \geq 1$, $\mathbf{x}_0 \in \mathbb{R}^d$, $L, \sigma > 0$, and $H_0 \geq 0$. Assume the following.*

(i) **Function.** *The function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex and differentiable, and for all $\mathbf{x} \in \mathbb{R}^d$, $u \in \mathbb{R}$, and $1 \leq i \leq d$,*

$$f(\mathbf{x} + ue_i) \leq f(\mathbf{x}) + u\partial_i f(\mathbf{x}) + Lu^2/2, \quad (13)$$

$$f(\mathbf{x}) \geq f(\mathbf{x}_0) - H_0. \quad (14)$$

Here $(e_i)_j := \mathbf{1}_{\{i=j\}}$.

(ii) **Gradient oracle.** *Each query at $\mathbf{x}$ uses fresh independent randomness and costs at most $T \geq d$. Its output $\mathbf{G}(\mathbf{x})$ is unbiased with coordinatewise σ -sub-Gaussian noise: for all $\mathbf{x}, u, i$ as above,*

$$\mathbb{E}[\mathbf{G}(\mathbf{x})] = \nabla f(\mathbf{x}),$$

$$\mathbb{E}[\exp(u(\mathbf{G}_i(\mathbf{x}) - \partial_i f(\mathbf{x})))] \leq \exp(\sigma^2 u^2/2).$$

For any $0 < \eta \leq 1$ and $0 < \delta < 1$, a stochastic coordinate-descent algorithm has the following guarantees.

(a) **Output and queries.** *With probability at least $1 - \delta$, it returns $\mathbf{x}$ with $\|\nabla f(\mathbf{x})\|_{\infty} \leq \eta$. On every execution, every queried or returned point $\mathbf{y}$ satisfies*

$$\|\mathbf{y} - \mathbf{x}_0\|_1 = O\left(\frac{H_0}{\eta} + \frac{\eta}{L}\right). \quad (15)$$

(b) **Cost.** *On every execution, it uses $O(Q)$ oracle calls and total time $O(TQ)$, where*

$$Q := \left(1 + \frac{\sigma^2}{\eta^2}\right) \left(1 + \frac{LH_0}{\eta^2}\right) \log \left[\frac{2d}{\delta} \left(1 + \frac{LH_0}{\eta^2}\right) \right]. \quad (16)$$

For instance, if each coordinate of $\mathbf{G}(\mathbf{x})$ takes values in an interval of length at most 2σ , the noise condition holds by Hoeffding's lemma [Hoe63]; centered Gaussian errors of variance at most σ^2 are also allowed. The algorithm and its analysis are given in [Appendix A.4](#).

Remark 3.3. If the j th oracle output, conditional on the history, is within total variation distance ξ_j of its prescribed law, then the conclusion of [Theorem 3.2\(a\)](#) holds with probability at least $1 - \delta - \sum_j \xi_j$, and the remaining guarantees hold on every execution. Indeed, coupling each output with an exact one, the two runs agree except with probability at most $\sum_j \xi_j$.

4 Global Mixing via Local Curvature

In this section, we prove [Theorem 1.14](#). We begin by setting up some notation. We extend functions on Ω by zero on $2^{\mathcal{U}} \setminus \Omega$.

Definition 4.1 (Discrete Gradient). For each $u \in \mathcal{U}$ and $f : 2^{\mathcal{U}} \rightarrow \mathbb{R}$, let

$$(\nabla_u f)(S) := f(S + u) - f(S - u).$$

Further define the upper and lower discrete derivatives via

$$(\nabla_u^+ f)(S) := (\nabla_u f)(S) \cdot \mathbf{1}[u \notin S] \quad \text{and} \quad (\nabla_u^- f)(S) := (\nabla_u f)(S) \cdot \mathbf{1}[u \in S].$$

Note that $(\nabla f)(S), (\nabla^+ f)(S), (\nabla^- f)(S)$ are vectors in $\mathbb{R}^{\mathcal{U}}$.

It will also be notationally convenient to define a “downward” analog of spectral curvature. For each $S \in \text{supp}(\mu)$, let

$$D_S^{\mu, \kappa}(u, v) := A_{\mathcal{U} \setminus S}^{\mu, \kappa}(u, v) = \begin{cases} \frac{\mu(S-v)^2}{\mu(S)^2} - \kappa \cdot \frac{\mu(S-v)}{\mu(S)}, & \text{if } u, v \in S, u = v \\ \frac{\mu(S-u)}{\mu(S)} \cdot \frac{\mu(S-v)}{\mu(S)} - \frac{\mu(S-u-v)}{\mu(S)}, & \text{if } u, v \in S, u \neq v \\ 0, & \text{otherwise} \end{cases}$$

This is a $\mathcal{U} \times \mathcal{U}$ matrix supported only on S . The (downward) κ -spectral curvature is then the condition that $D_S^{\mu, \kappa} \geq 0$.

We will use the following Bochner-type identity to prove [Theorem 1.14](#).

Proposition 4.2. Under the hypotheses of [Theorem 1.14](#), for every function $f : 2^{\mathcal{U}} \rightarrow \mathbb{R}$, we have the identity,

$$\begin{aligned} \left\langle f, \left(\mathcal{L}^2 - (\eta + \kappa) \cdot \mathcal{L} \right) f \right\rangle_{\rho} &= \sum_{S \in \Omega} \rho(S) \cdot (\nabla^+ f)(S)^{\top} \cdot A_S^{\mu, \eta} \cdot (\nabla^+ f)(S) \\ &\quad + \sum_{S \in \Omega} \rho(S) \cdot (\nabla^- f)(S)^{\top} \cdot D_S^{\mu, \kappa} \cdot (\nabla^- f)(S) \\ &\quad + \frac{1}{Z} \sum_{S \in \Omega} \sum_{\substack{u, v \notin S: u \neq v \\ S+u+v \in \Omega}} \mu(S+u+v) \cdot v(S) \cdot (\Delta_{uv} f)(S)^2, \end{aligned}$$

where

$$(\Delta_{uv} f)(S) := (\nabla_u \nabla_v f)(S) = f(S+u+v) - f(S+u-v) - f(S-u+v) + f(S-u-v),$$

and $Z = \sum_{S \subseteq \mathcal{U}} \mu(S) \cdot v(S)$ is the normalizing constant for ρ .

[Proposition 4.2](#) is proved in [Section 4.1](#) below.

Proof of [Theorem 1.14](#). In [Proposition 4.2](#), the term $(\Delta_{uv} f)(S)^2$ is non-negative automatically; and, by the spectral-curvature conditions, the first two terms are also non-negative. Hence $\langle \mathcal{L} f, \mathcal{L} f \rangle_{\rho} \geq (\eta + \kappa) \cdot \langle f, \mathcal{L} f \rangle_{\rho}$ for all f . The chain generated by $-\mathcal{L}$ is reversible with respect to ρ and irreducible on Ω , since Ω is connected under single-element additions and deletions and every feasible transition has positive rate. Thus [Proposition 2.1](#) gives spectral gap $\gamma \geq \eta + \kappa$, which is the desired Poincaré inequality. $\square$

4.1 Proof of [Proposition 4.2](#)

Proof. We show that subtracting the two curvature terms leaves a weighted sum of squares of the mixed second differences $\Delta_{uv} f$, the off-diagonal entries of the discrete Hessian of f . Let $S \sim \rho$ and write $\mathbf{r}(S) := (r_u(S))_{u \in \mathcal{U}}$, where $r_u(S)$ is defined just before [Eq. \(3\)](#). Since $r_u(S)$ is signed, the vector of nonnegative transition rates is $|\mathbf{r}(S)| := (|r_u(S)|)_{u \in \mathcal{U}}$. Self-adjointness and [Eq. \(9\)](#) give

$$\begin{aligned} \langle f, \mathcal{L}^2 f \rangle_{\rho} &= \langle \mathcal{L} f, \mathcal{L} f \rangle_{\rho} = \mathbb{E}_{S \sim \rho} \left[(\nabla f)(S)^{\top} \mathbf{r}(S) \mathbf{r}(S)^{\top} (\nabla f)(S) \right], \\ \langle f, \mathcal{L} f \rangle_{\rho} &= \frac{1}{2} \cdot \mathbb{E}_{S \sim \rho} \left[(\nabla f)(S)^{\top} \text{diag}(|\mathbf{r}(S)|) (\nabla f)(S) \right]. \end{aligned}$$

Detailed balance pairs each addition with its reverse deletion, so

$$\begin{aligned}\langle f, \mathcal{L}f \rangle_\rho &= \mathbb{E} \left[(\nabla^+ f)(S)^\top \text{diag}(|\mathbf{r}(S)|) (\nabla^+ f)(S) \right] \\ &= \mathbb{E} \left[(\nabla^- f)(S)^\top \text{diag}(|\mathbf{r}(S)|) (\nabla^- f)(S) \right].\end{aligned}$$

Define the matrix

$$B_S := \text{offdiag}(\mathbf{r}(S)\mathbf{r}(S)^\top - A_S^{\mu,\eta} - D_S^{\nu,\kappa}),$$

where $\text{offdiag}(M)$ sets the diagonal entries of M to zero. Subtracting the two curvature terms from the left-hand side gives

$$\begin{aligned}\left\langle f, (\mathcal{L}^2 - (\eta + \kappa)\mathcal{L})f \right\rangle_\rho &= \mathbb{E} \left[(\nabla^+ f)(S)^\top A_S^{\mu,\eta} (\nabla^+ f)(S) \right] - \mathbb{E} \left[(\nabla^- f)(S)^\top D_S^{\nu,\kappa} (\nabla^- f)(S) \right] \\ &= \mathbb{E} \left[(\nabla f)(S)^\top B_S (\nabla f)(S) \right].\end{aligned}$$

The diagonal terms cancel using the preceding Dirichlet-form identities and the definitions of $A_S^{\mu,\eta}$ and $D_S^{\nu,\kappa}$. Expanding the expectation gives

$$\mathbb{E} \left[(\nabla f)(S)^\top B_S (\nabla f)(S) \right] = \sum_{X \in \Omega} \sum_{u,v \in \mathcal{U}: u \neq v} \rho(X) B_X(u,v) (\nabla_u f)(X) (\nabla_v f)(X).$$

Whenever $S, T = S + u + v \in \Omega$ for distinct $u, v \notin S$, both $S + u$ and $S + v$ also belong to Ω . Indeed, if $\mu(S + u) = 0$, then $A_S^{\mu,\eta}(u, u) = 0$ but $A_S^{\mu,\eta}(u, v) = -\mu(T)/\mu(S) \neq 0$, contradicting positive semidefiniteness. The other intermediate set is handled symmetrically.

Observation 4.3. For $S \in \Omega$ and distinct $u, v \notin S$ with $T := S + u + v \in \Omega$, set $w := \mu(T)\nu(S)/Z$. Then

$$\rho(S)B_S(u,v) = \rho(T)B_T(u,v) = -\rho(S+u)B_{S+u}(u,v) = -\rho(S+v)B_{S+v}(u,v) = w.$$

We regroup by $S = X - u - v$, writing $T = S + u + v$. For each $X \in \{S, S + u, S + v, T\} \cap \Omega$, direct substitution gives $\rho(X)B_X(u,v) = \pm\mu(T)\nu(S)/Z$. This vanishes unless both $S, T \in \Omega$, since the two measures have common support. In that case, both intermediate sets are supported by the argument above. Thus

$$\mathbb{E} \left[(\nabla f)(S)^\top B_S (\nabla f)(S) \right] = \sum_{\substack{S \in \Omega \\ T = S + u + v \in \Omega}} \sum_{\substack{u,v \notin S: u \neq v \\ T = S + u + v \in \Omega}} \sum_{X \in \{S, S + u, S + v, T\}} \rho(X) B_X(u,v) (\nabla_u f)(X) (\nabla_v f)(X).$$

The gradient $\nabla_u f$ is unchanged when u is added or removed, and similarly for $\nabla_v f$. Thus [Observation 4.3](#) gives, for each group,

$$\begin{aligned}\sum_{X \in \{S, S + u, S + v, T\}} \rho(X) B_X(u,v) (\nabla_u f)(X) (\nabla_v f)(X) \\ = w \cdot ((\nabla_u f)(T) - (\nabla_u f)(S)) ((\nabla_v f)(T) - (\nabla_v f)(S)) = w \cdot (\Delta_{uv} f)(S)^2.\end{aligned}$$

The last equality holds because each of the two differences equals $(\nabla_u \nabla_v f)(S) = (\Delta_{uv} f)(S)$. Substituting into the regrouped sum finishes the proof. $\square$

5 Subset Convolution for Homogeneous Log-Concave Measures

In this section, we prove the following formal version of [Theorem 2.2](#).

Theorem 5.1. Let $\mu : \binom{\mathcal{U}}{a} \rightarrow \mathbb{R}_{\geq 0}$ and $\nu : \binom{\mathcal{U}}{b} \rightarrow \mathbb{R}_{\geq 0}$, extended by zero to $2^{\mathcal{U}}$, where $a, b \geq 0$ are integers with $a + b = n := |\mathcal{U}| \geq 1$. Suppose g_μ, g_ν are log-concave. Given value oracles for μ, ν , the coefficient ranges $\mathcal{R}_\mu, \mathcal{R}_\nu$ in [Eq. \(2\)](#), and sets S_0, T_0 with $\mu(S_0) > 0$ and $\nu(T_0) > 0$, there is an FPRAS for the subset-convolution value $(\mu * \nu)(\mathcal{U})$, namely

$$Z_\cap := [\mathbf{x}^{\mathcal{U}}] (g_\mu(\mathbf{x}) g_\nu(\mathbf{x})) = \sum_{S \subseteq \mathcal{U}} \mu(S) \nu(\mathcal{U} \setminus S), \quad \mathbf{x}^{\mathcal{U}} := \prod_{u \in \mathcal{U}} x_u. \quad (17)$$

For every $0 < \varepsilon \leq 1$, it uses at most

$$O\left(n^{10}(n + \log(\mathcal{R}_\mu \mathcal{R}_\nu))^2 \varepsilon^{-2} \log^2 \frac{2(n + \log(\mathcal{R}_\mu \mathcal{R}_\nu))}{\varepsilon}\right) \quad (18)$$

arithmetic operations and value-oracle calls on every execution.

Throughout this section, μ, ν satisfy the hypotheses of [Theorem 5.1](#), with degrees a, b and $a + b = n$. We extend the measures to downward-closed set families by translating their generating polynomials.

Definition 5.2 (Independent-Set Pair Law). Let μ, ν be as in [Theorem 5.1](#), $\mathbf{t} = (t_u)_{u \in \mathcal{U}} \in (0, 1]^\mathcal{U}$, and $\mathbf{z} \in \mathbb{R}_{>0}^\mathcal{U}$. Write $\mathbf{zx} := (z_u x_u)_{u \in \mathcal{U}}$ and, for $R \subseteq \mathcal{U}$, $\mathbf{z}^R := \prod_{u \in R} z_u$ and $\mathbf{t}^R := \prod_{u \in R} t_u$. Define the weights $\hat{\mu}_z, \hat{\nu}_z$ through their generating polynomials by

$$g_{\hat{\mu}_z}(\mathbf{x}) := g_\mu(\mathbf{1} + \mathbf{zx}), \quad g_{\hat{\nu}_z}(\mathbf{x}) := g_\nu(\mathbf{1} + \mathbf{zx}).$$

Write $(\hat{\mu}_z \otimes \hat{\nu}_z)(S, T) := \hat{\mu}_z(S) \hat{\nu}_z(T)$ and $\psi_t(S, T) := \mathbf{t}^{S \cap T}$. The independent-set pair law on $2^\mathcal{U} \times 2^\mathcal{U}$ is the normalized Hadamard product

$$\hat{\pi}_{t,z} := (\hat{\mu}_z \otimes \hat{\nu}_z) \odot \psi_t.$$

Remark 5.3. Explicitly, for $S, T \subseteq \mathcal{U}$,

$$\hat{\mu}_z(S) = \mathbf{z}^S \sum_{A \subseteq S} \mu(A), \quad \hat{\nu}_z(T) = \mathbf{z}^T \sum_{B \subseteq T} \nu(B).$$

For fixed $\mathbf{t}, \mathbf{z}$, the law $\hat{\pi}_{t,\lambda z}$ converges as $\lambda \rightarrow \infty$ to the following bases pair law; see [Section 5.1](#).

Definition 5.4 (Bases Pair Law). With $\mu, \nu, \mathbf{t}, \mathbf{z}$ as in [Definition 5.2](#), set $\Omega := \binom{\mathcal{U}}{a} \times \binom{\mathcal{U}}{b}$. The bases pair law is $\hat{\pi}_{t,z}$ conditioned on Ω . Its partition function and probability mass function are

$$Z_{t,z} := \sum_{(S,T) \in \Omega} \mu(S) \nu(T) \mathbf{t}^{S \cap T} \mathbf{z}^S \mathbf{z}^T, \\ \pi_{t,z}(S, T) := \frac{\mu(S) \nu(T) \mathbf{t}^{S \cap T} \mathbf{z}^S \mathbf{z}^T}{Z_{t,z}}.$$

A scalar penalty $t \in (0, 1]$ denotes the constant vector $t\mathbf{1}$ in the notation for these laws and their partition functions. The three subsections below implement the framework described in [Section 2.3](#).

5.1 Poincaré Inequality for the Bases Pair Law $\pi_{t,z}$

Lemma 5.5. Let $\mathbf{t} \in (0, 1]^\mathcal{U}$, $\mathbf{z} \in \mathbb{R}_{>0}^\mathcal{U}$, and let $\pi_{t,z}$ be the bases pair law of [Definition 5.4](#). On $\text{supp } \pi_{t,z}$, consider the exchange chain with rates

$$q((S, T), (S - u + v, T)) := t_u^{-1[u \in T]} \frac{z_v \mu(S - u + v)}{\sum_{w \in \mathcal{U} \setminus (S - u)} z_w \mu(S - u + w)}, \quad (19)$$

where $u \in S$ and $v \notin S$. The rates updating T are obtained by interchanging (S, μ) and (T, ν) ; all other off-diagonal rates are zero. This chain is reversible with respect to $\pi_{t,z}$ and satisfies

$$\text{Var}_{\pi_{t,z}}[f] \leq \mathcal{E}_q(f, f), \quad \forall f : \text{supp } \pi_{t,z} \rightarrow \mathbb{R}. \quad (20)$$

Applying [Eq. \(12\)](#) to the exchange chain gives the following sampling bound.

Corollary 5.6. Under the hypotheses of [Lemma 5.5](#) with a common scalar penalty t , let $z_{\min} := \min_u z_u$ and $z_{\max} := \max_u z_u$. Given a pair $(S_0, T_0) \in \text{supp } \pi_{t,z}$, the coefficient ranges $\mathcal{R}_\mu, \mathcal{R}_\nu$ in [Eq. \(2\)](#), and $0 < \varepsilon < 1$, one can sample within total variation distance ε of $\pi_{t,z}$ using

$$O\left(\frac{n^2}{t} \left(n + \log(\mathcal{R}_\mu \mathcal{R}_\nu) + n \log \frac{z_{\max}}{z_{\min}} + \min\{a, b\} \log \frac{1}{t} + \log \frac{1}{\varepsilon}\right)\right)$$

arithmetic operations and calls to the value oracles for μ, ν .

Proof. Apply Eq. (12) with $\gamma = 1$ and $R = n/t$: there are $a + b = n$ possible deletions, each with total rate at most $1/t$. Each step of the lazy discrete-time chain costs $O(n)$ operations and oracle calls: choose a present copy uniformly and, if it is $u \in S$, with probability $t^{1-1[u \in T]}/2$ resample it using the reinsertion weights in Eq. (19), allowing reinsertion of u ; the update of T is symmetric. Since there are at most 4^n pairs, each of total size n ,

$$\pi_{\min} \geq \frac{t^{\min\{a,b\}}}{4^n \mathcal{R}_\mu \mathcal{R}_\nu} \left(\frac{z_{\min}}{z_{\max}} \right)^n.$$

Substituting this bound and multiplying by the cost per step proves the claim. $\square$

To prove Lemma 5.5, we first apply Theorem 1.14 to the independent-set pair law $\hat{\pi}_{t,\lambda z}$ of Definition 5.2. The induced chain on pairs of bases inherits the Poincaré bound, and letting $\lambda \rightarrow \infty$ gives the desired exchange chain.

Poincaré inequality for $\hat{\pi}_{t,\lambda z}$. We will first use Theorem 1.14 to show the Poincaré inequality on $\hat{\pi}_{t,\lambda z}$. For convenience, for $\lambda > 0$, define the measure

$$w_\lambda(S, T) := \hat{\mu}_{\lambda z}(S) \hat{\nu}_{\lambda z}(T),$$

whose support is identical to

$$\text{supp } \hat{\pi}_{t,\lambda z} = \text{supp } \hat{\pi}_{t,z} = \text{supp } \hat{\mu}_{\lambda z} \times \text{supp } \hat{\nu}_{\lambda z}.$$

Restrict ψ_t from Definition 5.2 to this downward closed support, and extend both measures by zero outside this support. Recall from Definition 5.2 that

$$\hat{\pi}_{t,\lambda z} = w_\lambda \odot \psi_t. \quad (21)$$

Corollary 5.7. Let $t \in (0, 1]^{\mathcal{U}}$, $z \in \mathbb{R}_{>0}^{\mathcal{U}}$, and $\lambda > 0$. Let $\mathcal{L}$ be the Laplacian in Eq. (3) for the factors w_λ, ψ_t , and let $\hat{q}_\lambda$ be its transition rates. For every $F : \text{supp } \hat{\pi}_{t,\lambda z} \rightarrow \mathbb{R}$, we have

$$\text{Var}_{\hat{\pi}_{t,\lambda z}}[F] \leq \mathcal{E}_{\hat{q}_\lambda}(F, F) = \langle F, \mathcal{L}F \rangle_{\hat{\pi}_{t,\lambda z}}. \quad (22)$$

In order to apply Theorem 1.14 to $\hat{\pi}_{t,\lambda z}$, we have the following observations.

Observation 5.8. w_λ has 0-spectral curvature.

Proof. The generating polynomial of w_λ is

$$g_{w_\lambda}(\mathbf{x}, \mathbf{y}) = g_\mu(\mathbf{1} + \lambda z \mathbf{x}) g_\nu(\mathbf{1} + \lambda z \mathbf{y}).$$

Since $p(\mathbf{x}, \mathbf{y}) := g_\mu(\mathbf{x}) g_\nu(\mathbf{y})$ is log-concave, Proposition 1.22 with $\alpha = 1$ gives zero curvature for the coefficient measure of $p(\mathbf{1} + \mathbf{x}, \mathbf{1} + \mathbf{y})$. Positive rescaling of the variables preserves zero curvature: each curvature matrix A becomes DAD , where D is a diagonal matrix with positive entries, so positive semidefiniteness is preserved. Applying the rescaling $(\mathbf{x}, \mathbf{y}) \mapsto (\lambda z \mathbf{x}, \lambda z \mathbf{y})$ therefore gives the assertion for $g_{w_\lambda}(\mathbf{x}, \mathbf{y}) = p(\mathbf{1} + \lambda z \mathbf{x}, \mathbf{1} + \lambda z \mathbf{y})$. $\square$

Observation 5.9. ψ_t^* has 1-spectral curvature.

Proof. By definition, the dual measure ψ_t^* is

$$\psi_t^*(S, T) := \psi_t(\mathcal{U} \setminus S, \mathcal{U} \setminus T) = t^{\mathcal{U} \setminus (S \cup T)} \mathbf{1}[(\mathcal{U} \setminus S, \mathcal{U} \setminus T) \in \text{supp } \hat{\pi}_{t,z}].$$

For $Y = (S, T) \in \text{supp } \psi_t^*$, downward closure of $\text{supp } \hat{\pi}_{t,z}$ ensures that all additions to Y remain supported. Substitution into Definition 1.10 gives, for $u, v \in \mathcal{U}$ and $i, j \in \{1, 2\}$,

$$A_Y^{\psi_t^*, 1}((u, i), (v, j)) = \begin{cases} (1 - t_u)/t_u^2, & u = v, u \notin S \cup T, \\ 0, & \text{otherwise.} \end{cases}$$

Here, we use $(u, 1)$ and $(u, 2)$ to denote that the element u is added to S and T , respectively. Thus $A_Y^{\psi_t^*, 1}$ is a direct sum of zero blocks and blocks $\frac{1-t_u}{t_u^2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, all positive semidefinite. Hence ψ_t^* has upward curvature one, and Theorem 1.14 applies with $\eta = 0$ and $\kappa = 1$. $\square$

Proof of Corollary 5.7. Apply Theorem 1.14 to $w_\lambda \odot \psi_t$ with $(\eta, \kappa) = (0, 1)$, using Observations 5.8 and 5.9. Normalizing the factors does not change their weight ratios. $\square$

Poincaré inequality of $\pi_{t,z}$ by letting $\lambda \rightarrow \infty$. We recall the induced-chain construction and its Poincaré comparison.

Definition 5.10 (Induced chain). Let $(X_s)_{s \geq 0}$ be an irreducible continuous-time Markov chain on a finite state space Ω , with off-diagonal rates $r(x, y)$, and let $\emptyset \neq B \subseteq \Omega$. Its *induced chain on B* is the chain obtained by deleting all time spent outside B . Writing $\tau_B := \inf\{s \geq 0 : X_s \in B\}$, its off-diagonal rates are

$$r^B(x, y) := r(x, y) + \sum_{u \in \Omega \setminus B} r(x, u) \Pr[X_{\tau_B} = y \mid X_0 = u] \quad (x, y \in B, x \neq y).$$

Proposition 5.11 ([LP17, Theorem 13.16]). *In the setting of Definition 5.10, suppose the chain is reversible with stationary law π and has spectral gap at least $\gamma > 0$. Then the induced chain is reversible with respect to $\pi_B(x) := \pi(x) / \pi(B)$ for $x \in B$, and for every $f : B \rightarrow \mathbb{R}$,*

$$\gamma \text{Var}_{\pi_B}[f] \leq \mathcal{E}_{r^B}(f, f).$$

The proof of [LP17, Theorem 13.16] is for discrete-time Markov chains but extends directly to the continuous-time setting. For the reader's convenience, we include a proof in [Appendix A.8](#).

Proof of Lemma 5.5. We pass to the induced chain on pairs of bases and show that its rates converge to [Eq. \(19\)](#). Singleton support is immediate. Recall from [Eq. \(3\)](#) that the chain with generator $-\mathcal{L}$ has rates

$$\begin{aligned} \hat{q}_\lambda((S, T), (S + u, T)) &= \frac{w_\lambda(S + u, T)}{w_\lambda(S, T)} = \frac{\hat{\mu}_{\lambda z}(S + u)}{\hat{\mu}_{\lambda z}(S)} & (u \notin S), \\ \hat{q}_\lambda((S, T), (S - u, T)) &= \frac{\psi_t(S - u, T)}{\psi_t(S, T)} = t_u^{-1[u \in T]} & (u \in S). \end{aligned}$$

The rates updating T are symmetric; unsupported additions are omitted and all other off-diagonal rates are zero.

Let q_λ^{ind} be the rates of the chain induced on $\text{supp } \pi_{t,z}$ from Definition 5.10. Conditioning $\hat{\pi}_{t,\lambda z}$ on pairs of bases cancels the common factor λ^n , giving exactly $\pi_{t,z}$. By [Proposition 5.11](#) and [Corollary 5.7](#), the induced chain is therefore reversible with stationary law $\pi_{t,z}$ and has spectral gap at least one.

It remains to identify the limiting rates. After deleting $u \in S$ from a pair of bases (S, T) , put $S' := S - u$. At (S', T) , additions have positive total rate proportional to λ , whereas the total deletion rate is at most $n / \min_i t_i$. Thus the next jump is an addition with probability $1 - O(\lambda^{-1})$. Every such addition returns to a pair of bases, and its conditional reinsertion law is

$$\Pr[\text{reinsert } v \mid \text{next jump is an addition}] = \frac{z_v \mu(S' + v)}{\sum_{w \notin S'} z_w \mu(S' + w)} \quad (v \notin S').$$

Multiplying by the initial deletion rate $t_u^{-1[u \in T]}$ gives [Eq. \(19\)](#) in the limit. The same argument applies to deletions from T ; excursions with a further deletion have probability $O(\lambda^{-1})$. Hence $q_\lambda^{\text{ind}}(X, Y) \rightarrow q(X, Y)$ for every distinct pair $X, Y \in \text{supp } \pi_{t,z}$.

Reversibility passes to the limit. Since the state space is finite, passing to the limit in the induced chains' Poincaré inequalities gives, for every f ,

$$\text{Var}_{\pi_{t,z}}[f] \leq \lim_{\lambda \rightarrow \infty} \mathcal{E}_{q_\lambda^{\text{ind}}}(f, f) = \mathcal{E}_q(f, f). \quad \square$$

5.2 Choosing External Fields by Stochastic Gradient Descent

We now use the Poincaré bound to choose fields that make complementary pairs occur with constant probability when $Z_\cap > 0$. We first prove [Lemma 2.5](#), restated below, and then apply stochastic gradient descent to find logarithmic coordinates $\mathbf{y}$ with small $\nabla \Phi(\mathbf{y})$, then set $\mathbf{z} = e^{\mathbf{y}}$.

Lemma 2.5. *Let $\gamma > 0$. Suppose that for every $\mathbf{t} \in (0, 1]^{\mathcal{U}}$ and $\mathbf{z} \in \mathbb{R}_{>0}^{\mathcal{U}}$, there is an irreducible continuous-time Markov chain reversible with respect to $\pi_{t,z}$, with spectral gap at least γ . Each move updates only one of the two sets S, T . Whenever $u \in S \oplus T$, the total rate of moves removing its unique copy is at most 1. Then for every scalar $t \in (0, 1]$ and every $\mathbf{z} \in \mathbb{R}_{>0}^{\mathcal{U}}$, with $\mathbf{y} = \log \mathbf{z}$,*

$$\pi_{t,z}(T \neq \mathcal{U} \setminus S) \leq n \left(\|\nabla \Phi(\mathbf{y})\|_\infty + \frac{4\sqrt{t}}{\gamma} \right). \quad (6)$$

Proof. Fix u , and for $j \in \{0, 1, 2\}$, let p_j be the probability under $\pi_{t,z}$ that u appears in exactly j of the sets S, T . Since $p_2 - p_0 = \partial_u \Phi(\mathbf{y})$,

$$p_0 + p_2 \leq |p_2 - p_0| + 2\sqrt{p_0 p_2} = |\partial_u \Phi(\mathbf{y})| + 2\sqrt{p_0 p_2},$$

so a union bound over u reduces the lemma to

$$p_0 p_2 \leq \frac{4t}{\gamma^2} p_1^2.$$

Assume $p_0 p_2 > 0$, and set $s := \sqrt{t p_0 / p_2}$. Remove the penalty at u and multiply z_u by s : let t', z' agree with t, z except at u , where $t'_u := 1$ and $z'_u := s z_u$. Then

$$\pi_{t',z'}(S, T) \propto \pi_{t,z}(S, T) \cdot s^{\mathbf{1}[u \in S] + \mathbf{1}[u \in T]} t^{-\mathbf{1}[u \in S \cap T]},$$

so the probabilities q_0, q_1, q_2 under $\pi_{t',z'}$ that u lies in exactly 0, 1, 2 sets satisfy

$$(q_0, q_1, q_2) \propto (p_0, s p_1, s^2 p_2 / t) = (p_0, s p_1, p_0), \quad \text{so } q_0 = q_2 \leq \frac{1}{2}.$$

Let $\mathcal{L}'$ be the Laplacian supplied by the hypothesis for $\pi_{t',z'}$. By assumption, the total rate of moves from one copy of u to none is at most 1, and no move goes from two copies to none. By reversibility, the Dirichlet form of $g(S, T) := \mathbf{1}[u \notin S \cup T]$ is therefore at most q_1 . The Poincaré inequality gives

$$\gamma q_0 (1 - q_0) = \gamma \text{Var}_{\pi_{t',z'}}[g] \leq \langle g, \mathcal{L}' g \rangle_{\pi_{t',z'}} \leq q_1, \quad \text{hence} \quad \frac{\gamma}{2} \leq \frac{q_1}{q_0} = \frac{\sqrt{t} p_1}{\sqrt{p_0 p_2}}. \quad \square$$

For the bases pair law, we apply [Lemma 2.5](#) to the exchange chain of [Lemma 5.5](#), which has gap 1 for every t, z and updates only one set at a time. If $u \in S \setminus T$, its total removal rate is

$$\sum_{v \notin S} q((S, T), (S - u + v, T)) = \frac{\sum_{v \notin S} z_v \mu(S - u + v)}{\sum_{w \notin (S - u)} z_w \mu(S - u + w)} \leq 1.$$

The same bound holds for $u \in T \setminus S$, so the hypotheses of [Lemma 2.5](#) hold with $\gamma = 1$. Thus making complementary pairs frequent reduces to finding an external field z with small $\nabla \Phi(\log z)$. We use stochastic coordinate descent as follows.

Corollary 5.12. *Under the hypotheses of [Theorem 5.1](#), put*

$$t := \frac{1}{256n^2}, \quad B := 2n + \lceil \log_2(\mathcal{R}_\mu \mathcal{R}_\nu) \rceil.$$

Given value oracles for μ, ν , the coefficient ranges $\mathcal{R}_\mu, \mathcal{R}_\nu$ in [Eq. \(2\)](#), and a supported pair (S_0, T_0) , there is a randomized algorithm that returns $z \in \mathcal{R}_{>0}^{\mathcal{U}}$ or reports failure. If $Z_\cap > 0$, with probability at least $31/32$ it returns a field satisfying

$$\pi_{t,z}(T = \mathcal{U} \setminus S) \geq \frac{1}{2}.$$

On every execution, it uses $O(n^{10} B^2 \log(2B))$ arithmetic operations and value-oracle calls, and every queried or returned field satisfies

$$\|\log z\|_1 = O(nB). \quad (23)$$

Proof. By [Lemma 2.5](#), it suffices to find $\mathbf{y}$ with $\|\nabla \Phi(\mathbf{y})\|_\infty \leq \eta := 1/(4n)$ and return $z := \exp(\mathbf{y})$: then $n(\eta + 4\sqrt{t}) = 1/2$. We apply [Theorem 3.2](#) with $d = n, L = \sigma = 1, H_0 = B$, and $\mathbf{x}_0 = \mathbf{0}$. For $(S, T) \sim \pi_{t,\exp(\mathbf{y})}$, the vector $\mathbf{G} := \mathbf{1}_S + \mathbf{1}_T - \mathbf{1}$ satisfies

$$\begin{aligned} \nabla \Phi(\mathbf{y}) &= \mathbb{E}[\mathbf{G}], & \nabla^2 \Phi(\mathbf{y}) &= \text{Cov}(\mathbf{G}) \succeq 0, \\ \partial_u^2 \Phi(\mathbf{y}) &= \text{Var}[G_u] \leq 1, & \mathbf{G} &\in [-1, 1]^n. \end{aligned}$$

Thus Φ is convex and coordinatewise 1-smooth, and Hoeffding's lemma gives coordinatewise 1-sub-Gaussian oracle noise. When $Z_\cap > 0$, every complementary pair contributes its unchanged weight to $\exp(\Phi(\mathbf{y}))$. Writing $M := \max_S \mu(S) \max_T \nu(T)$, we therefore have

$$\Phi(\mathbf{y}) \geq \log Z_\cap \geq \log \frac{M}{\mathcal{R}_\mu \mathcal{R}_\nu}, \quad \Phi(\mathbf{0}) \leq \log(4^n M).$$

Thus the required gap bound is at most B , so take $H_0 := B$; no minimizing point need exist.

Take $\delta = 1/64$ in [Theorem 3.2](#). The algorithm uses at most $Q_{\text{cd}} = O(n^4 B \log(2B))$ gradient samples and $O(T' Q_{\text{cd}})$ operations, where T' bounds the cost of one gradient sample. It also gives $\|\mathbf{y}\|_1 = O(B/\eta + \eta) = O(nB)$ for every queried or returned point, which proves [Eq. \(23\)](#). We stop and report failure before exceeding any of these bounds. Thus they also hold when $Z_\cap = 0$; when $Z_\cap > 0$, this stopping rule does not change the algorithm with exact samples.

Supply each gradient sample using [Corollary 5.6](#), with fresh randomness and total variation error at most $1/(64Q_{\text{cd}})$. By [Remark 3.3](#), these sampling errors add at most $1/64$ to the failure probability. Hence the algorithm finds the required field with probability at least $1 - 1/64 - 1/64 = 31/32$ when $Z_\cap > 0$.

For the running time, the field bound gives $n \log(z_{\max}/z_{\min}) = O(n^2 B)$ at every query. Since $t^{-1} = 256n^2$ and $B = \Theta(n + \log(\mathcal{R}_\mu \mathcal{R}_\nu))$, [Corollary 5.6](#) gives $T' = O(n^6 B)$ operations and value-oracle calls per gradient sample. Multiplying by Q_{cd} proves the claimed bound; the $O(nQ_{\text{cd}})$ additional operations cost less. The field updates can be implemented using arithmetic operations only; see [Appendix A.5](#). $\square$

5.3 Wrapping Up

Proof of Theorem 5.1. The cases $a = 0$ or $b = 0$ are evaluated directly, so assume $n \geq 2$. We choose fields making complementary pairs frequent, then estimate the two unknown factors in

$$Z_\cap = \frac{Z_{t,z}}{z^\mathcal{U}} \times \pi_{t,z}(T = \mathcal{U} \setminus S). \quad (24)$$

Let B, t be as in [Corollary 5.12](#), and put $\beta := \varepsilon/4$ and $\delta_0 := 1/64$. Apply [Corollary 5.12](#); if it reports failure, return zero. Otherwise fix the returned field $\mathbf{z}$. When $Z_\cap > 0$, with probability at least $1 - 2\delta_0$, this field satisfies $\pi_{t,z}(T = \mathcal{U} \setminus S) \geq 1/2$.

Estimating the normalizer. For any returned field $\mathbf{z}$, we apply [Lemma 3.1](#) to the $2n$ membership bits of (S, T) , with weight

$$W(S, T) := \mu(S)\nu(T)t^{|S \cap T|} \mathbf{z}^S \mathbf{z}^T.$$

The supplied pair (S_0, T_0) is a valid starting state.

Each conditional law is obtained by restricting μ, ν to the prescribed inclusions and exclusions, with the same $t, \mathbf{z}$. We keep the original ground set and the forced elements in each set, so the degrees remain a, b . Inclusion replaces a generating polynomial by its coordinate derivative times that variable; exclusion sets the variable to zero. These operations preserve log-concavity, and the coefficient ranges do not increase. For every nonempty conditional law, both measures are nonzero. Thus [Corollary 5.6](#) applies to every conditional law.

[Lemma 3.1](#), with relative error β and failure probability δ_0 , therefore gives an estimate $\hat{Z}$ of $Z_{t,z}$ using $O(n^3 \beta^{-2} \log(2n/\delta_0))$ conditional samples, each within total variation distance $\beta/(32n)$ of its law. The sample counts and running times are bounded on every execution.

Combining the estimates. Independently draw $m = \lceil 8\beta^{-2} \log(2/\delta_0) \rceil$ pairs from $\pi_{t,z}$, each to total variation error $\beta/4$, and let $\hat{p}$ be the fraction that are complementary. Conditional on the returned fields, Hoeffding's inequality gives $|\hat{p} - \pi_{t,z}(T = \mathcal{U} \setminus S)| \leq \beta/2$ except with probability δ_0 : the sampling bias is at most $\beta/4$, as is the fluctuation of the empirical average. Return

$$\hat{Z}_\cap := \hat{p} \hat{Z} / \mathbf{z}^\mathcal{U}.$$

When the returned field satisfies $\pi_{t,z}(T = \mathcal{U} \setminus S) \geq 1/2$ and both preceding estimation bounds hold, this is a product of two estimates, each with relative error at most β . These conditions hold jointly with probability at least $1 - 4\delta_0 > 3/4$, so

$$(1 - \beta)^2 \leq \hat{Z}_\cap / Z_\cap \leq (1 + \beta)^2,$$

which implies relative error ε . If $Z_\cap = 0$, every supported pair fails the complementary-pair test. Thus every execution returns zero, either at the field-finding step or because $\hat{p} = 0$.

Running time. Finding the field costs $O(n^{10}B^2 \log(2B))$ by [Corollary 5.12](#). Throughout the self-reduction, the coefficient ranges do not increase, and $n \log(z_{\max}/z_{\min}) = O(n^2B)$ by [Eq. \(23\)](#). Each restricted oracle uses one original query and $O(n)$ operations to check the prescribed memberships. Thus [Corollary 5.6](#), with $t^{-1} = 256n^2$, bounds the cost of a conditional sample by

$$O\left(n^5 \left(n^2B + \log \frac{n}{\varepsilon}\right)\right).$$

Self-reduction uses $O(n^3\varepsilon^{-2} \log(2n))$ such samples. Estimating the complementary-pair probability takes $O(\varepsilon^{-2})$ additional samples and is covered by the same bound. Since $B = \Theta(n + \log(\mathcal{R}_\mu \mathcal{R}_\nu))$, the total cost is

$$O\left(n^{10}B^2\varepsilon^{-2} \log^2 \frac{2B}{\varepsilon}\right),$$

as required by [Eq. \(18\)](#). □

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A Deferred Proofs

A.1 Proofs of the convolution theorems

We prove [Theorems 1.6](#) and [1.16](#) by a common reduction to the coefficient $[x^{\mathcal{U}}](g_\mu g_\nu)$, followed by separate arguments for the two hypotheses. We first record three lemmas about supports and homogeneous components; proofs of the first two are deferred to [Appendix A.2](#).

Lemma A.1. *Let ω be a nonzero nonnegative measure on $2^{\mathcal{U}}$ whose support Ω is connected under single-element additions and deletions, and put $V := \bigcup_{S \in \Omega} S$. If ω has upward spectral curvature $\kappa > 0$, then Ω is upward closed in 2^V . In particular, if Ω is also downward closed, then $\Omega = 2^V$, where $V = \{u : \omega(\{u\}) > 0\}$.*

Lemma A.2. *Let $g_\omega(\mathbf{x}) = \sum_{A \subseteq \mathcal{U}} \omega(A) \mathbf{x}^A$ be nonzero and log-concave, with nonnegative coefficients, and put $n := |\mathcal{U}|$. Given a value oracle for ω , a set $A_0 \in \text{supp } \omega$, and disjoint $C, O \subseteq \mathcal{U}$, an algorithm using $O(n^3)$ queries can:*

- *certify that no $A \in \text{supp } \omega$ contains C and avoids O ; or*
- *find, for every size at which such a set exists, a set $A \in \text{supp } \omega$ containing C and avoiding O .*

Lemma A.3 ([\[RSJ19, Corollary 28\]](#)). *Let g be a multiaffine log-concave polynomial with nonnegative coefficients. For each degree k , the sum of the degree- k terms of g is either zero or log-concave.*

Reduce to the coefficient $[x^{\mathcal{U}}](g_\mu g_\nu)$. Put $C := \{u : \alpha_u = 2\}$, $W := \{u : \alpha_u = 1\}$, and $O := \{u : \alpha_u = 0\}$. In a pair contributing to Z_α , both sets contain C and avoid O , while each element of W belongs to exactly one of them. Define measures μ', ν' on 2^W by

$$\mu'(A) := \mu(C \cup A), \quad \nu'(A) := \nu(C \cup A), \quad A \subseteq W.$$

Then

$$Z_\alpha = \sum_{A \subseteq W} \mu'(A) \nu'(W \setminus A).$$

If $W = \emptyset$, return $Z_\alpha = \mu(C)\nu(C)$ directly. Henceforth assume $W \neq \emptyset$. We check that these measures satisfy the assumptions needed for the reduction.

Under [Theorem 1.16](#), if $\mu(C)\nu(C) = 0$, then $Z_\alpha = 0$ by downward closure. Otherwise, both new measures are nonzero, have downward-closed supports, and satisfy

$$A_S^{\mu', \eta} = (A_{C \cup S}^{\mu, \eta})_{W \times W} \succeq 0, \quad S \in \text{supp } \mu'.$$

The same identity holds for ν' . Thus μ' and ν' have upward spectral curvatures η and κ , respectively.

When $\gamma > 0$, for $S \in \text{supp } \mu'$ and $u \in W \setminus S$,

$$\mu'(S + u) = \mu(C \cup S + u) \leq \chi \gamma \mu(C \cup S) = \chi \gamma \mu'(S).$$

The same calculation applies to ν' , so the bound in [Theorem 1.16](#) holds for both new measures.

Under [Theorem 1.6](#), obtain $g_{\mu'}$ from g_μ by differentiating once with respect to each variable indexed by C , then setting the variables indexed by O to zero. The first operation keeps the terms containing C and removes their C factors; the second removes the terms containing any element of O . For multiaffine polynomials, both operations preserve log-concavity whenever the resulting polynomial is not zero. Hence $g_{\mu'}$ is either zero or log-concave, and the same holds for $g_{\nu'}$.

To obtain starting sets, apply [Lemma A.2](#) using the supplied sets S_0, T_0 . This finds sets in $\text{supp } \mu$ and $\text{supp } \nu$ that contain C and avoid O , or reports that one does not exist. In the latter case, $Z_\alpha = 0$. Otherwise, removing C from the returned sets gives starting sets in $\text{supp } \mu'$ and $\text{supp } \nu'$.

In both cases, evaluating $\mu'(A)$ requires only the oracle query $\mu(C \cup A)$, and similarly for ν' . Also, $\mathcal{R}_{\mu'} \leq \mathcal{R}_\mu$ and $\mathcal{R}_{\nu'} \leq \mathcal{R}_\nu$, since each new measure uses only values of its original measure. We now write μ, ν for μ', ν' and $\mathcal{U}, n$ for $W, |W|$. It therefore suffices to estimate

$$Z_\cap = \sum_{A \subseteq \mathcal{U}} \mu(A) \nu(\mathcal{U} \setminus A).$$

Proof of Theorem 1.6. By the reduction above, it suffices to estimate $Z_\cap$, using Theorem 5.1, proved in Section 5. Write $g_{\mu,k}$ for the sum of the degree- k terms of g_μ , and similarly for ν . By Lemma A.3, each of these polynomials is either zero or log-concave. Then

$$Z_\cap = \sum_{k=0}^n [x^\mathcal{U}] (g_{\mu,k}(x) g_{\nu,n-k}(x)).$$

Each layer inherits its value oracle and coefficient-range bound; Lemma A.2 supplies sets of positive weight in all nonempty layers using $O(n^3)$ queries. Apply Theorem 5.1 to each pair of nonempty layers, with relative error ε and failure probability $1/(4(n+1))$, using $O(\log(2n))$ repetitions. Summing the estimates and taking a union bound proves the guarantee. The at most $n+1$ applications of Eq. (18), including the repetitions needed to boost the success probability, cost

$$O\left(n^{11}(n + \log(\mathcal{R}_\mu \mathcal{R}_\nu))^2 \varepsilon^{-2} \log(2n) \log^2 \frac{2(n + \log(\mathcal{R}_\mu \mathcal{R}_\nu))}{\varepsilon}\right).$$

This is the bound in Theorem 1.6. $\square$

Proof of Theorem 1.16. By the reduction above, it suffices to estimate $Z_\cap$. If $\max\{\kappa, \eta\} = 0$, then $\eta = \kappa = 0$ and Proposition 1.24 makes both generating polynomials log-concave; the empty sets supply starting sets of positive weight. Thus Theorem 1.6 applies, and its cost is the first term in the stated bound. It remains to treat the case $\max\{\eta, \kappa\} > 0$, whose cost is the second term.

The case $\max\{\eta, \kappa\} > 0$. Interchange μ and ν if necessary so that ν has upward spectral curvature $\kappa = \gamma > 0$. Since $\text{supp } \nu$ is downward closed and hence connected, the final conclusion of Lemma A.1 gives $\text{supp } \nu = 2^V$, where $V := \{u : \nu(\{u\}) > 0\}$. As in the reduction above, we may assume without loss of generality that $V = \mathcal{U}$, by pinning $\mathcal{U} \setminus V$ into μ and restricting to V (returning zero if the pinning is infeasible, and evaluating the case $V = \emptyset$ directly). The measure μ still has downward-closed support, and $\text{supp } \nu = 2^\mathcal{U}$.

To obtain a positive gap even when $\eta + \kappa = 0$, decrease the field of μ by setting

$$t := 1 - \frac{\varepsilon}{16n}, \quad \tilde{\mu}(A) := t^{|A|} \mu(A).$$

This multiplies each ratio $\mu(S+u)/\mu(S)$ by t and each ratio $\mu(S+u+v)/\mu(S)$, for distinct $u, v \notin S$, by t^2 . Substituting into Definition 1.10 with parameter $t\eta$ gives

$$A_S^{\tilde{\mu}, t\eta} = t^2 A_S^{\mu, \eta} \succeq 0.$$

Thus $\tilde{\mu}$ has upward spectral curvature $t\eta$, and

$$\delta := t\eta + \kappa = t(\eta + \kappa) + (1-t)\kappa \geq \frac{\varepsilon\gamma}{16n}.$$

Let $\tilde{Z}_\cap$ be the convolution value with μ replaced by $\tilde{\mu}$. Since $t^n \geq 1 - \varepsilon/16$,

$$(1 - \varepsilon/16)Z_\cap \leq t^n Z_\cap \leq \tilde{Z}_\cap \leq Z_\cap.$$

Thus an estimate of $\tilde{Z}_\cap$ with relative error at most $\varepsilon/4$ also estimates $Z_\cap$ with relative error at most ε .

Apply Theorem 1.14 with first factor $\tilde{\mu}$ and second factor $\nu^*(A) = \nu(\mathcal{U} \setminus A)$, restricted to $\Omega := \text{supp } \mu$. The second factor needs downward curvature κ , which follows from

$$D_A^{\nu^*, \kappa} = A_{\mathcal{U} \setminus A}^{\nu, \kappa} \succeq 0.$$

Restricting ν^* to Ω leaves these matrices unchanged, since every deletion of a set in Ω remains in Ω . The resulting chain samples $\rho(A) \propto \tilde{\mu}(A)\nu^*(A)$, has gap at least δ , and has rates

$$q(A, A+u) = t \frac{\mu(A+u)}{\mu(A)}, \quad q(A, A-u) = \frac{\nu((\mathcal{U} \setminus A) + u)}{\nu(\mathcal{U} \setminus A)}.$$

Only the addition rates are multiplied by $t < 1$. Both rates are at most $B := \chi\gamma$. Normalizing the rates by $2nB$ gives a lazy chain with gap at least

$$\frac{\delta}{2nB} \geq \frac{\varepsilon\gamma/(16n)}{2n\chi\gamma} = \frac{\varepsilon}{32\chi n^2}.$$

The factor γ cancels because the rate bound also contains γ . The minimum stationary mass is at least $t^n/(2^n\mathcal{R}_\mu\mathcal{R}_\nu)$, and each step costs $O(1)$. It starts from the supported empty set. The mixing bound Eq. (12) gives a sample with total-variation error τ in

$$O\left(\frac{n^2\chi}{\varepsilon} [n + \log(\mathcal{R}_\mu\mathcal{R}_\nu) + \log(1/\tau)]\right)$$

steps. The same bounds hold for the conditional distributions obtained by fixing coordinates.

Lemma 3.1, applied to the n membership bits with weight $\tilde{\mu}(A)\nu(\mathcal{U} \setminus A)$ and starting state $\emptyset$, estimates $\tilde{Z}_\cap$ to relative error $\varepsilon/4$, with success probability at least $3/4$, using $O(n^3\varepsilon^{-2}\log(2n))$ conditional samples of accuracy $\tau = \varepsilon/(64n)$. With fixed walk lengths, this gives total cost

$$O\left(n^5\chi\varepsilon^{-3}\log(2n) \left[n + \log(\mathcal{R}_\mu\mathcal{R}_\nu) + \log\frac{2n}{\varepsilon}\right]\right)$$

arithmetic operations and value-oracle calls on every execution. The tilt bound gives the required relative estimate of $Z_\cap$. $\square$

A.2 Proofs of the support lemmas

We now prove [Lemmas A.1](#) and [A.2](#).

Proof of Lemma A.1. Fix $u \in V$. We show that u can be added to every supported set that does not already contain it. The key observation is that, along an edge between two such sets, u can be added at both endpoints or at neither.

To see this, let $S, S+v \in \Omega$ with distinct $u, v \notin S$, and put

$$a = \frac{\omega(S+u)}{\omega(S)}, \quad b = \frac{\omega(S+v)}{\omega(S)} > 0, \quad c = \frac{\omega(S+u+v)}{\omega(S)}.$$

The principal submatrix of $A_S^{\omega,\kappa}$ on u, v is

$$M = \begin{pmatrix} a^2 - \kappa a & ab - c \\ ab - c & b^2 - \kappa b \end{pmatrix} \succeq 0.$$

If $a = 0$, the first diagonal entry is zero, so positive semidefiniteness forces the off-diagonal entry to vanish, giving $c = 0$. If $a > 0$, positive semidefiniteness also gives

$$0 \leq (b, -a)M(b, -a)^T = ab(2c - \kappa(a+b)),$$

so $c \geq \kappa(a+b)/2 > 0$. Thus $a > 0$ if and only if $c > 0$, proving the observation.

Now take any $S \in \Omega$ with $u \notin S$. Since $u \in V$ and Ω is connected, there is a path in Ω from S to a set containing u . Stop at the first such set. The last step adds u , so u can be added to the preceding set. Every earlier edge has both endpoints omitting u ; by the observation, working backwards along the path shows that u can also be added to S . Since $u \in V$ was arbitrary, Ω is upward closed in 2^V .

If Ω is also downward closed, then $\emptyset \in \Omega$. Upward closure now gives $\Omega = 2^V$, and the description of V as the set of elements u with $\omega(\{u\}) > 0$ follows. $\square$

Proof of Lemma A.2. Let $P := \text{conv}\{\mathbf{1}_A : A \in \text{supp } \omega\}$, where $\mathbf{1}_A$ is the incidence vector of A . This convex hull is the *Newton polytope* of g_ω . Its vertices are precisely the incidence vectors of sets in $\text{supp } \omega$, and an edge is a one-dimensional face. By [\[Ove20, Proposition 10.4\]](#), every edge is parallel to a coordinate vector e_i or a difference $e_i - e_j$. Consequently, a linear objective with no improving addition, deletion, or exchange within $\text{supp } \omega$ is globally optimal: these moves include every incident edge. Starting at A_0 , minimize $|C \setminus A| + |A \cap O|$ by such moves. At most n improvements, each found with $O(n^2)$ queries, either produce a feasible set or certify infeasibility. The convex hull of the feasible incidence vectors is a face of P and inherits its edge directions. Greedy additions and deletions within this face therefore reach the maximum and minimum cardinalities, visiting every intervening size in $O(n^2)$ further queries. $\square$

A.3 Optimality of the curvature threshold

Proof of Remark 1.18. We reduce counting proper two-colourings of a 3-uniform hypergraph to subset convolution. Deciding whether such a colouring exists is NP-complete [DRS05]. Fix $\delta > 0$ and a rational constant $0 < c < \min\{1, \delta/2\}$. For a 3-uniform hypergraph $\mathcal{H} = (\mathcal{U}, \mathcal{F})$ with $n := |\mathcal{U}| \geq 1$, put $a := c/n$ and define

$$\begin{aligned}\mathcal{I} &:= \{S \subseteq \mathcal{U} : F \not\subseteq S \text{ for every } F \in \mathcal{F}\}, \\ \mu(S) = \nu(S) &:= a^{|S|} \mathbf{1}[S \in \mathcal{I}].\end{aligned}$$

The two colour classes form a proper colouring exactly when both belong to $\mathcal{I}$. Consequently,

$$[\mathbf{x}^{\mathcal{U}}](g_\mu(\mathbf{x})g_\nu(\mathbf{x})) = a^n \#\{\text{proper two-colourings of } \mathcal{H}\}.$$

Since a^n is known, it remains to verify the promised curvature and oracle bounds.

Fix $S \in \mathcal{I}$, and let $J_S := \{u \in \mathcal{U} \setminus S : S + u \in \mathcal{I}\}$ be its feasible additions. On J_S , define the symmetric matrix

$$B_S(u, v) := \mathbf{1}[u \neq v, S + u + v \notin \mathcal{I}].$$

Each row of B_S has at most $n - 1$ nonzero entries, all equal to one, so $B_S \succeq -(n - 1)I$. Substituting the weight ratios into Definition 1.10 gives

$$A_S^{\mu, -c}|_{J_S \times J_S} = a^2(I + B_S) + caI = a^2((n + 1)I + B_S) \succeq 2a^2I.$$

All other rows and columns vanish by downward closure. Thus both measures have upward curvature $-c$, and we may take $\eta = \kappa = -c$, giving $\eta + \kappa = -2c > -\delta$. Their common support is downward closed, their rational weights are computable in polynomial time, and

$$\log(\mathcal{R}_\mu \mathcal{R}_\nu) \leq 2n \log(1/a) = 2n \log(n/c) = O(n \log n).$$

An FPRAS on these instances would therefore give a bounded-error polynomial-time test for hypergraph two-colourability. By reducing the decision queries in the usual SAT self-reduction to this problem, amplifying their success probabilities, and verifying the resulting assignment, we obtain an RP algorithm for SAT. Hence $\text{NP} = \text{RP}$. $\square$

A.4 The coordinate-descent bound

Proof of Theorem 3.2. We average the noisy gradients and take a fixed step in any coordinate whose empirical derivative is large. Choose

$$\frac{\eta}{16L} \leq s \leq \frac{\eta}{8L}, \tag{25}$$

$$K := \lceil 128LH_0\eta^{-2} \rceil + 1, \quad m_b := \left\lceil 32\sigma^2\eta^{-2} \log(2d(K + 1)/\delta) \right\rceil. \tag{26}$$

Start at $\mathbf{x} := \mathbf{x}_0$. At each iteration, average m_b fresh oracle vectors to obtain $\hat{\mathbf{g}} := m_b^{-1} \sum_{j=1}^{m_b} \mathbf{G}^{(j)}(\mathbf{x})$. Return $\mathbf{x}$ if $\|\hat{\mathbf{g}}\|_\infty \leq \eta/2$; otherwise declare failure if K updates have already occurred, and otherwise choose i with $|\hat{g}_i| > \eta/2$ and update

$$\mathbf{x} \leftarrow \mathbf{x} - s \text{sign}(\hat{g}_i) \mathbf{e}_i, \quad \text{sign}(u) := u/|u| \quad (u \neq 0).$$

There are at most $K + 1$ batches and K updates, so every queried or returned point is $\mathbf{x} = \mathbf{x}_0 + s\mathbf{k}$ for an integer vector $\mathbf{k}$ satisfying

$$\|\mathbf{k}\|_1 \leq K, \quad \|\mathbf{x} - \mathbf{x}_0\|_1 \leq Ks \leq 16H_0/\eta + \eta/(4L). \tag{27}$$

These deterministic bounds give Eq. (15). Moreover, $(K + 1)m_b = O(Q)$ for Q in Eq. (16); averaging and selecting a coordinate cost $O(dm_b)$ per batch, so $T \geq d$ gives the total cost $O(TQ)$ on every history.

For correctness, conditional sub-Gaussian concentration gives

$$\Pr[|\hat{g}_i - \partial_i f(\mathbf{x})| > \eta/4 \mid \text{past}] \leq 2e^{-m_b \eta^2 / (32\sigma^2)} \leq \frac{\delta}{d(K+1)}.$$

Thus, with probability at least $1 - \delta$, every coordinate estimate in every batch is accurate within $\eta/4$. On this event, any returned point satisfies $\|\nabla f(\mathbf{x})\|_\infty \leq 3\eta/4 < \eta$. Each updated coordinate has $|\partial_i f(\mathbf{x})| > \eta/4$ and the same sign $\tau := \text{sign}(\hat{g}_i)$ as its estimate. By [Eq. \(13\)](#) and [Eq. \(25\)](#),

$$\begin{aligned} f(\mathbf{x} - s\tau \mathbf{e}_i) &\leq f(\mathbf{x}) - s|\partial_i f(\mathbf{x})| + Ls^2/2 \\ &\leq f(\mathbf{x}) - \frac{\eta^2}{64L} + \frac{\eta^2}{128L} = f(\mathbf{x}) - \frac{\eta^2}{128L}. \end{aligned} \quad (28)$$

After K updates this would give $f(\mathbf{x}) \leq f(\mathbf{x}_0) - K\eta^2/(128L) < f(\mathbf{x}_0) - H_0$, contradicting the lower bound. Hence the algorithm returns before making K updates. $\square$

A.5 Arithmetic for the counting algorithm

The field-finding and counting algorithms use only arithmetic operations. Ceilings of binary logarithms are computed by repeated doubling; other logarithms in sample counts can be replaced by these upper bounds, changing only absolute constants. In the coordinate-descent application, $L = \sigma = 1$. Take $s = \log(1 + \eta/8)$, which lies in $[\eta/16, \eta/8]$, and maintain z_u directly: an update multiplies or divides one field coordinate z_u by $1 + \eta/8$. Each update costs $O(1)$ operations, and neither s nor η needs to be evaluated. The self-reduction uses membership tests, empirical frequencies, and products and quotients of oracle values, fields, and integer powers of t , so it requires no further logarithms or exponentials.

Remark A.4 (Bit complexity). For rational input parameters and polynomial-time value oracles returning rationals of polynomial encoding length, the algorithm in [Theorem 1.6](#) has an implementation whose running time is polynomial in the input length and ε^{-1} on every execution. Indeed, the update rule above gives $z_u = (1 + 1/(32n))^{k_u}$ for integers k_u , and the field bound [Eq. \(23\)](#) implies $\sum_u |k_u| = O(n^2 B)$. Each field coordinate can therefore be stored using polynomially many bits. The probabilities for choosing which element to add back are ratios of sums of at most n weighted oracle values, and empirical frequencies are ratios of sample counts. The final estimate uses products and ratios of these numbers, fields, and powers of t . All these numbers have polynomially many bits.

We simulate the required rational probabilities using fair coin flips; see [\[NP05, Proposition 13\]](#). Each binary choice can be simulated with probability at most 2^{-r} of requiring more than r bits. If N is a polynomial bound on the total number of binary choices in the homogeneous algorithm, aborting and returning zero when any choice needs more than $\lceil \log_2(64N) \rceil$ bits adds at most $1/64$ to its failure probability. The success probability proved in [Section 5](#) is at least $15/16$, so it remains above $3/4$. Conditioning, amplification, and summation over homogeneous layers preserve polynomial bit complexity. In particular, the rational six-vertex instances in [Appendix C.7](#) satisfy these oracle assumptions.

A.6 Fractional log-concavity and spectral curvature

Proof of [Proposition 1.22](#). Expanding $p(\mathbf{1} + \mathbf{x})$ using its nonnegative coefficients shows that $\text{supp } \hat{\mu}$ is downward closed. For $S \in \text{supp } \hat{\mu}$, set $q := \partial_S p$, so $q(\mathbf{1}) = \hat{\mu}(S) > 0$. As verified below, q is α -fractionally log-concave. The coefficient ratios in [Definition 1.10](#) and the chain rule therefore give

$$\begin{aligned} A_S^{\hat{\mu}, 1-\alpha^{-1}} &= -\nabla^2 \log q(\mathbf{1}) - (1 - \alpha^{-1}) \text{diag}(\nabla \log q(\mathbf{1})) \\ &= -\alpha^{-2} \nabla_x^2 \log q(x_1^\alpha, \dots, x_n^\alpha) \Big|_{x=1} \succeq 0, \end{aligned}$$

which proves the asserted curvature bound.

It remains to verify that nonzero coordinate derivatives preserve fractional log-concavity. We use the limiting argument of [\[Ali+21, Section 1.5\]](#). Writing $\mathbf{x}^\alpha := (x_1^\alpha, \dots, x_n^\alpha)$, multiaffinity gives

$$(\partial_i p)(\mathbf{x}^\alpha) = \lim_{s \rightarrow \infty} s^{-\alpha} p((\mathbf{x} + s\mathbf{e}_i)^\alpha).$$

Each function on the right is a positive multiple of a translate of $x \mapsto p(x^\alpha)$, hence is log-concave. The log-concavity inequality is preserved under the limit. Iterating proves the assertion for every nonzero $\partial_S p$. $\square$

A.7 Examples of spectral curvature

We show that every $\kappa \in \mathbb{R}$ is the largest admissible curvature of some distribution. For a positive value, take the product measure $\mu(S) \propto \lambda^{|S|}$ with $\lambda > 0$: the curvature matrix on coordinates outside S is $\lambda(\lambda - \kappa)I$, so the largest admissible κ is λ . For a nonpositive value, take $\mathcal{U} = \{1, 2\}$ and

$$(\mu(\emptyset), \mu(\{1\}), \mu(\{2\}), \mu(\{1, 2\})) = \frac{(1, 1, 1, M)}{3 + M}, \quad M \geq 2.$$

The smallest eigenvalue of $A_{\emptyset}^{\mu, \kappa}$ is $2 - M - \kappa$, and the singleton matrices are positive semidefinite exactly when $\kappa \leq M$. Thus the largest admissible κ is $2 - M$, giving every nonpositive value.

A.8 Poincaré inequality for induced chains

Proof of Proposition 5.11. Reversing paths whose intermediate states lie outside B and applying detailed balance gives $\pi(x)r^B(x, y) = \pi(y)r^B(y, x)$, proving reversibility with respect to π_B . To transfer the Poincaré inequality, extend $f : B \rightarrow \mathbb{R}$ to Ω by

$$h(x) := \mathbb{E}[f(X_{\tau_B}) \mid X_0 = x].$$

Then $h = f$ on B . We claim that

$$(\mathcal{L}_r h)(x) = \begin{cases} 0, & x \notin B, \\ (\mathcal{L}_{r^B} f)(x), & x \in B. \end{cases}$$

Assuming these identities, we have $\mathcal{E}_r(h, h) = \pi(B)\mathcal{E}_{r^B}(f, f)$, while the law of total variance gives $\mathbf{Var}_{\pi}[h] \geq \pi(B)\mathbf{Var}_{\pi_B}[f]$. Applying the original Poincaré inequality yields

$$\begin{aligned} \pi(B)\mathcal{E}_{r^B}(f, f) &= \mathcal{E}_r(h, h) \\ &\geq \gamma \mathbf{Var}_{\pi}[h] \geq \gamma \pi(B)\mathbf{Var}_{\pi_B}[f]. \end{aligned}$$

Canceling $\pi(B) > 0$ proves the desired inequality.

It remains to verify the two identities. For $x \notin B$, expanding the definition gives

$$\begin{aligned} (\mathcal{L}_r h)(x) &= \sum_{y \neq x} r(x, y)(h(x) - h(y)) \\ &= \left(\sum_{y \neq x} r(x, y) \right) h(x) - \sum_{y \neq x} r(x, y)h(y) = 0, \end{aligned}$$

where the last equality is justified as follows. Put $q_x := \sum_{z \neq x} r(x, z) > 0$, and let $\tau := \inf\{s > 0 : X_s \neq X_0\}$ be the first jump time. Starting from x , the first jump goes to $y \neq x$ with probability $r(x, y)/q_x$. Since $x \notin B$, the chain has not hit B before τ . If $X_\tau = y \in B$, then $\tau_B = \tau$ and $f(X_{\tau_B}) = f(y) = h(y)$. If $y \notin B$, the strong Markov property at τ says that the remaining trajectory has the law of the chain started at y , so the conditional expectation of $f(X_{\tau_B})$ is again $h(y)$. Consequently,

$$h(x) = \mathbb{E}[h(X_\tau) \mid X_0 = x] = \sum_{y \neq x} \frac{r(x, y)}{q_x} h(y).$$

Multiplying by q_x gives the cancellation above.

For $x \in B$, substituting $h(u) = \sum_{y \in B} \mathbf{Pr}[X_{\tau_B} = y \mid X_0 = u] f(y)$ for $u \notin B$ gives

$$\begin{aligned} (\mathcal{L}_r h)(x) &= \sum_{y \in B \setminus \{x\}} r(x, y)(f(x) - f(y)) + \sum_{u \notin B} r(x, u)(f(x) - h(u)) \\ &= \sum_{y \in B \setminus \{x\}} r^B(x, y)(f(x) - f(y)) = (\mathcal{L}_{r^B} f)(x). \end{aligned} \quad \square$$

B Background for Applications

We recall the background used in [Appendix C](#); see [\[Oxl11\]](#) for standard matroid facts.

B.1 Further matroid terminology and examples

We use the definitions and notation from [Section 3.2](#). A set is *spanning* if it contains a basis. An element e is a *loop* if $\text{rk}_{\mathcal{M}}(\{e\}) = 0$; distinct nonloops e, f are *parallel* if $\text{rk}_{\mathcal{M}}(\{e, f\}) = 1$. Together with equality, this partitions the nonloops into parallel classes.

In a *partition matroid*, $\mathcal{U} = \sqcup_i E_i$ and independence means $|I \cap E_i| \leq b_i$ for prescribed nonnegative integer bounds. In a *graphic matroid*, independent edge sets are forests and bases are spanning forests, one tree per connected component. In a *column matroid*, independent sets index linearly independent columns of a matrix over a field.

B.2 Operations and oracles

The *restriction* $\mathcal{M}|_A$ keeps the independent sets contained in A ; deletion of A is restriction to $\mathcal{U} \setminus A$. The *contraction* $\mathcal{M}/A$ has ground set $\mathcal{U} \setminus A$. The ranks of the dual and contraction satisfy

$$\begin{aligned} \text{rk}_{\mathcal{M}^*}(T) &= |T| + \text{rk}_{\mathcal{M}}(\mathcal{U} \setminus T) - \text{rk}_{\mathcal{M}}(\mathcal{U}), \\ \text{rk}_{\mathcal{M}/A}(S) &= \text{rk}_{\mathcal{M}}(S \cup A) - \text{rk}_{\mathcal{M}}(A) \quad (S \subseteq \mathcal{U} \setminus A). \end{aligned} \quad (29)$$

Thus T is independent in $\mathcal{M}^*$ exactly when its complement is spanning. If B_A is a basis of $\mathcal{M}|_A$, contraction independence is tested by $B_A \cup S \in \mathcal{I}(\mathcal{M})$, even when A is dependent.

The *direct sum* of matroids on disjoint ground sets has independent sets whose restriction to each component is independent. The *union* of matroids on a common ground set has independent sets admitting a partition $I = I_1 \sqcup \dots \sqcup I_k$, with $I_i \in \mathcal{I}(\mathcal{M}_i)$.

The rank formulas above, together with the greedy rank computation in [Section 3.2](#), give independence oracles for duals and contractions. With separate independence oracles, matroid intersection finds a maximum common independent set in polynomial time [\[Edm79\]](#), and matroid union has a polynomial-time independence oracle [\[BF10, Section 4.1\]](#).

B.3 Submodularity and integral polymatroids

A set function $f : 2^{[m]} \rightarrow \mathbb{R}$ is *submodular* if

$$f(A) + f(B) \geq f(A \cup B) + f(A \cap B) \quad (A, B \subseteq [m]).$$

It is *monotone* if $A \subseteq B$ implies $f(A) \leq f(B)$, and *normalized* if $f(\emptyset) = 0$. Matroid rank functions are a special case of normalized monotone submodular functions. For such an integer-valued f , the *polymatroid base polytope* is

$$B(f) := \left\{ \mathbf{x} \in \mathbb{R}_{\geq 0}^m : \sum_{i=1}^m x_i = f([m]), \quad \sum_{i \in A} x_i \leq f(A) \quad \forall A \subseteq [m] \right\}.$$

Its vertices are integral [\[FM19, Section 2\]](#). For $f = \text{rk}_{\mathcal{M}}$, its integer points are exactly the basis indicators.

B.4 Log-concave and Lorentzian polynomials

Positive element weights preserve the log-concavity of the basis polynomial recalled in [Section 3.2](#), by variable rescaling:

$$\sum_{B \in \mathcal{B}(\mathcal{M})} \left(\prod_{e \in B} w_e \right) \mathbf{x}^B = g_{\mathcal{M}}((w_e x_e)_{e \in \mathcal{U}}).$$

Zero weights are handled by deletion; if the original rank drops, the weighted polynomial is zero. Products on disjoint variable sets also preserve log-concavity and multiaffinity.

A nonzero homogeneous polynomial with nonnegative coefficients is *Lorentzian* if every iterated coordinate partial derivative, including order zero, is either zero or log-concave on the positive

orthant [BH20, Theorem 2.30]. For a polynomial $p(z_1, \dots, z_m)$, choose positive integers c_i with $\deg_{z_i} p \leq c_i$ and new variables $x_{i,1}, \dots, x_{i,c_i}$. Its *polarization* applies the rule

$$z_i^k \mapsto \frac{1}{\binom{c_i}{k}} \sum_{\substack{J \subseteq [c_i] \\ |J|=k}} \prod_{j \in J} x_{i,j}, \quad 0 \leq k \leq c_i,$$

to each variable in each monomial and extends linearly. This operation preserves the Lorentzian property [BH20, Proposition 3.1].

A nonzero, finitely supported weight $\omega : \mathbb{Z}_{\geq 0}^m \rightarrow \mathbb{R}_{\geq 0}$ is *log-M-concave* if, for some integer r , $\sum_i k_i = r$ for every $\mathbf{k} \in \text{supp } \omega$, and, for every $\mathbf{x}, \mathbf{y} \in \text{supp } \omega$ and every i with $x_i > y_i$, there is a j with $x_j < y_j$ such that

$$\omega(\mathbf{x})\omega(\mathbf{y}) \leq \omega(\mathbf{x} - \mathbf{e}_i + \mathbf{e}_j)\omega(\mathbf{y} + \mathbf{e}_i - \mathbf{e}_j). \quad (30)$$

Here $\mathbf{e}_i$ is the i th coordinate unit vector. For such a weight ω , the polynomial $\sum_{\mathbf{k}} \omega(\mathbf{k}) \mathbf{z}^{\mathbf{k}} / \mathbf{k}!$ is Lorentzian, where $\mathbf{k}! = \prod_i k_i!$ [BH20, Corollary 3.16].

C Applications

We apply the subset-convolution framework to counting and approximation problems involving graphs, determinants, and partition functions. The common step is to encode the relevant weights and constraints in two measures that satisfy the log-concavity or spectral-curvature hypotheses of our convolution theorems. The shared matroid and log-concavity preliminaries are collected in [Appendix B](#).

[Figure 2](#) summarizes the reductions to [Theorems 1.6](#) and [1.16](#). The corollaries below give the precise assumptions and guarantees. We group the applications by the reductions they share.

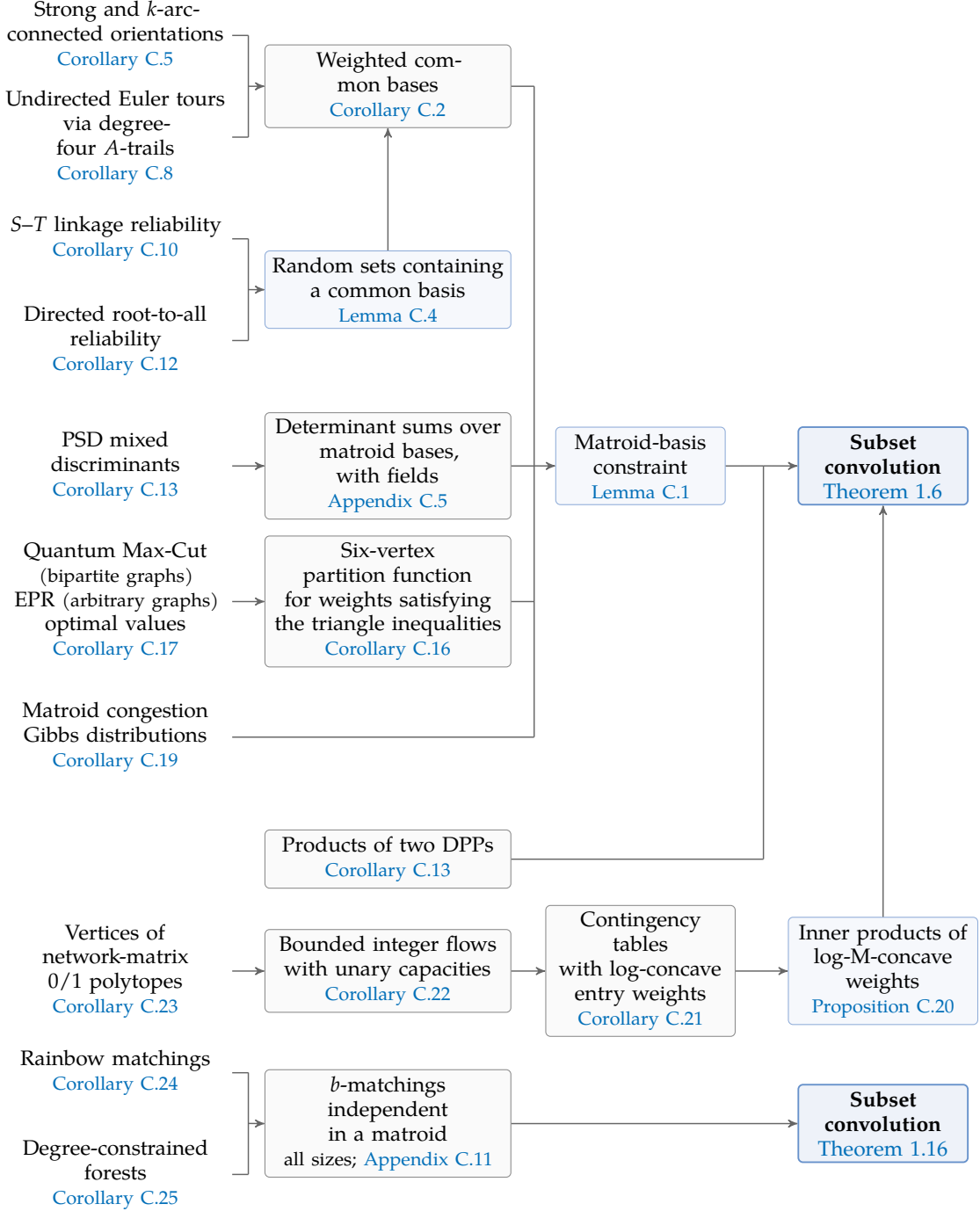


Figure 2: Reduction tree for the applications of [Theorems 1.6](#) and [1.16](#). Arrows point toward the problem or result used by each reduction. Some intermediate steps are omitted.

Unless stated otherwise, running-time statements below use the arithmetic model of [Theorems 1.6](#) and [1.16](#); we omit bit-complexity analysis for simplicity. The EPR-in-BPP conclusion still uses the bit-complexity bound in [Remark A.4](#). We first record two reductions used throughout the appendix. Products of polynomially many rational input weights, determinants of rational input matrices, and factorials of polynomially bounded integers have polynomial bit length. We use this observation below to bound the logarithmic ranges of positive coefficients.

Lemma C.1. *Let $\omega : 2^{\mathcal{U}} \rightarrow \mathbb{R}_{\geq 0}$ have a log-concave generating polynomial, an evaluation oracle, a given set in $\text{supp } \omega$, and a supplied bound on $\mathcal{R}_\omega$, as in [Theorem 1.6](#). For any matroid $\mathcal{M}$ on $\mathcal{U}$ given by an independence oracle, there is an FPRAS for*

$$\sum_{S \in \mathcal{B}(\mathcal{M})} \omega(S),$$

with running time polynomial in $|\mathcal{U}|$, $\log \mathcal{R}_\omega$, and the inverse accuracy, counting oracle calls as single operations.

Proof. Set $\nu(T) := \mathbf{1}[T \in \mathcal{B}(\mathcal{M})]$. Its generating polynomial is the basis polynomial of $\mathcal{M}^*$, hence is log-concave, and $\mathcal{R}_\nu = 1$. The complement of a greedy basis initializes ν , and $[x^{\mathcal{U}}](g_\omega g_\nu)$ is the desired sum. Apply [Theorem 1.6](#). $\square$

Corollary C.2. *Let $\mathcal{M}, \mathcal{N}$ be matroids of the same rank r on $\mathcal{U}$, given by independence oracles, and let w_e be nonnegative rational element weights. There is an FPRAS for*

$$\sum_{S \in \mathcal{B}(\mathcal{M}) \cap \mathcal{B}(\mathcal{N})} \prod_{e \in S} w_e.$$

Proof. Delete zero-weight elements; if either rank drops below r , return zero. The weight $\omega(S) := \mathbf{1}[S \in \mathcal{B}(\mathcal{M})] \prod_{e \in S} w_e$ has a log-concave generating polynomial by positive variable rescaling. A greedy basis initializes it, and its logarithmic coefficient range is polynomial in the weight encoding length. Apply [Lemma C.1](#) with the matroid $\mathcal{N}$. $\square$

To count random subsets that contain a common basis, we use the following consequence of Rado's theorem [[Rad42](#)].

Theorem C.3 (Rado matroids [[Oxl11](#), Theorem 11.2.12]). *Let G be a finite bipartite graph with left vertex set L and right vertex set R , and let $\mathcal{R}$ be a matroid on R . There is a matroid $\mathcal{L}$ on L such that $A \subseteq L$ is an independent set of $\mathcal{L}$ exactly when G has a matching whose left endpoints are exactly A and whose right endpoints form an independent set of $\mathcal{R}$.*

Lemma C.4. *Let $\mathcal{M}, \mathcal{N}$ be matroids of the same rank on $\mathcal{U}$, given by independence oracles. Retain each element u independently with rational probability $p_u \in [0, 1]$. There is an FPRAS for the probability that the retained set contains a common basis.*

Proof. We encode the event by weighted common bases of two new matroids. Put $m := |\mathcal{U}|$, and let $\mathcal{U}^+ = \{u^+ : u \in \mathcal{U}\}$ and $\mathcal{U}^- = \{u^- : u \in \mathcal{U}\}$ be two disjoint copies of $\mathcal{U}$. Let $\mathcal{H}$ be the direct sum of copies of $\mathcal{M}$ and $\mathcal{N}^*$ on $\mathcal{U}^+$ and $\mathcal{U}^-$, respectively; it has rank m . Form a bipartite graph with left vertex set $\mathcal{X} := \{x_u^0, x_u^1 : u \in \mathcal{U}\}$ and right vertex set $\mathcal{U}^+ \cup \mathcal{U}^-$. The new elements x_u^0, x_u^1 represent u being absent or retained, respectively. Join x_u^0 only to u^- , and join x_u^1 to both u^- and u^+ .

Apply [Theorem C.3](#) with $L = \mathcal{X}$, $R = \mathcal{U}^+ \cup \mathcal{U}^-$, and $\mathcal{R} = \mathcal{H}$, and denote the resulting matroid $\mathcal{L}$ by $\mathcal{Q}$. A subset is independent in $\mathcal{Q}$ exactly when it can be matched to an independent set of $\mathcal{H}$. Let $\mathcal{P}$ be the partition matroid on $\mathcal{X}$ allowing at most one of x_u^0, x_u^1 for each u . Both matroids have rank m : any basis of $\mathcal{H}$ can be matched to distinct left vertices by matching u^+ to x_u^1 and u^- to x_u^0 .

The retained set $F \subseteq \mathcal{U}$ has the unique representation $X_F := \{x_u^1 : u \in F\} \cup \{x_u^0 : u \notin F\}$. Since $|X_F| = m = \text{rk}(\mathcal{H})$, X_F is independent in $\mathcal{Q}$ exactly when it can be matched to a basis of $\mathcal{H}$. A matching covering X_F uses exactly one of u^+ and u^- for each element u , so its matched right vertices form a set $B^+ \cup (\mathcal{U} \setminus B)^-$ with $B \subseteq F$. This set is a basis of $\mathcal{H}$ exactly when B is a common basis of $\mathcal{M}, \mathcal{N}$. Consequently,

$$X_F \in \mathcal{B}(\mathcal{P}) \cap \mathcal{B}(\mathcal{Q}) \iff F \text{ contains a common basis of } \mathcal{M}, \mathcal{N}.$$

Giving x_u^1, x_u^0 weights $p_u, 1 - p_u$ makes the weighted common-bases sum the required probability. Each F is counted once, regardless of the number of common bases it contains.

It remains to implement an independence oracle for $\mathcal{Q}$. We do so by matroid intersection on restrictions and contractions of $\mathcal{M}$ and $\mathcal{N}$. For a queried set $Y \subseteq \mathcal{X}$, let S_2 contain the elements u for which both x_u^0, x_u^1 belong to Y , let S_0 contain those for which neither belongs to Y , and let $S_1 := \{u : x_u^1 \in Y, x_u^0 \notin Y\}$. The possible sets of right endpoints of a matching with left endpoint set Y are exactly

$$(S_2 \cup B)^+ \cup (\mathcal{U} \setminus (S_0 \cup B))^- , \quad B \subseteq S_1.$$

Here B consists of the elements $u \in S_1$ for which x_u^1 is matched to $u^+ \in \mathcal{U}^+$. By duality, the displayed set is independent in $\mathcal{H}$ exactly when $S_2 \cup B$ is independent in $\mathcal{M}$ and $S_0 \cup B$ contains a basis of $\mathcal{N}$.

Reject if S_2 is dependent in $\mathcal{M}$. Otherwise, apply matroid intersection to $(\mathcal{M}/S_2)|_{S_1}$ and $(\mathcal{N}/S_0)|_{S_1}$, and accept exactly when they have a common independent set of size $k := \text{rk}_{\mathcal{N}}(\mathcal{U}) - \text{rk}_{\mathcal{N}}(S_0)$, where $\text{rk}_{\mathcal{N}}$ is the rank function of $\mathcal{N}$. The required oracles use only the original independence oracles: test $I \subseteq S_1$ in $\mathcal{M}/S_2$ by testing $S_2 \cup I$ in $\mathcal{M}$, and test it in $\mathcal{N}/S_0$ by testing $B_0 \cup I$ in $\mathcal{N}$, where B_0 is a basis of $\mathcal{N}|_{S_0}$ found greedily. This gives a polynomial-time oracle for $\mathcal{Q}$, so [Corollary C.2](#) applies. $\square$

C.1 Strong orientations

Let $G = (V, E)$ be a loopless undirected multigraph, with $|V| \geq 2$, and let $k \geq 1$ be an integer. An orientation of the edges of G is *k-arc-connected* if for every nonempty proper subset $X \subsetneq V$, there are at least k arcs directed from $V \setminus X$ into X . Give each direction of each labeled edge a nonnegative rational weight $w_{e,v}$, where v is its head. Define

$$Z_k(G; w) := \sum_{D \text{ a } k\text{-arc-connected orientation}} \prod_{e \in E} w_{e, \text{head}_D(e)}.$$

The case $k = 1$ counts strong orientations, and zero weights allow fixed directions.

Corollary C.5. *There is an FPRAS for $Z_k(G; w)$, even when k is part of the input in binary.*

The following form of the classical common-bases representation [[Goe12](#), Lecture 16] follows from the indegree characterization in [[FM19](#)] and the matroid construction of McDiarmid [[McD73](#)]; see also [[BCF23](#), Theorem 2.1].

Lemma C.6 (Common bases for strong orientations). *Let $G = (V, E)$ be a loopless undirected multigraph with $|V| \geq 2$, and let $k \geq 1$ be an integer. Suppose every nontrivial undirected cut has size at least $2k$. Put $m := |E|$, and define*

$$\mathcal{U} := \{(e, v) : e \in E, v \in e\}, \quad \mathcal{U}_X := \{(e, v) \in \mathcal{U} : v \in X\}.$$

For an orientation D , let $d_D^-(v)$ be the number of edges directed into v . The function

$$f(X) := \max_{D \text{ a } k\text{-arc-connected orientation}} \sum_{v \in X} d_D^-(v) \quad (X \subseteq V)$$

is integer-valued, monotone, and submodular, with $f(\emptyset) = 0$ and $f(V) = m$. Let $\mathcal{M}$ be the partition matroid on $\mathcal{U}$ allowing at most one incidence per edge. There is a matroid $\mathcal{N}$ on $\mathcal{U}$ with rank function

$$\text{rk}_{\mathcal{N}}(Y) = \min_{X \subseteq V} (f(X) + |Y \setminus \mathcal{U}_X|) \quad (Y \subseteq \mathcal{U}).$$

Both matroids have rank m , and the map

$$D \mapsto A_D := \{(e, \text{head}_D(e)) : e \in E\}$$

is a bijection from the k -arc-connected orientations of G to $\mathcal{B}(\mathcal{M}) \cap \mathcal{B}(\mathcal{N})$.

Proof of Corollary C.5. We apply [Corollary C.2](#) to the matroids in [Lemma C.6](#). First, by the orientation theorem of Nash-Williams [[Nas60](#), Theorem 1], a k -arc-connected orientation exists exactly when every nontrivial undirected cut has size at least $2k$. Check this by a minimum cut, returning zero if it fails, and put $m := |E|$.

Give incidence (e, v) weight $w_{e,v}$. The bijection in [Lemma C.6](#) preserves the product of weights, so the weighted common-bases sum is exactly $Z_k(G; w)$. It remains to implement the two independence oracles. For $\mathcal{M}$, check that the queried set contains at most one incidence per edge.

For $\mathcal{N}$, given any $Y \subseteq \mathcal{U}$, compute $\text{rk}_{\mathcal{N}}(Y)$ using the lemma's rank formula and accept exactly when $\text{rk}_{\mathcal{N}}(Y) = |Y|$. For fixed Y , the term

$$|Y \setminus \mathcal{U}_X| = |Y| - \sum_{v \in X} |Y \cap \mathcal{U}_{\{v\}}|$$

is a constant minus a sum of fixed vertex weights. Adding it to $f(X)$ therefore preserves submodularity, so the rank formula can be evaluated by submodular minimization [[Sch00](#)]. To supply the required values $f(X)$, assign cost zero to directions whose heads lie in X and cost one to the others. The minimum cost of a k -arc-connected orientation is $m - f(X)$, and it can be found in polynomial time [[IK05](#)]. Thus every value query, and hence each independence query, takes polynomial time.

With these oracles, [Corollary C.2](#) gives the claimed FPRAS. If k is part of the input in binary, the successful minimum-cut test bounds k by m , so the oracle implementations still take polynomial time in the input length. $\square$

C.2 Undirected Euler tours

Let G be a connected 4-regular multigraph. Each edge has two ends, called *half-edges*; a loop contributes two distinct half-edges at the same vertex. A *transition* at a vertex pairs its four incident half-edges into two pairs; there are three choices. Entering along either half-edge of a pair is followed by leaving along the other. A *closed trail* is a closed walk that does not repeat edges. Choosing a transition at every vertex partitions the edges into closed trails. We count the choices that give a single closed trail, or equivalently Euler tours up to cyclic shift and reversal. For each vertex v and each of its three transitions τ , give the whole pairing τ a nonnegative rational weight $w_v(\tau)$. These three weights may be chosen arbitrarily at each vertex. The total weight of Euler tours is

$$\sum_P \prod_{v \in V(G)} w_v(P_v),$$

where the sum ranges over all Euler tours P , and P_v is the transition used by P at v . Thus the weights are attached to local pairings, not to individual edge directions.

Corollary C.7 (Weighted Euler tours). *Let G be a connected 4-regular multigraph with nonnegative rational weights on its local transitions. There is an FPRAS for the total weight of Euler tours in G .*

Proof. We encode the weighted Euler tours in G as weighted common bases of two matroids and apply [Corollary C.2](#). Fix an Euler tour C , viewed as a cyclic sequence of vertex visits. Each vertex occurs twice because G is 4-regular. Two distinct vertices u, v alternate along C if, after ignoring visits to all other vertices, their cyclic order is u, v, u, v . The *interlace matrix* $A \in \mathbb{F}_2^{V(G) \times V(G)}$ is defined by $A_{uv} = 1$ when u and v alternate, and $A_{uv} = 0$ otherwise; in particular, $A_{uu} = 0$ [[Tra09](#), p. 2]. Choose a direction for traversing C and orient every edge accordingly. At a vertex v , label the two incoming half-edges a_1, a_2 and the two outgoing half-edges b_1, b_2 so that C enters through a_i and leaves through b_i on its i th visit, for $i = 1, 2$. Define

$$\begin{aligned} \phi_v &:= \{\{a_1, b_1\}, \{a_2, b_2\}\}, \\ \chi_v &:= \{\{a_1, b_2\}, \{a_2, b_1\}\}, \\ \psi_v &:= \{\{a_1, a_2\}, \{b_1, b_2\}\}. \end{aligned}$$

Only ϕ_v is used by C . The alternative χ_v also pairs each incoming half-edge with an outgoing half-edge, but switches which outgoing half-edge follows each incoming one. The transition ψ_v pairs the two incoming half-edges together and the two outgoing half-edges together. Associate these choices with the labeled columns

$$\phi_v \longleftrightarrow e_v, \quad \chi_v \longleftrightarrow A_v, \quad \psi_v \longleftrightarrow e_v + A_v \quad \text{of } [I \mid A \mid I + A] \text{ over } \mathbb{F}_2.$$

For a transition choice P , let K_P contain its selected columns. The extended Cohn–Lempel equality [[Tra09](#), Theorem 4] gives

$$\#\{\text{closed trails of } P\} = 1 + \text{nullity } K_P.$$

Indeed, deleting the identity choices leaves the principal interlace matrix with a diagonal one at each ψ choice, which is the matrix in that equality. Hence P is an Euler tour precisely when it is a common basis of the column matroid and the partition matroid choosing one column from each vertex triple. Giving each column its transition weight preserves the weight and counts each tour once. Gaussian elimination and checking that at most one column is selected from each vertex triple give the independence oracles, so [Corollary C.2](#) applies. $\square$

Ge and Štefankovič [[GŠ10](#), Theorem 4] give an approximation-preserving reduction from counting Euler tours in general undirected graphs to counting A -trails in 4-regular graphs with prescribed cyclic orders.

In a 4-regular multigraph, suppose that a cyclic order of the four half-edges is specified separately at each vertex. An A -trail is an Euler tour such that, at every visit to a vertex, the entering and leaving half-edges are adjacent in that vertex's cyclic order. Either direction through an adjacent pair is allowed. For cyclic order h_1, h_2, h_3, h_4 , the allowed arrival–departure pairs are the unordered pairs

$$\{h_1, h_2\}, \quad \{h_2, h_3\}, \quad \{h_3, h_4\}, \quad \{h_4, h_1\}.$$

For example, entering through h_1 and leaving through h_2 is allowed, as is entering through h_2 and leaving through h_1 . The transition at that vertex must therefore be one of

$$\{\{h_1, h_2\}, \{h_3, h_4\}\} \quad \text{or} \quad \{\{h_1, h_4\}, \{h_2, h_3\}\}.$$

The remaining pairing $\{\{h_1, h_3\}, \{h_2, h_4\}\}$ is forbidden.

Counting A -trails is a special case of weighted Euler-tour counting: assign weight one to each allowed transition and zero to each forbidden transition. Every A -trail then has weight one, and every other Euler tour has weight zero. By [Corollary C.7](#), this count admits an FPRAS. Composing that algorithm with the reduction gives the following corollary.

Corollary C.8 (Euler tours in general graphs). *There is an FPRAS for the unweighted number of Euler tours in general undirected graphs.*

C.3 Network reliability and disjoint linkages

Definition C.9 (S – T linkage). Let $G = (V, E)$ be a directed or undirected graph, and let $S, T \subseteq V$ be disjoint terminal sets with $|S| = |T| = k \geq 1$. An S – T linkage consists of k paths that share no vertices. Each path joins a vertex of S to a vertex of T , and every vertex of S and T is used as an endpoint exactly once. We do not specify which vertex of S is joined to which vertex of T . If G is directed, all paths must follow edge directions from S to T .

Corollary C.10 (S – T Network reliability). *Let $G = (V, E)$ be a directed or undirected graph, and let $S, T \subseteq V$ be disjoint terminal sets with $|S| = |T| = k \geq 1$. Retain each edge e independently with given rational probability $p_e \in [0, 1]$. There is an FPRAS for*

$$\text{Rel}_{S,T}(G; \mathbf{p}) := \sum_{\substack{F \subseteq E: \\ (V, F) \text{ admits an } S\text{--}T \text{ linkage}}} \prod_{e \in F} p_e \prod_{e \in E \setminus F} (1 - p_e). \quad (31)$$

Remark C.11. Taking $S = \{s\}$ and $T = \{t\}$ in an undirected graph recovers the FPRAS for s – t reliability in [[FFG26](#)] with worse running time.

Proof of Corollary C.10. We first encode directed linkage as the existence of a perfect matching in a random bipartite graph, then apply [Lemma C.4](#). An undirected edge gadget gives the undirected case.

For a directed graph, discard loops, arcs entering S , and arcs leaving T . Form a bipartite graph with vertex classes

$$V_L := \{\ell_v : v \in V \setminus T\}, \quad V_R := \{r_v : v \in V \setminus S\}.$$

Replace each remaining arc $u \rightarrow v$ by the edge $\ell_u r_v$. Keep this edge with the same probability as the original arc. Add an always-present edge $\ell_v r_v$ for each $v \notin S \cup T$. A linkage gives a perfect matching by selecting its path edges and the diagonal edges at unused vertices. Conversely, following a

perfect matching from each source reaches a distinct sink: entering a cycle or another source's path would give a vertex two selected incoming edges. Thus, for every retained edge set,

the directed graph admits an S – T linkage $\iff$ the bipartite graph has a perfect matching.

Let $\mathcal{M}, \mathcal{N}$ be the partition matroids on the bipartite edge set, allowing at most one edge at each left and right endpoint, respectively. If the full bipartite graph has no perfect matching, return zero. Otherwise both matroids have rank $|V| - k$, and a retained edge set contains a perfect matching exactly when it contains a common basis. Thus [Lemma C.4](#) approximates the desired probability, including the always-present diagonal edges.

Finally, for an undirected edge $e = \{u, v\}$, introduce new vertices a_e, b_e and arcs

$$u \rightarrow a_e, \quad v \rightarrow a_e, \quad a_e \rightarrow b_e, \quad b_e \rightarrow u, \quad b_e \rightarrow v.$$

Only $a_e \rightarrow b_e$ is random, with retention probability p_e ; the other four arcs are always present. Vertex-disjoint undirected paths lift to vertex-disjoint directed paths. Conversely, simple directed paths project to vertex-disjoint undirected paths, since returning through the gadget to the vertex where the path entered would repeat that vertex. This preserves the linkage event for every choice of retained original edges and uses one independent random variable per original edge. The directed result therefore also proves the undirected case. $\square$

C.4 Directed root-to-all reliability

Let $D = (V, E)$ be a directed multigraph with root r , and retain each arc e independently with rational probability $p_e \in [0, 1]$. For each $F \subseteq E$, let $\Lambda(V, F) \in \mathbb{R}^{V \times V}$ be the matrix whose entry $\Lambda(V, F)_{u,v}$, for $u \neq v$, is the maximum number of arc-disjoint directed paths from u to v in (V, F) . Set its diagonal entries to zero. For an integer $k \geq 1$, define

$$R_k(D, r; \mathbf{p}) := \sum_{\substack{F \subseteq E: \\ \Lambda(V, F)_{r,v} \geq k \\ \forall v \neq r}} \prod_{e \in F} p_e \prod_{e \notin F} (1 - p_e).$$

The paths to different target vertices need not be mutually disjoint. In particular, $k = 1$ asks that the root reach every vertex. Reversing the arcs gives the all-to-root version.

Corollary C.12. *There is an FPRAS for $R_k(D, r; \mathbf{p})$, even when k is part of the input in binary.*

Proof. We express the event as containment of a common basis and apply [Lemma C.4](#). Delete loops and arcs entering r , which do not affect this event. The one-vertex case has probability one. Otherwise put $q := k(|V| - 1)$, and return zero if $q > |E|$. Let $\mathcal{M}$ be the union of k copies of the graphic matroid of the underlying undirected multigraph. A set is independent in $\mathcal{M}$ exactly when it can be partitioned into k forests after ignoring the arc directions. Let $\mathcal{N}$ be the partition matroid allowing at most k arcs entering each nonroot vertex. The polynomial-time matroid-union algorithm [[BF10](#), Section 4.1] gives an independence oracle for $\mathcal{M}$, using cycle detection to test independence in each graphic matroid. For $\mathcal{N}$, we count the queried arcs entering each nonroot vertex and check that every count is at most k . If either rank is less than q , the event is impossible; otherwise both ranks equal q .

For each $F \subseteq E$, we have

$$\Lambda(V, F)_{r,v} \geq k \text{ for all } v \neq r \iff F \text{ contains a common basis of } \mathcal{M}, \mathcal{N}.$$

Edmonds's arborescence-packing theorem [[Edm73](#)] (see also [[BF10](#), Theorem 1.1]) equates the left-hand condition with the existence in F of k arc-disjoint spanning out-arborescences rooted at r , that is, directed spanning trees in which r reaches every vertex. The standard matroid representation [[BF10](#), Section 4.1] identifies the unions of k such arborescences with the common bases of $\mathcal{M}, \mathcal{N}$.

Applying [Lemma C.4](#) now gives the claimed approximation. The preliminary bound $k(|V| - 1) \leq |E|$ ensures polynomial size even when k is given in binary. $\square$

C.5 Mixed discriminants and determinantal point processes

The matrices defining the counting problems below are rational and positive semidefinite. For a matrix L indexed by $\mathcal{U}$, write $L[S]$ for its principal submatrix on S , with the convention $\det L[\emptyset] = 1$. The determinantal point process with matrix L assigns probability $\det L[S] / \det(I + L)$ to $S \subseteq \mathcal{U}$.

Corollary C.13. *There is an FPRAS for each of the following quantities:*

1. For a rational positive semidefinite matrix L indexed by $\mathcal{U}$ and a matroid $\mathcal{M}$ on $\mathcal{U}$ given by an independence oracle,

$$Z_{\mathcal{M}}(L) := \sum_{S \in \mathcal{B}(\mathcal{M})} \det L[S].$$

2. For rational positive semidefinite matrices A, B indexed by the same set $\mathcal{U}$,

$$Z(A, B) := \sum_{S \subseteq \mathcal{U}} \det A[S] \det B[S].$$

3. For rational positive semidefinite $d \times d$ matrices $A_1, \dots, A_d$, the mixed discriminant

$$\Delta(A_1, \dots, A_d) := [x_1 \cdots x_d] \det \left(\sum_{i=1}^d x_i A_i \right).$$

Before proving [Corollary C.13](#), we recall the following standard determinant identity.

Proposition C.14 (Cauchy–Binet formula [[Sta13](#), Theorem 9.4]). *Let $A \in \mathbb{R}^{d \times n}$ and $B \in \mathbb{R}^{n \times d}$, where $n \geq d$. For $S \subseteq [n]$ with $|S| = d$, let $A_{:,S}$ be the $d \times d$ matrix formed by the columns of A indexed by S , and let $B_{S,:}$ be the $d \times d$ matrix formed by the rows of B indexed by S , both in increasing order. Then*

$$\det(AB) = \sum_{\substack{S \subseteq [n] \\ |S|=d}} \det(A_{:,S}) \det(B_{S,:}).$$

Proof of [Corollary C.13](#). The first two reductions use the principal-minor identities (see [[BBL09](#), Section 3.2])

$$\begin{aligned} p_L(\mathbf{x}) &:= \sum_S \det L[S] \mathbf{x}^S = \det(I + LD_{\mathbf{x}}) = \det(I + L^{1/2} D_{\mathbf{x}} L^{1/2}), \\ q_L(\mathbf{x}) &:= \sum_T \det L[\mathcal{U} \setminus T] \mathbf{x}^T = \det(L + D_{\mathbf{x}}), \end{aligned}$$

where $D_{\mathbf{x}} := \text{diag}(\mathbf{x})$. To obtain the first expansion, use multilinearity of the determinant in its columns: choosing columns from $LD_{\mathbf{x}}$ at indices in S and from I elsewhere contributes $\det L[S] \mathbf{x}^S$. Similarly, choosing columns from $D_{\mathbf{x}}$ at indices in T and from L elsewhere contributes $\det L[\mathcal{U} \setminus T] \mathbf{x}^T$, giving the second expansion. The last equality for p_L follows from Sylvester's identity $\det(I + AB) = \det(I + BA)$.

Both polynomials are log-concave. Indeed, the matrices $I + L^{1/2} D_{\mathbf{x}} L^{1/2}$ and $L + D_{\mathbf{x}}$ are affine functions of $\mathbf{x}$ and positive definite when $\mathbf{x} > \mathbf{0}$, even if L is singular. The function $X \mapsto \log \det X$ is concave on positive definite matrices [[BV04](#), Section 3.1.5, p. 74]. Composing a concave function with an affine map preserves concavity, so $\log p_L(\mathbf{x})$ and $\log q_L(\mathbf{x})$ are concave.

For Item 1, apply [Lemma C.1](#) with $\omega(S) := \det L[S]$, initialized at $\emptyset$.

For Item 2, use

$$Z(A, B) = [\mathbf{x}^{\mathcal{U}}] (p_A(\mathbf{x}) q_B(\mathbf{x}))$$

and apply [Theorem 1.6](#), with initial sets $\emptyset$ and $\mathcal{U}$. This is the normalizing constant for the product of two DPPs [[OM21](#)]; dividing by $\det(I + A) \det(I + B)$ gives their agreement probability.

For Item 3, use the partition-matroid reduction of [[Cel+17](#), Section 4.1]. Rational LDL^T decompositions give $A_i = \sum_{e \in E_i} c_e \mathbf{v}_e \mathbf{v}_e^T$ with $c_e > 0$ and $|E_i| \leq d$. Here E_i indexes the rank-one terms of A_i ; we take these index sets to be pairwise disjoint. If $E_i = \emptyset$ for some i , then $A_i = 0$ and $\Delta = 0$. Otherwise put $E := \bigcup_{i=1}^d E_i$, let V have columns $\mathbf{v}_e$, and let $\mathcal{M}$ be the partition matroid on E whose bases contain exactly one element from each E_i . Thus a basis selects one rank-one term from each matrix A_i .

To compute the coefficient defining Δ , let $D(\mathbf{x})$ be the diagonal matrix indexed by E with $D(\mathbf{x})_{e,e} = c_e x_i$ for $e \in E_i$. Then $\sum_{i=1}^d x_i A_i = VD(\mathbf{x})V^\top$. For $S \subseteq E$ with $|S| = d$, write V_S for the $d \times d$ matrix formed by the columns of V indexed by S . Applying [Proposition C.14](#) with $A = VD(\mathbf{x})$ and $B = V^\top$ gives

$$\det \left(\sum_{i=1}^d x_i A_i \right) = \sum_{\substack{S \subseteq E \\ |S|=d}} \det(V_S)^2 \left(\prod_{e \in S} c_e \right) \prod_{i=1}^d x_i^{|S \cap E_i|}.$$

Here $\det(V_S)^2 = \det(V_S^\top V_S) = \det((V^\top V)[S])$. The monomial $\prod_{i=1}^d x_i^{|S \cap E_i|}$ equals $x_1 \cdots x_d$ exactly when S contains one element of each E_i , that is, when S is a basis of $\mathcal{M}$. Extracting this coefficient therefore gives

$$\Delta(A_1, \dots, A_d) = \sum_{S \in \mathcal{B}(\mathcal{M})} \det((V^\top V)[S]) \prod_{e \in S} c_e.$$

Apply [Lemma C.1](#) to this determinant weight: the factors c_e are external fields, so log-concavity persists, and $\emptyset$ is again an initial set. All coefficient queries and decompositions use rational arithmetic of polynomial bit length. The square roots in p_L serve only to prove log-concavity. $\square$

C.6 The six-vertex model

Let $G = (V, E)$ be a loopless 4-regular multigraph, with the four incident edges at each vertex labeled 1, 2, 3, 4. We consider Eulerian orientations D of the edges, meaning that exactly two edges are directed into each vertex. Let $I_v(D) \subseteq \{1, 2, 3, 4\}$ be the labels of the two edges entering v . The weight $w_v(I)$ assigned to an incoming pair I is a_v for $I = \{1, 2\}$ or $\{3, 4\}$, b_v for $I = \{1, 3\}$ or $\{2, 4\}$, and c_v for $I = \{1, 4\}$ or $\{2, 3\}$. The partition function is

$$\sum_{\substack{D \text{ an Eulerian} \\ \text{orientation of } G}} \prod_{v \in V(G)} w_v(I_v(D)).$$

These are joint weights for the incoming pair; no factorization into individual edge-direction weights is assumed [[CLL19](#)].

Remark C.15. An Eulerian orientation specifies the direction of each edge, with equal indegree and outdegree at every vertex. If G is connected, this guarantees the existence of a directed Euler tour; otherwise, each connected component admits one. The six-vertex partition function includes one term per orientation, regardless of how many directed Euler tours it admits.

The Euler-tour weights in [Appendix C.2](#) depend on the whole pairing of the four incident half-edges, whereas the six-vertex weights depend only on the incoming pair. For example, if edges 1, 2 enter a vertex and 3, 4 leave it, the compatible local pairings

$$\{\{1, 3\}, \{2, 4\}\} \quad \text{and} \quad \{\{1, 4\}, \{2, 3\}\}$$

may receive different transition weights in [Appendix C.2](#). Both have incoming pair $\{1, 2\}$ under this orientation, so the six-vertex weight is a_v in either case.

Corollary C.16. *For every loopless 4-regular multigraph G , there is an FPRAS for its six-vertex partition function when the local pair weights a_v, b_v, c_v are nonnegative rationals satisfying*

$$a_v \leq b_v + c_v, \quad b_v \leq a_v + c_v, \quad c_v \leq a_v + b_v \quad \text{for every } v \in V(G).$$

Proof. Use the incidences $\mathcal{U} := \{(v, e) : v \in e\}$ as the ground set, and let $\mathcal{M}$ be the partition matroid choosing one incidence of each edge. Assign to $S \subseteq \mathcal{U}$ the product of its local incoming-pair weights, with weight zero unless S has two incidences at every vertex. Then bases of $\mathcal{M}$ with positive weight are exactly the six-vertex orientations, with their prescribed weights. We apply [Lemma C.1](#) to this weight.

Its generating polynomial is $\prod_v q_v$, where

$$q_v(\mathbf{x}) = a_v(x_1 x_2 + x_3 x_4) + b_v(x_1 x_3 + x_2 x_4) + c_v(x_1 x_4 + x_2 x_3).$$

An all-zero local triple gives partition function zero. Otherwise the Hessian of q_v has eigenvalues $a_v + b_v + c_v, a_v - b_v - c_v, -a_v + b_v - c_v$, and $-a_v - b_v + c_v$. The triangle inequalities leave at most

one positive eigenvalue, which is the log-concavity criterion for a nonzero homogeneous quadratic with nonnegative coefficients [BH20, Section 2]. Hence the product is log-concave. Choosing any positive incoming pair at each vertex gives an initial set, and coefficient queries are products of the given local weights. Thus Lemma C.1 applies, including equality in the triangle inequalities. $\square$

The earlier FPRAS of [CLL19] requires the stronger inequalities $a_v^2 \leq b_v^2 + c_v^2$ and their permutations.

C.7 Bipartite Quantum Max-Cut and the EPR model

The Pauli matrices are

$$X := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y := \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

with $i^2 = -1$. A ket $|\psi\rangle$ denotes a column vector, and the bra $\langle\psi|$ is its conjugate transpose. For $a, b \in \{0, 1\}$, set

$$|0\rangle := \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle := \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad |ab\rangle := |a\rangle \otimes |b\rangle,$$

where $\otimes$ denotes the tensor product. Thus

$$|\Phi^+\rangle := \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

and the local matrix

$$P := |\Phi^+\rangle\langle\Phi^+| \in \mathbb{C}^{4 \times 4}$$

is the orthogonal projector onto its span.

For a vertex set $V = \{1, \dots, n\}$, the operators act on $\bigotimes_{v=1}^n \mathbb{C}^2$, with one factor for each vertex. The subscript u means that a matrix acts on the factor at vertex u and as the identity on every other factor; explicitly,

$$X_u := I_2^{\otimes(u-1)} \otimes X \otimes I_2^{\otimes(n-u)},$$

where I_2 is the 2×2 identity matrix and zero tensor powers are omitted. Define Y_u, Z_u similarly, and let $I = I_2^{\otimes n}$. Thus $X_u X_v$ applies X at both vertices u, v . For example, when $n = 3$, $X_1 X_3 = X \otimes I_2 \otimes X$.

For $a \in \{0, 1\}$, the ket $|a\rangle_u$ denotes $|a\rangle$ in the copy of $\mathbb{C}^2$ at vertex u . For $u < v$, the notation

$$|\Phi^+\rangle_{uv} := \frac{|0\rangle_u \otimes |0\rangle_v + |1\rangle_u \otimes |1\rangle_v}{\sqrt{2}}$$

denotes a vector on these two vertex factors. Let $P_{uv} \in \mathbb{C}^{2^n \times 2^n}$ act as P on the factors at u, v and as the identity on all remaining factors. For example, when $n = 3$,

$$P_{12} = P \otimes I_2, \quad P_{23} = I_2 \otimes P.$$

For a loopless graph $G = (V, E)$ with nonnegative rational edge weights w_e , let

$$Q_{\text{cut}} := \sum_{e=uv} w_e \frac{I - X_u X_v - Y_u Y_v - Z_u Z_v}{4}, \quad Q_{\text{EPR}} := \sum_{e=uv} w_e P_{uv}.$$

For a Hermitian matrix M and $\beta > 0$, define its *partition function* by

$$Z_\beta(M) := \text{Tr } e^{\beta M}.$$

We consider approximating this quantity and the largest eigenvalue of M . For background on the two problems, see [TSC24; MS26]. Write $W := \sum_e w_e$.

Corollary C.17. *Given a loopless graph G with nonnegative rational edge weights, there is an FPRAS for $\lambda_{\max}(Q_{\text{EPR}})$. For rational $\beta > 0$, there is also an FPRAS for $Z_\beta(Q_{\text{EPR}})$ when βW is polynomially bounded in the input length. More precisely, a randomized algorithm approximates this partition function with relative error at most ε in time polynomial in the input length, βW , and ε^{-1} .*

If G is bipartite, the same claims hold for $\lambda_{\max}(Q_{\text{cut}})$ and $Z_\beta(Q_{\text{cut}})$.

Proof. We first prove the EPR claims by reducing them to the six-vertex partition function in Appendix C.6. We then deduce the Quantum Max-Cut claims by an explicit change of basis.

The EPR matrix: eigenvalue estimation. Fix $0 < \varepsilon \leq 1$. If $W = 0$, the largest eigenvalue is zero and $Z_\beta(Q_{\text{EPR}}) = 2^n$. Otherwise delete zero-weight edges, put $p_e := w_e/W$, and set $Q := Q_{\text{EPR}}/W = \sum_{e=uv} p_e P_{uv}$. Each P_{uv} is an orthogonal projector, and $\sum_e p_e = 1$, so $0 \preceq Q \preceq I$. For the unit vector $|\xi\rangle = |0\rangle^{\otimes n}$, we have $\langle \xi | P_{uv} | \xi \rangle = 1/2$ for every edge uv . Consequently,

$$\frac{1}{2} = \frac{1}{2} \sum_e p_e = \langle \xi | Q | \xi \rangle \leq \lambda_{\max}(Q) \leq 1.$$

For $\beta > 0$, let $\lambda_1 = \lambda_{\max}(Q) \geq \dots \geq \lambda_{2^n}$ be the eigenvalues of Q . The eigenvalues of $e^{\beta Q}$ are $e^{\beta \lambda_i}$, so

$$e^{\beta \lambda_1} \leq Z_\beta(Q) = \sum_{i=1}^{2^n} e^{\beta \lambda_i} \leq 2^n e^{\beta \lambda_1}.$$

The lower bound keeps only the $i = 1$ term; the upper bound replaces each term by $e^{\beta \lambda_1}$. Taking logarithms and dividing by β gives

$$\lambda_{\max}(Q) \leq \frac{\log Z_\beta(Q)}{\beta} \leq \lambda_{\max}(Q) + \frac{n \log 2}{\beta}.$$

Thus $\beta := 8(n+1)/\varepsilon$ and an estimate of $Z_\beta(Q)$ with relative error at most $1/4$ suffice for additive error $\varepsilon/4$ in $\lambda_{\max}(Q)$, including the error from evaluating the logarithm to sufficiently many bits. Since $\lambda_{\max}(Q) \geq 1/2$, this gives relative error at most ε after multiplying by W .

The EPR matrix: a six-vertex instance. We next transform Q into a matrix A with the same eigenvalues, so that $Z_\beta(Q) = \text{Tr } e^{\beta A}$, and approximate this trace using [Corollary C.16](#). Put

$$K := \frac{I_2 - iX}{\sqrt{2}}, \quad |\Psi^+\rangle := \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \quad R := |\Psi^+\rangle\langle\Psi^+|.$$

The matrix K is unitary and $(K \otimes K)|\Phi^+\rangle = -i|\Psi^+\rangle$. For $e = uv$, let $R_e := R_{uv}$ be the extension of R to the full space by the same convention as P_{uv} . Thus

$$A := K^{\otimes n} Q (K^*)^{\otimes n} = \sum_e p_e R_e, \quad \text{Tr } e^{\beta A} = Z_\beta(Q).$$

Here K^* is the conjugate transpose of K .

Fix rational $\beta > 0$ and a target relative error $0 < \eta \leq 1$. Set $m := \lceil \max\{1, 4\beta, 20\beta^2/\eta\} \rceil$ and $\delta := \beta/m$. Label the edges $1, \dots, |E|$ in any order, and multiply their matrices in this order and then in reverse order:

$$B := \left(\prod_{e=1}^{|E|} \left(I + \frac{\delta p_e}{2} R_e \right) \right) \left(\prod_{e=|E|}^1 \left(I + \frac{\delta p_e}{2} R_e \right) \right).$$

We claim the following bound:

$$B \succ 0, \quad \|\log B - \delta A\| \leq 5\delta^2, \tag{32}$$

where $\|\cdot\|$ denotes the operator norm. The matrix logarithm $\log B$ is obtained by applying the scalar logarithm to the positive eigenvalues of B in an orthonormal eigenbasis. The intuition comes from the scalar approximation $\log(1 + \delta x) - \delta x = O(\delta^2 x^2)$ as $\delta x \rightarrow 0$: the linear terms sum to δA , and we expect an error quadratic in δ . We will verify [Eq. \(32\)](#) at the end of the proof, accounting for the order of the matrix factors. Assuming this estimate, our choice of m gives

$$\|m \log B - \beta A\| \leq 5m\delta^2 = \frac{5\beta^2}{m} \leq \frac{\eta}{4}.$$

Let $\mu_1 \geq \dots \geq \mu_{2^n}$ and $\nu_1 \geq \dots \geq \nu_{2^n}$ be the eigenvalues of $m \log B$ and βA , respectively. These matrices are Hermitian, so the norm bound implies

$$\beta A - \frac{\eta}{4} I \preceq m \log B \preceq \beta A + \frac{\eta}{4} I.$$

The min-max principle for eigenvalues therefore gives

$$v_i - \frac{\eta}{4} \leq \mu_i \leq v_i + \frac{\eta}{4} \quad (1 \leq i \leq 2^n).$$

Exponentiating these scalar inequalities and summing over i gives

$$e^{-\eta/4} \sum_i e^{v_i} \leq \sum_i e^{\mu_i} \leq e^{\eta/4} \sum_i e^{v_i}.$$

Since

$$\sum_i e^{\mu_i} = \text{Tr} e^{m \log B} = \text{Tr}(B^m), \quad \sum_i e^{v_i} = \text{Tr} e^{\beta A} = Z_\beta(Q),$$

we obtain

$$e^{-\eta/4} Z_\beta(Q) \leq \text{Tr}(B^m) \leq e^{\eta/4} Z_\beta(Q).$$

For each edge e , write $s_e := \delta p_e/2$ and $F_e := I + s_e R_e$ for its matrix in B . Writing I_4 for the 4×4 identity matrix, the local table of F_e in row and column order 00, 01, 10, 11 is

$$I_4 + s_e R = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 + s_e/2 & s_e/2 & 0 \\ 0 & s_e/2 & 1 + s_e/2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The product B^m uses each edge $2m$ times. Each use gives one *node* carrying the displayed 4×4 table. At a node for $e = uv$, two edge ends specify the first and second bits of the row label, corresponding to u, v ; another two specify the first and second bits of the column label. For each original vertex u , connect its column end at each node to its row end at the next node involving u , following the order in B^m and connecting the last such node back to the first. This defines the tensor network.

A configuration σ assigns a bit 0 or 1 to every network edge. The four incident bits select one entry at each node t ; call this entry its weight $w_t(\sigma)$. The *partition function* sums the product of node weights over all configurations:

$$Z_{\text{TN}} := \sum_{\sigma} \prod_{\text{nodes } t} w_t(\sigma).$$

Expanding the trace sums the same products of node weights. Matrices for edges not incident to u leave its bit unchanged, while each isolated vertex of G contributes two choices for its bit. Hence, if G has h isolated vertices,

$$\text{Tr}(B^m) = 2^h Z_{\text{TN}}.$$

This tensor network has the same partition function as a six-vertex model; see Fig. 3 for an illustrated example.

There are $2m|E|$ nodes in the network, with rational weights of polynomial encoding length. An estimate of $\text{Tr}(B^m)$ with relative error at most $\eta/4$, obtained from Corollary C.16, together with the displayed trace bound, gives an estimate of $Z_\beta(Q)$ with relative error at most η : indeed, $(1 + \eta/4)e^{\eta/4} \leq 1 + \eta$ and $(1 - \eta/4)e^{-\eta/4} \geq 1 - \eta$. The running time is polynomial in the input length, β , and η^{-1} . Since

$$Z_\beta(Q_{\text{EPR}}) = Z_{\beta W}(Q),$$

this proves the partition-function claim with running time polynomial in the input length, βW , and η^{-1} . Taking $\eta = 1/4$ and $\beta = 8(n+1)/\varepsilon$ for the normalized matrix Q also completes the largest-eigenvalue approximation described above.

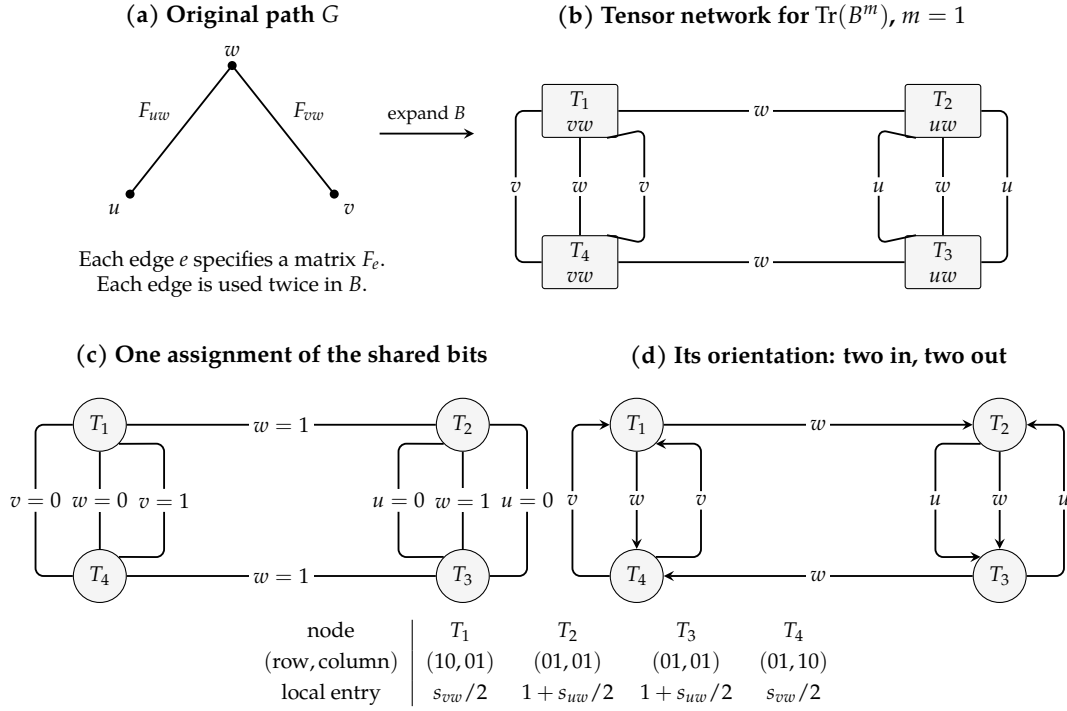


Figure 3: From a two-edge path to a six-vertex instance. (a)–(b) The edge sequence vw, uw, uw, vw in B gives nodes $T_1, \dots, T_4$, each carrying a 4×4 table. Each network edge carries the bit of the original vertex named by its label. (c)–(d) Keep row bits, flip column bits, and point each arrow toward its end marked 1. Every nonzero table entry has equally many 1's in its row and column labels, so two arrows enter each node. This correspondence is reversible and preserves the node weights. Label the row ends 1, 2 and the column ends 4, 3, respectively. The local six-vertex weights are $(a, b, c) = (1, 1 + s_e/2, s_e/2)$, satisfying $b = a + c$ and hence the triangle inequalities. Each node has degree four; there are no loops because each original edge is used at least twice. The table lists the four selected local entries; their product is this assignment's contribution to $\text{Tr}(B)$. For $\text{Tr}(B^m)$, repeat this edge sequence m times before closing the connections.

The Quantum Max-Cut matrix. Now suppose G is bipartite, with $V = V_0 \sqcup V_1$. We show that Q_{cut} has the same eigenvalues as Q_{EPR} on the same weighted graph. Define the unitary matrix

$$U := \bigotimes_{v \in V} U_v, \quad U_v := \begin{cases} I_2, & v \in V_0, \\ Y, & v \in V_1. \end{cases}$$

Since $YXY = -X$, $YYY = Y$, and $YZY = -Z$, conjugation by U gives

$$U Q_{\text{cut}} U^* = \sum_{e=uv} w_e \frac{I + X_u X_v - Y_u Y_v + Z_u Z_v}{4} = \sum_{e=uv} w_e P_{uv} = Q_{\text{EPR}}.$$

The second equality follows by multiplying the displayed Pauli matrices. Therefore

$$\lambda_{\max}(Q_{\text{cut}}) = \lambda_{\max}(Q_{\text{EPR}}), \quad Z_{\beta}(Q_{\text{cut}}) = Z_{\beta}(Q_{\text{EPR}}).$$

The EPR algorithms prove both remaining claims.

Verification of Eq. (32). We return to the EPR product B on an arbitrary graph. Let F be the first parenthesized product in the definition of B . Each factor is Hermitian and positive definite, with eigenvalues 1 and $1 + \delta p_e/2$. Thus F is invertible, the reverse product is F^* , and $B = FF^* \succ 0$.

To bound $B - I - \delta A$, list the matrices $\delta p_e R_e/2$ in their order of occurrence in B as $H_1, \dots, H_N$, where $N = 2|E|$, and put $h_j := \|H_j\|$. Each edge occurs twice, so

$$B = \prod_{j=1}^N (I + H_j), \quad \sum_{j=1}^N H_j = \delta A, \quad \sum_{j=1}^N h_j = \delta \sum_e p_e = \delta.$$

Expand the product by choosing either I or H_j at each position, keeping the chosen matrices in their original order. The term with no chosen H_j is I , while the terms with exactly one chosen H_j sum to $\sum_j H_j = \delta A$. Subtracting these terms therefore leaves

$$B - I - \delta A = \sum_{k=2}^N \sum_{1 \leq i_1 < \dots < i_k \leq N} H_{i_1} \cdots H_{i_k}.$$

The norm of a matrix product is at most the product of the norms of its matrices:

$$\|H_{i_1} \cdots H_{i_k}\| \leq \|H_{i_1}\| \cdots \|H_{i_k}\| = h_{i_1} \cdots h_{i_k}.$$

Applying the triangle inequality to the displayed remainder now yields

$$\begin{aligned} \|B - I - \delta A\| &\leq \sum_{k=2}^N \sum_{1 \leq i_1 < \dots < i_k \leq N} h_{i_1} \cdots h_{i_k} \\ &= \prod_{j=1}^N (1 + h_j) - 1 - \sum_{j=1}^N h_j \\ &\leq e^\delta - 1 - \delta. \end{aligned}$$

The equality uses the scalar expansion of $\prod_j (1 + h_j)$, with its constant term 1 and linear terms $\sum_j h_j$ removed. The last inequality uses $1 + h_j \leq e^{h_j}$ and $\sum_j h_j = \delta$. Similarly, subtracting only I keeps all terms with $k \geq 1$, so

$$\begin{aligned} \|B - I\| &\leq \sum_{k=1}^N \sum_{1 \leq i_1 < \dots < i_k \leq N} h_{i_1} \cdots h_{i_k} \\ &= \prod_{j=1}^N (1 + h_j) - 1 \leq e^\delta - 1. \end{aligned}$$

Set $D := B - I$. Since $\delta \leq 1/4$, the scalar exponential series gives

$$e^\delta - 1 \leq \frac{\delta}{1 - \delta} \leq 2\delta, \quad e^\delta - 1 - \delta \leq \frac{\delta^2}{2(1 - \delta)} \leq \delta^2.$$

Hence $\|D\| \leq 2\delta \leq 1/2$ and $\|D - \delta A\| \leq \delta^2$. Applying the scalar series for $\log(1 + x)$ in an orthonormal eigenbasis of D gives the absolutely convergent matrix series

$$\begin{aligned} \log B &= D + \sum_{k=2}^{\infty} \frac{(-1)^{k+1}}{k} D^k, \\ \|\log B - D\| &\leq \sum_{k=2}^{\infty} \frac{\|D\|^k}{k} \leq \frac{\|D\|^2}{2(1 - \|D\|)} \leq 4\delta^2. \end{aligned}$$

Combining the two remainder bounds proves

$$\|\log B - \delta A\| \leq \|\log B - D\| + \|D - \delta A\| \leq 5\delta^2.$$

This proves [Eq. \(32\)](#) and completes the proof. □

C.8 Matroid-congestion Gibbs distributions

Let $p, q \geq 1$. There are p labeled players, each choosing a basis B_i of a matroid $\mathcal{M}_i$ on the same ground set E of size q , given by an independence oracle. A *profile* $B = (B_1, \dots, B_p)$ records all players' choices. The load of an element $e \in E$ is the number of bases containing it, $\ell_e(B) := |\{i : e \in B_i\}|$. For each element, supply a table of nonnegative rational weights $a_e(0), \dots, a_e(p)$ that are positive exactly for $l_e \leq k \leq u_e$, where $0 \leq l_e \leq u_e \leq p$ are integers. Assume these weights are log-concave: $a_e(k)^2 \geq a_e(k-1)a_e(k+1)$ for $1 \leq k < p$. The partition function is

$$Z(a) := \sum_{B_i \in \mathcal{B}(\mathcal{M}_i), 1 \leq i \leq p} \prod_{e \in E} a_e(\ell_e(B)). \quad (33)$$

Thus a profile has positive weight exactly when $l_e \leq \ell_e(B) \leq u_e$ for every element e .

Corollary C.18. *For the matroids and rational log-concave tables above, there is an FPRAS for $Z(a)$ with running time polynomial in pq , the table encoding length, and ε^{-1} .*

Proof. We apply [Lemma C.1](#) on the set $\mathcal{U} := [p] \times E$ of ordered pairs (i, e) : the first entry $i \in [p]$ is a player index, and the second entry $e \in E$ is an element. Including (i, e) records that player i uses element e . Let $\mathcal{M}$ be the matroid whose bases are precisely $\{(i, e) : 1 \leq i \leq p, e \in B_i\}$ as each B_i ranges over the bases of $\mathcal{M}_i$. Thus its rank is $R := \sum_i \text{rk}(\mathcal{M}_i)$. Put $V_e := [p] \times \{e\}$ and define

$$\omega(X) := \mathbf{1}[|X| = R] \prod_e a_e(|X \cap V_e|).$$

Each basis of $\mathcal{M}$ specifies one profile, so $Z(a) = \sum_{X \in \mathcal{B}(\mathcal{M})} \omega(X)$.

To show that the generating polynomial of ω is log-concave, we use the following sufficient condition [[BH20](#), Corollary 3.16 and Theorem 2.30]: for every X, Y with $\omega(X), \omega(Y) > 0$ and every $a \in X \setminus Y$, some $b \in Y \setminus X$ satisfies

$$\omega(X)\omega(Y) \leq \omega(X - a + b)\omega(Y + a - b). \quad (34)$$

Here $X - a + b$ means $(X \setminus \{a\}) \cup \{b\}$, and similarly for $Y + a - b$. The sets X, Y in this condition need only have positive ω -weight; they are not required to be bases of $\mathcal{M}$. For each element $g \in E$, write $x_g = |X \cap V_g|$ and $y_g = |Y \cap V_g|$. Fix $a \in V_e$ for some element e . If $x_e \leq y_e$, the fact that $a \in (X \setminus Y) \cap V_e$ gives

$$|(Y \setminus X) \cap V_e| = y_e - x_e + |(X \setminus Y) \cap V_e| \geq 1.$$

Thus we can choose $b \in (Y \setminus X) \cap V_e$ simply by counting; no matroid property is needed. Exchanging a and b leaves every load unchanged, so $\omega(X - a + b) = \omega(X)$ and $\omega(Y + a - b) = \omega(Y)$. Otherwise $x_e > y_e$. Since $\sum_g x_g = |X| = |Y| = \sum_g y_g$, there must be a different element $f \neq e$ with $x_f < y_f$. Choose $b \in (Y \setminus X) \cap V_f$. Since $V_f = [p] \times \{f\}$, the first entry of b is a player index and its second entry is the fixed element f . This choice is possible because Y contains more pairs in V_f than X . Exchanging a and b changes the loads at e from (x_e, y_e) to $(x_e - 1, y_e + 1)$, and at f from (x_f, y_f) to $(x_f + 1, y_f - 1)$. These loads stay within their allowed intervals, and the log-concavity of a_e, a_f gives [Eq. \(34\)](#).

To find a set of positive weight, choose integers $k_e \in [l_e, u_e]$ with $\sum_e k_e = R$ and select k_e elements from each V_e . Such integers exist exactly when $\sum_e l_e \leq R \leq \sum_e u_e$; otherwise $Z(a) = 0$. We evaluate $\omega(X)$ by checking $|X| = R$, counting $|X \cap V_e|$ for each element, and multiplying the corresponding weights. The ratio of its largest to smallest positive weight satisfies

$$\mathcal{R}_\omega \leq \prod_e \frac{\max_{l_e \leq k \leq u_e} a_e(k)}{\min_{l_e \leq k \leq u_e} a_e(k)}.$$

Thus [Lemma C.1](#) applies. □

Weights specified by costs. For the same matroids and integers l_e, u_e as in [Corollary C.18](#), suppose we are given a rational $\beta \geq 0$ and *rational costs* $c_e(1) \leq \dots \leq c_e(p)$, where $c_e(k)$ is the cost to each player using element e at load k . Define the tables by

$$a_e(k) := \begin{cases} \exp(-\beta \sum_{j=1}^k c_e(j)), & l_e \leq k \leq u_e, \\ 0, & \text{otherwise,} \end{cases}$$

with the empty sum equal to zero. We use the same partition function $Z(a)$ as in [Eq. \(33\)](#). When $Z(a) > 0$, the *Gibbs distribution* assigns probability $\prod_e a_e(\ell_e(B)) / Z(a)$ to profile B . When $l_e = 0$ and $u_e = p$ for every e , sampling from this distribution is the question posed in [[Kle23](#), Section 1.3].

If each real weight can be evaluated exactly in one operation, this is already an application of the proof of [Corollary C.18](#): the ratios $a_e(k) / a_e(k-1) = e^{-\beta c_e(k)}$, for $l_e < k \leq u_e$, are nonincreasing in k , so the tables are log-concave. Sampling follows by using approximate counts to make the players' choices one element at a time. The next corollary handles the fact that these weights need not be rational: we approximate them by rational log-concave tables and control the resulting error. To state the running time, put

$$\Lambda := \sum_e \log \frac{\max_{l_e \leq k \leq u_e} a_e(k)}{\min_{l_e \leq k \leq u_e} a_e(k)}.$$

This value is rational and can be computed exactly from the costs. The weight-ratio bound in the preceding proof gives $\log \mathcal{R}_\omega \leq \Lambda$, so even with exact evaluation the running-time bound depends polynomially on the numerical value of Λ . Maxima below range over $l_e \leq k \leq u_e$.

Corollary C.19. *For the matroids and rational numbers $c_e(1) \leq \dots \leq c_e(p)$ and $\beta \geq 0$ specified above, restrict the profiles B to those satisfying $l_e \leq \ell_e(B) \leq u_e$ for every $e \in E$. A randomized algorithm returns an estimate of $Z(a)$ with relative error at most ε , with probability at least $3/4$, in the form $(\prod_e \max_k a_e(k)) \widehat{Z}$, where $\widehat{Z}$ is rational and the exponential factor is specified by its exact rational exponent. If $Z(a) > 0$, there is also a sampler within total variation distance ε of the Gibbs distribution. The running times are polynomial in pq , the input encoding length, ε^{-1} , and the numerical value of Λ . If the costs are nonnegative and every $l_e = 0$, then $\max_k a_e(k) = a_e(0) = 1$ for every e , so the estimate is rational.*

Proof. It suffices to approximate the partition function and sample after dividing each table by its largest entry. Indeed, put

$$\bar{a}_e(k) := \frac{a_e(k)}{\max_j a_e(j)}.$$

This leaves the Gibbs distribution unchanged, and $Z(a) = (\prod_e \max_k a_e(k)) Z(\bar{a})$. The logarithm of this factor is rational and can be computed exactly from the given costs. We replace $\bar{a}_e$ by rational log-concave tables with nearly the same weights. For $0 < \varepsilon \leq 1$, set $\tau := \varepsilon / [64q(p+1)^2]$ and $r := 1 + \tau$. For each $l_e \leq k \leq u_e$, write $\bar{a}_e(k) = e^{-x}$, where $x = -\log \bar{a}_e(k) \in [0, \Lambda]$ is rational and computable exactly from the costs. Standard algorithms for evaluating the exponential function [BZ10, Sections 4.3–4.4] compute a positive rational approximation $\tilde{a}_e(k)$ satisfying

$$\left| \log \frac{\tilde{a}_e(k)}{\bar{a}_e(k)} \right| \leq \tau/16$$

in time polynomial in the input encoding length, Λ , and $\log(\tau^{-1})$.

The rational approximation may lose log-concavity. To ensure log-concavity, we make a further small adjustment: define $b_e(k) := \tilde{a}_e(k) r^{-k^2}$ on the allowed interval and zero outside it. For every $l_e < k < u_e$,

$$\log \frac{b_e(k)^2}{b_e(k-1)b_e(k+1)} \geq 2 \log r - \tau/4 > 0.$$

Thus b_e is log-concave, and every profile B with $l_e \leq \ell_e(B) \leq u_e$ for every $e \in E$ obeys

$$\left| \log \frac{\prod_e b_e(\ell_e(B))}{\prod_e \bar{a}_e(\ell_e(B))} \right| \leq q\tau/16 + qp^2 \log r \leq \varepsilon/64.$$

Consequently the sums $Z(b)$ and $Z(\bar{a})$ differ by a factor at most $e^{\varepsilon/64}$ in either direction. When these sums are positive, dividing the profile weights by their respective sums gives two probability distributions whose total variation distance is at most $e^{\varepsilon/32} - 1 < \varepsilon/16$. Applying Corollary C.18 with accuracy $\varepsilon/4$ gives an estimate of $Z(\bar{a})$ with relative error at most ε . Multiplying by $\prod_e \max_k a_e(k)$ gives the required estimate of $Z(a)$. For sampling, decide for each pair (i, e) whether $e \in B_i$, using approximate counts to estimate the total weight of each choice [JVV86]. Fixing these choices deletes or contracts elements of the player matroids. For each element, it also shifts the load index by the number of players already required to use it and removes infeasible entries. The resulting tables remain log-concave, so Corollary C.18 applies at every step. Sampling from the distribution defined by the rational tables b_e within total variation distance $\varepsilon/4$ then gives the desired approximate Gibbs sample. The rational tables have encoding length polynomial in Λ , the input encoding length, and ε^{-1} , which gives the stated running time. $\square$

C.9 Integer polymatroid intersections, contingency tables, and bounded flows

We first reduce weighted intersections of integer families to subset convolution. Each integer coordinate is encoded by a subset of a block of labeled elements; factorial weights compensate for the multiple encodings of the same integer vector. This gives an approximate counting algorithm for intersections of integral polymatroid bases. We next obtain an FPRAS for contingency tables with log-concave entry weights by separating their row-sum and column-sum constraints. A weight-preserving bijection then gives the result for bounded integer flows. In Appendix C.10, the flow result yields an FPRAS for vertices of network-matrix polytopes.

We use log-M-concave weights and polarization as defined in [Appendix B.4](#). A nonempty finite family $\mathcal{A} \subseteq \mathbb{Z}_{\geq 0}^m$ is M-convex if its indicator weight is log-M-concave: all vectors have the same coordinate sum, and both exchanged vectors in [Eq. \(30\)](#) belong to $\mathcal{A}$. In particular, the integer points of an integral polymatroid base polytope form such a family [[BH20](#)].

Proposition C.20 (Counting weighted intersections of integer families). *Let $\mathbf{c} \in \mathbb{Z}_{\geq 1}^m$ be given in unary, meaning that each c_i is represented by c_i copies of the symbol 1. Let $\mu, \nu : [\mathbf{0}, \mathbf{c}] \cap \mathbb{Z}^m \rightarrow \mathbb{R}_{\geq 0}$ be log-M-concave weights, extended by zero outside the box, given by evaluation oracles. Suppose we are given initial vectors $\mathbf{x}_0, \mathbf{y}_0$ with $\mu(\mathbf{x}_0) > 0$ and $\nu(\mathbf{y}_0) > 0$, and upper bounds on their coefficient ranges whose logarithms are polynomial in the input length. Then there is an FPRAS for*

$$Z := \sum_{\mathbf{k} \in [\mathbf{0}, \mathbf{c}] \cap \mathbb{Z}^m} \mu(\mathbf{k}) \nu(\mathbf{k})$$

in the arithmetic model of [Theorem 1.6](#). In particular, membership oracles and one point in each of two M-convex families $\mathcal{A}, \mathcal{D}$ in the box suffice to approximate $|\mathcal{A} \cap \mathcal{D}|$.

Proof. We construct two subset weights whose convolution is $Z/c!$, where $\mathbf{k}! := \prod_i k_i!$. We then show that their generating polynomials are log-concave and apply [Theorem 1.6](#). If the coordinate sums of the two initial vectors differ, the supports are disjoint and $Z = 0$, so assume these sums agree.

Encoding the integer vectors. Take disjoint sets $\mathcal{U}_i$ with $|\mathcal{U}_i| = c_i$ and put $\mathcal{U} := \sqcup_i \mathcal{U}_i$. For $S \subseteq \mathcal{U}$, write $k_i(S) := |S \cap \mathcal{U}_i|$ and define

$$\begin{aligned} \hat{\mu}(S) &:= \mu(\mathbf{k}(S)) \frac{(c - \mathbf{k}(S))!}{c!}, \\ \hat{\nu}(S) &:= \nu(c - \mathbf{k}(S)) \frac{(c - \mathbf{k}(S))!}{c!}. \end{aligned} \tag{35}$$

In the convolution, $\hat{\mu}(S)$ and $\hat{\nu}(\mathcal{U} \setminus S)$ use the same integer vector $\mathbf{k}(S)$. Exactly $\binom{c_i}{k_i} := \prod_i \binom{c_i}{k_i}$ subsets encode $\mathbf{k}$, and the factorial weights cancel this multiplicity:

$$(\hat{\mu} * \hat{\nu})(\mathcal{U}) = \sum_{\mathbf{0} \leq \mathbf{k} \leq \mathbf{c}} \binom{\mathbf{c}}{\mathbf{k}} \mu(\mathbf{k}) \nu(\mathbf{k}) \frac{(c - \mathbf{k})! \mathbf{k}!}{(c!)^2} = \frac{Z}{c!}. \tag{36}$$

Log-concavity of the generating polynomials. We use the weighted exchange condition in [Eq. \(30\)](#) for ν ; log-concavity of its generating polynomial alone would not provide this condition. Put $\bar{\nu}(\mathbf{k}) := \nu(c - \mathbf{k})$ for integer vectors $\mathbf{k}$ satisfying $0 \leq k_i \leq c_i$ for every i , and set $\bar{\nu}(\mathbf{k}) = 0$ otherwise. We first verify that $\bar{\nu}$ is also log-M-concave. Take $\mathbf{x}, \mathbf{y} \in \text{supp } \bar{\nu}$ with $x_i > y_i$. Then $\mathbf{c} - \mathbf{y}, \mathbf{c} - \mathbf{x} \in \text{supp } \nu$ and $(\mathbf{c} - \mathbf{y})_i > (\mathbf{c} - \mathbf{x})_i$. Apply [Eq. \(30\)](#) to these two vectors in this order. It gives an index j with $(\mathbf{c} - \mathbf{y})_j < (\mathbf{c} - \mathbf{x})_j$, equivalently $x_j < y_j$, and

$$\begin{aligned} \bar{\nu}(\mathbf{x}) \bar{\nu}(\mathbf{y}) &= \nu(\mathbf{c} - \mathbf{y}) \nu(\mathbf{c} - \mathbf{x}) \\ &\leq \nu(\mathbf{c} - \mathbf{y} - \mathbf{e}_i + \mathbf{e}_j) \nu(\mathbf{c} - \mathbf{x} + \mathbf{e}_i - \mathbf{e}_j) \\ &= \bar{\nu}(\mathbf{x} - \mathbf{e}_i + \mathbf{e}_j) \bar{\nu}(\mathbf{y} + \mathbf{e}_i - \mathbf{e}_j). \end{aligned}$$

Moreover, if the vectors in $\text{supp } \nu$ have coordinate sum r , then those in $\text{supp } \bar{\nu}$ have coordinate sum $\sum_i c_i - r$. Thus $\bar{\nu}$ satisfies both parts of the definition.

By [[BH20](#), Corollary 3.16], the exponential generating polynomial of a log-M-concave weight is Lorentzian. Applying this separately to μ and $\bar{\nu}$ gives two Lorentzian polynomials:

$$\begin{aligned} g_\mu^{\text{exp}}(\mathbf{x}) &:= \sum_{\mathbf{0} \leq \mathbf{k} \leq \mathbf{c}} \frac{\mu(\mathbf{k})}{\mathbf{k}!} \mathbf{x}^{\mathbf{k}}, \\ g_{\bar{\nu}}^{\text{exp}}(\mathbf{x}) &:= \sum_{\mathbf{0} \leq \mathbf{k} \leq \mathbf{c}} \frac{\bar{\nu}(\mathbf{k})}{\mathbf{k}!} \mathbf{x}^{\mathbf{k}} = \sum_{\mathbf{0} \leq \mathbf{k} \leq \mathbf{c}} \frac{\nu(\mathbf{k})}{(c - \mathbf{k})!} \mathbf{x}^{c - \mathbf{k}}. \end{aligned}$$

Apply polarization from [Appendix B.4](#), with the new variables for coordinate i indexed by $\mathcal{U}_i$. Since $1/(k!(c_i)) = (c_i - k)!/c_i!$, the resulting polynomials are exactly $g_{\hat{\mu}}$ and $g_{\hat{\nu}}$ from [Eq. \(35\)](#). These are multiaffine log-concave polynomials because polarization preserves the Lorentzian property [[BH20](#), Proposition 3.1 and Theorem 2.30].

Oracle access and running time. Choose $S_0, T_0 \subseteq \mathcal{U}$ with $|S_0 \cap \mathcal{U}_i| = (x_0)_i$ and $|T_0 \cap \mathcal{U}_i| = c_i - (y_0)_i$ for every i . These satisfy $\hat{\mu}(S_0) > 0$ and $\hat{\nu}(T_0) > 0$. The input length is at least $\sum_i c_i = |\mathcal{U}|$ under the stated encoding. Factorials can be precomputed and the weights $\hat{\mu}$ and $\hat{\nu}$ evaluated in polynomial time. Their coefficient ranges increase by at most a factor of $c!$, whose logarithm is also polynomial in the input length. Apply [Theorem 1.6](#) and multiply the estimate by $c!$, using [Eq. \(36\)](#), to obtain the claimed FPRAS. $\square$

The proposition allows a separate log-concave weight on each coordinate. Indeed, if $\mathcal{A}$ is M-convex and each a_i is log-concave with interval support, then, whenever nonzero,

$$\omega(\mathbf{k}) := \mathbf{1}_{\{\mathbf{k} \in \mathcal{A}\}} \prod_i a_i(k_i)$$

satisfies [Eq. \(30\)](#). For the exchange indices i, j , log-concavity gives $a_i(x_i)a_i(y_i) \leq a_i(x_i - 1)a_i(y_i + 1)$ and the analogous inequality with j increased in x and decreased in y . Thus the proposition counts weighted integer points in the intersection of two integral polymatroid base polytopes.

A *contingency table* is a nonnegative integer matrix with specified row and column sums. The next corollary allows both entrywise upper bounds and a separate log-concave weight on each entry.

Corollary C.21 (Contingency tables with log-concave entry weights). *The input consists of row sums $\mathbf{r} \in \mathbb{Z}_{\geq 0}^m$, column sums $\mathbf{c} \in \mathbb{Z}_{\geq 0}^n$, entrywise upper bounds $u_{ij} \in \mathbb{Z}_{\geq 0}$, and nonzero nonnegative log-concave tables $w_{ij}(0), \dots, w_{ij}(u_{ij})$ with interval support. There is an FPRAS for*

$$Z_{\mathbf{r}, \mathbf{c}} := \sum_{\substack{X \in \mathbb{Z}_{\geq 0}^{m \times n}, \\ \sum_j X_{ij} = r_i \ (i \in [m]) \\ \sum_i X_{ij} = c_j \ (j \in [n]) \\ X_{ij} \leq u_{ij}}} \prod_{i=1}^m \prod_{j=1}^n w_{ij}(X_{ij})$$

in the arithmetic model of [Theorem 1.6](#), provided that $\sum_{i,j} u_{ij}$ and $\log R$ are polynomially bounded in the input length, where

$$R := \prod_{i=1}^m \prod_{j=1}^n \frac{\max_{k: w_{ij}(k) > 0} w_{ij}(k)}{\min_{k: w_{ij}(k) > 0} w_{ij}(k)}.$$

Taking $w_{ij} \equiv 1$ counts contingency tables with the stated row sums, column sums, and entrywise upper bounds.

Proof. We apply [Proposition C.20](#) to the intersection of the row-sum family and the column-sum family, assigning all entry weights to the first family.

First, find a table with positive weight, or certify that $Z_{\mathbf{r}, \mathbf{c}} = 0$. This is a standard integral-flow feasibility problem on the bipartite graph of rows and columns: the edge (i, j) carries X_{ij} , its lower and upper bounds are the endpoints of the positive interval of w_{ij} , and the required totals at the row and column vertices are $\mathbf{r}$ and $\mathbf{c}$. It can be solved in polynomial time. Assume henceforth that such a table X_0 exists. If all $u_{ij} = 0$, evaluate the single possible table directly. Otherwise, regard matrices as vectors of the entries with $u_{ij} > 0$; the omitted entries remain fixed at zero.

Let $\mathcal{A}$ and $\mathcal{D}$ be the families satisfying $0 \leq X_{ij} \leq u_{ij}$ and, respectively, the row sums and the column sums. Both are M-convex. Indeed, if $X_{ij} > Y_{ij}$, the equal sum in that row (respectively, column) gives another entry where X is smaller than Y . Exchanging one unit between these two entries in opposite directions in X and Y preserves the required sums and entry bounds. Set

$$\mu(X) := \mathbf{1}_{\{X \in \mathcal{A}\}} \prod_{i=1}^m \prod_{j=1}^n w_{ij}(X_{ij}), \quad \nu(X) := \mathbf{1}_{\{X \in \mathcal{D}\}}.$$

The preceding observation about products of coordinate weights shows that these weights are log-M-concave, with coefficient ranges at most R and 1. Their evaluations take polynomial time, and X_0 has positive weight under both. Their inner product is $Z_{\mathbf{r}, \mathbf{c}}$. The coordinate upper bounds have polynomial sum, so they can be represented in unary. Thus [Proposition C.20](#) gives the claimed FPRAS. $\square$

Corollary C.22 (Bounded integer flows). *Let B be the incidence matrix of a directed graph (V, E) , with outgoing entries $+1$ and incoming entries -1 . The input consists of a vector $\mathbf{h} \in \mathbb{Z}^V$, capacities $d_a \in \mathbb{Z}_{\geq 0}$, and nonzero nonnegative log-concave tables $w_a(0), \dots, w_a(d_a)$ with interval support. Each h_v specifies the required outgoing flow minus incoming flow at vertex v . There is an FPRAS for*

$$Z_{\mathbf{h}} := \sum_{\substack{q \in \mathbb{Z}^E, Bq = \mathbf{h} \\ 0 \leq q_a \leq d_a}} \prod_{a \in E} w_a(q_a)$$

in the arithmetic model of Theorem 1.6, provided that $\sum_a d_a$ and $\log R$ are polynomially bounded in the input length, where

$$R := \prod_{a \in E} \frac{\max_{k: w_a(k) > 0} w_a(k)}{\min_{k: w_a(k) > 0} w_a(k)}.$$

Taking $w_a \equiv 1$ counts all feasible integer flows.

Proof. We give a weight-preserving bijection between flows and contingency tables, then apply Corollary C.21. Write $D := \sum_a d_a$ for the total flow capacity. If $\sum_v h_v \neq 0$, there is no feasible flow, so return zero.

From flows to square matrices. We construct an $N \times N$ nonnegative integer matrix X , where $N = |V| + |E|$ and the rows and columns are indexed by the disjoint union $V \sqcup E$. Giving each edge its own index keeps parallel edges and loops distinct. The required row sums R_i and column sums C_i are

$$\begin{aligned} R_v &:= \sum_{a \text{ out of } v} d_a, & C_v &:= R_v - h_v & (v \in V), \\ R_a &:= d_a, & C_a &:= d_a & (a \in E). \end{aligned}$$

If any $C_v < 0$, return zero: a feasible flow would satisfy $h_v \leq \sum_{a \text{ out of } v} q_a \leq R_v$. The row and column sums are now nonnegative and satisfy $\sum_i R_i = \sum_i C_i = 2D$. Given a flow q , start with $X = 0$ and set

$$\begin{aligned} X_{u,a} &= X_{a,v} := q_a, & X_{a,a} &:= d_a - q_a & (a = u \rightarrow v), \\ X_{v,v} &:= R_v - \sum_{a \text{ out of } v} q_a & & & (v \in V). \end{aligned}$$

All entries not assigned above remain zero. In particular,

$$\begin{aligned} X_{u,v} &= 0 & (u, v \in V, u \neq v), \\ X_{a,b} &= 0 & (a, b \in E, a \neq b). \end{aligned}$$

For an edge $a = u \rightarrow v$, the mixed entries also satisfy $X_{w,a} = 0$ for $w \in V \setminus \{u\}$ and $X_{a,w} = 0$ for $w \in V \setminus \{v\}$. Allow each of $X_{u,a}, X_{a,v}, X_{a,a}$ to lie in $[0, d_a]$, and each $X_{v,v}$ to lie in $[0, R_v]$. These bounds hold for the matrix just constructed, and flow conservation gives

$$\sum_{j \in V \sqcup E} X_{i,j} = R_i, \quad \sum_{j \in V \sqcup E} X_{j,i} = C_i \quad (i \in V \sqcup E).$$

Conversely, consider a matrix with these entry bounds, fixed zero entries, and row and column sums. For each edge $a = u \rightarrow v$, its row and column equations give

$$X_{a,v} + X_{a,a} = d_a = X_{u,a} + X_{a,a},$$

so $X_{u,a} = X_{a,v}$ defines a unique flow q_a . Subtracting the column equation at vertex v from its row equation gives outgoing flow minus incoming flow equal to $R_v - C_v = h_v$. All diagonal entries are then uniquely determined, proving the bijection.

Applying the contingency-table result. Assign weight $w_a(X_{u,a})$ to the entry $X_{u,a}$ and weight one to the other entries. The bijection preserves weights, and every entry-weight table is log-concave with interval support. The product of their coefficient ranges is R , and their upper bounds sum to

$$3 \sum_{a \in E} d_a + \sum_{v \in V} R_v = 4D.$$

Thus Corollary C.21 gives an FPRAS for $Z_{\mathbf{h}}$. □

C.10 Vertices of network-matrix polytopes

We now apply [Corollary C.22](#) to count vertices of network-matrix polytopes. The vertices correspond to integer flows; the key point is that their binary coordinates let us bound the total flow capacity by a polynomial in the matrix dimensions.

A *network matrix* records signed paths in a directed tree. More precisely, let T be a directed tree with m edges, and let E consist of n additional labeled directed edges on the same vertex set. Write V for this vertex set, so $|V| = m + 1$. For each directed edge $e = (u \rightarrow v)$, its incidence entry in row w is $b_{w,e} = \mathbf{1}_{\{w=u\}} - \mathbf{1}_{\{w=v\}}$. Choose any vertex $r \in V$ and define

$$B_T = (b_{w,e})_{w \in V \setminus \{r\}, e \in E(T)} \in \mathbb{R}^{m \times m},$$

$$B_E = (b_{w,e})_{w \in V \setminus \{r\}, e \in E} \in \mathbb{R}^{m \times n}.$$

Thus we omit the row for vertex r from both incidence matrices and keep all other vertex rows in the same order. To see that the square matrix B_T is invertible, let $\mathbf{z} \in \mathbb{R}^{V \setminus \{r\}}$ satisfy $B_T^\top \mathbf{z} = \mathbf{0}$, and put $z_r = 0$. For every tree edge $e = (u \rightarrow v)$,

$$0 = (B_T^\top \mathbf{z})_e = z_u - z_v.$$

Every vertex is joined to r by a path when edge directions are ignored, so these equalities give $z_w = z_r = 0$ for every $w \in V$. Thus $B_T^\top \mathbf{z} = \mathbf{0}$ has only the zero solution, proving that B_T is invertible. The matrix $A = B_T^{-1} B_E$ is a network matrix. Given only a network matrix A , one can find in polynomial time a directed tree T and additional directed edges E such that $A = B_T^{-1} B_E$ [[BC80](#)]. Network matrices are totally unimodular, so in particular their entries lie in $\{0, \pm 1\}$. Consequently, for integral bounds $\ell, \mathbf{u}$, the vertices of

$$P = \{\mathbf{x} \in [0, 1]^n : \ell \leq A\mathbf{x} \leq \mathbf{u}\}$$

are exactly its binary points: integrality gives one direction, and every binary point of a polytope contained in the cube is a vertex.

The following result answers the network-matrix case posed in [[GJ23](#), Problem 4.3]. It follows by combining their reduction to integer flows [[GJ23](#), Section 4] with [Corollary C.22](#).

Corollary C.23. *There is an FPRAS for the number of vertices of P above, with running time polynomial in the input length and ε^{-1} in the arithmetic model of [Theorem 1.6](#). This includes a polytope $\{\mathbf{x} : A\mathbf{x} \leq \mathbf{b}\}$ promised to have only binary vertices, for a network matrix A and integral $\mathbf{b}$.*

Proof. We recall the reduction of Guo and Jerrum [[GJ23](#), Section 4] and apply [Corollary C.22](#) with unit weights. We verify the bijection from vertices to flows and bound the total capacity in terms of the dimensions of A .

The vertices of P are exactly its binary points. For each such point $\mathbf{x}$, set $\mathbf{y} = A\mathbf{x}$. The identity $A = B_T^{-1} B_E$ gives

$$\mathbf{y} = A\mathbf{x} \iff B_T \mathbf{y} - B_E \mathbf{x} = \mathbf{0}.$$

For each labeled edge $e = (u \rightarrow v) \in E$, let $\overleftarrow{e} = (v \rightarrow u)$ denote the same edge with its direction reversed, and set $\overleftarrow{E} := \{\overleftarrow{e} : e \in E\}$. On the graph with edge set $E(T) \sqcup \overleftarrow{E}$, write f_a for the flow on edge a : set $f_t = y_t$ on each tree edge and $f_{\overleftarrow{e}} = x_e$ on the reversal of each edge $e \in E$. The equation on the right expresses *flow conservation*: total outgoing flow equals total incoming flow at each vertex whose row appears in the matrices,

$$\sum_{a \text{ out of } v} f_a = \sum_{a \text{ into } v} f_a \quad (v \in V \setminus \{r\}).$$

Here r is the vertex whose row was deleted when defining B_T and B_E above. The same equality also holds at r : every edge contributes once to an outgoing sum and once to an incoming sum, so the sum of outgoing minus incoming flow over all vertices is zero. Since this difference vanishes at every $v \neq r$, it must vanish at r as well. The tree flows are uniquely determined by $\mathbf{x}$, so this is a bijection between vertices of P and integer flows satisfying $\ell_t \leq y_t \leq u_t$ and $0 \leq x_e \leq 1$.

We next shift the tree flows to make all flows nonnegative and bound their capacities. Let a_t^+, a_t^- count the $+1, -1$ entries in row t of A . Since $y_t = (A\mathbf{x})_t$, every binary $\mathbf{x}$ satisfies $-a_t^- \leq y_t \leq a_t^+$.

We may therefore replace the original constraint $\ell_t \leq y_t \leq u_t$ on the flow through tree edge t by $L_t \leq y_t \leq U_t$, without changing the feasible flows, where

$$L_t := \max\{\ell_t, -a_t^-\}, \quad U_t := \min\{u_t, a_t^+\}.$$

If any $L_t > U_t$, the count is zero. Otherwise put $q_t = y_t - L_t$ on tree edges and $q_e = x_e$ on reversed edges. Let B be the full vertex-edge incidence matrix of this graph, with outgoing entries $+1$ and incoming entries -1 . The equivalent constraints are

$$Bq = h, \quad 0 \leq q_a \leq d_a, \quad h = -B(L, 0),$$

where $d_t = U_t - L_t \leq a_t^+ + a_t^- \leq n$ and $d_e = 1$. In particular, $D := \sum_a d_a \leq mn + n$. Thus [Corollary C.22](#), with $w_a \equiv 1$ and $R = 1$, gives the required approximation using polynomially many arithmetic operations in $m + n$ and ε^{-1} .

Finally, if $Q = \{x : Ax \leq b\}$ is a polytope with only binary vertices, then $Q \subseteq [0, 1]^n$, since a polytope is the convex hull of its vertices. Adding the cube bounds therefore does not change Q . The lower bounds $\ell_t = -a_t^-$ hold throughout the cube, so taking these bounds and $u = b$ puts Q in the form of P . $\square$

C.11 Rainbow matchings and degree-constrained forests

Let $G = (V, E)$ be a simple graph with rational edge weights $0 \leq w_e \leq 1$, and write $w^S := \prod_{e \in S} w_e$. For $S \subseteq E$, let $\deg_S(v) := |\{e \in S : v \in e\}|$ be the degree of v in the subgraph (V, S) . A matching is *rainbow* if its edges have distinct colors; the coloring need not be proper.

Corollary C.24. *Given any edge coloring of G , there is an FPRAS for*

$$\sum_{\substack{M \subseteq E: \\ M \text{ a rainbow matching}}} w^M.$$

In particular, taking every $w_e = 1$ counts all rainbow matchings.

For nonnegative integer bounds b_v , a *degree-constrained forest* is an acyclic edge set $F \subseteq E$ with $\deg_F(v) \leq b_v$ for every vertex v .

Corollary C.25. *There is an FPRAS for*

$$\sum_{\substack{F \subseteq E \text{ a forest:} \\ \deg_F(v) \leq b_v \ \forall v}} w^F.$$

In particular, taking $b_v = 2$ and $w_e = 1$ counts all linear forests, that is, all unions of vertex-disjoint paths, with isolated vertices allowed.

Both sums range over all sizes, including the empty edge set.

Proof of Corollaries C.24 and C.25. For the fixed graph G and any matroid $\mathcal{N}$ on its edge set E , given by an independence oracle, we approximate

$$Z_0 := \sum_{\substack{S \in \mathcal{I}(\mathcal{N}): \\ \deg_S(v) \leq b_v \ \forall v}} w^S.$$

For rainbow matchings, take $b_v = 1$ and let $\mathcal{N}$ be the partition matroid allowing at most one edge of each color. For forests, take the graphic matroid. We first replace Z_0 by a weighted sum Z_q that gives a small weight to dependent sets, and bound the resulting error. We then express Z_q through a convolution coefficient and apply [Theorem 1.16](#).

Delete zero-weight edges and edges incident to a vertex v with $b_v = 0$, restricting $\mathcal{N}$ accordingly. Write $n := |E|$; the case $n = 0$ has $Z_0 = 1$. Define the measure

$$\mu(S) := \mathbf{1}[\deg_S(v) \leq b_v \ \forall v] w^S.$$

Its support is downward closed.

For $0 < \varepsilon \leq 1$, put $q := \varepsilon/(8n)$ and define

$$Z_q := \sum_{S \in \text{supp } \mu} \mu(S) q^{|S| - \text{rk}_{\mathcal{N}}(S)}.$$

Independent sets contribute exactly Z_0 , since their exponent is zero. To bound the additional contribution, for each $S \in \text{supp } \mu$ choose a basis $I(S)$ of $\mathcal{N}|S$ by fixed-order greedy. Then $I(S) \subseteq S$ satisfies the degree bounds, $|S \setminus I(S)| = |S| - \text{rk}_{\mathcal{N}}(S)$, and $\mu(S) \leq \mu(I(S))$ because $w_e \leq 1$. The map $S \mapsto (I(S), S \setminus I(S))$ is injective. Therefore

$$Z_0 \leq Z_q \leq \sum_{I \in \text{supp } \mu \cap \mathcal{I}(\mathcal{N})} \mu(I) \sum_{T \subseteq E \setminus I} q^{|T|} \leq (1+q)^n Z_0 \leq (1+\varepsilon/4) Z_0,$$

where $(1+q)^n \leq e^{\varepsilon/8} \leq 1 + \varepsilon/4$. Thus an estimate of Z_q with relative error at most $\varepsilon/2$ also estimates Z_0 with relative error at most ε .

To approximate Z_q , define

$$\mu_q(S) := q^{|S|} \mu(S), \quad \nu_q(T) := q^{|T| - \text{rk}_{\mathcal{N}^*}(T)}.$$

For every $T \subseteq E$, the matroid dual-rank identity is

$$\text{rk}_{\mathcal{N}^*}(T) = |T| + \text{rk}_{\mathcal{N}}(E \setminus T) - \text{rk}_{\mathcal{N}}(E).$$

Taking $T = E \setminus S$ gives $\nu_q(E \setminus S) = q^{\text{rk}_{\mathcal{N}}(E) - \text{rk}_{\mathcal{N}}(S)}$. Hence

$$[\mathbf{x}^E](g_{\mu_q} g_{\nu_q}) = \sum_{S \subseteq E} \mu_q(S) \nu_q(E \setminus S) = q^{\text{rk}_{\mathcal{N}}(E)} Z_q.$$

We now verify the hypotheses of [Theorem 1.16](#). First, μ has upward spectral curvature -1 . Fix $S \in \text{supp } \mu$ and put $W_S := \{e \in E \setminus S : S + e \in \text{supp } \mu\}$. Let B be the unsigned incidence matrix of W_S on the vertices v with $b_v - \deg_S(v) = 1$, where at most one further incident edge can be added. Let d_e be its column sums, and put $D := \text{diag}(w_e)$ on these edges. Two available edges cannot be added together precisely when they share such a vertex. Since G is simple, they share at most one vertex. The principal submatrix indexed by W_S is

$$(A_S^{\mu, -1})_{W_S \times W_S} = DB^T BD + \text{diag}_e(w_e - (d_e - 1)w_e^2) \succeq 0.$$

Indeed, $d_e \leq 2$ and $w_e \leq 1$. Unavailable edges contribute zero rows and columns. Also, every addition ratio of μ is at most one.

Consequently, μ_q has curvature $-q$, since $A_S^{\mu_q, -q} = q^2 A_S^{\mu, -1}$. The measure ν_q has full support and curvature q . To see this, consider the contraction $\mathcal{N}^*/T$: its loops have addition ratio q , and its nonloops have addition ratio one. In $A_T^{\nu_q, q}$ the loop rows vanish, each nonloop parallel class gives a block $(1-q)J$, where J is the all-ones matrix, and all other entries vanish. Thus both supports are downward closed and their curvatures sum to zero. Their addition ratios are at most q and one, respectively, so we may take $\gamma = q$ and $\chi = q^{-1}$ in [Theorem 1.16](#).⁵

Degree checks and greedy rank queries evaluate both measures using the dual-rank identity above. Both are positive at the empty set, and

$$\log(\mathcal{R}_{\mu_q} \mathcal{R}_{\nu_q}) \leq 2n \log(1/q) + \sum_{e \in E} \log(1/w_e).$$

This bound and $\chi = 8n/\varepsilon$ are polynomial in the input length and ε^{-1} . Apply [Theorem 1.16](#) with relative error $\varepsilon/2$ and divide its coefficient estimate by $q^{\text{rk}_{\mathcal{N}}(E)}$. This gives an estimate of Z_q with relative error at most $\varepsilon/2$, and hence an estimate of Z_0 with relative error at most ε . $\square$

⁵To meet the theorem's requirement that χ be polynomial in the ground-set size, add $\lceil 1/\varepsilon \rceil$ dummy elements forbidden by μ and made loops of $\mathcal{N}$. This leaves the target and the coefficient above unchanged, and makes $\chi = 8n/\varepsilon$ polynomial in the enlarged ground-set size. The enlargement is polynomial in n and ε^{-1} .

D Miscellaneous Aspects of Theorem 1.14

D.1 A Matroid Intersection Analog of the Monomer-Dimer Model

In this section, we consider a generalization of the classical monomer-dimer model over matchings of a graph to matroid intersection. We use Theorem 1.14 to prove a spectral gap for the standard heat-bath Glauber dynamics that decays smoothly as the parameter λ grows. In particular, the model does not exhibit a dynamical phase transition in λ . The motivation for studying this problem was discussed in Section 1.5.

Let $\mathcal{M}, \mathcal{N}$ be two matroids on a common ground set, and fix a parameter $\lambda \geq 0$. We consider the following probability measure over common independent sets of all sizes:

$$\mu_\lambda(S) \propto \lambda^{|S|}, \quad \forall S \in \mathcal{I}(\mathcal{M}) \cap \mathcal{I}(\mathcal{N}).$$

The discrete-time heat-bath Glauber dynamics is a Markov chain whose evolution can be described as follows: If the state at time step t is S_t , then we transition to S_{t+1} by

1. sampling a uniformly random element $u \in \mathcal{U}$, and
2. resampling the in/out status of u conditioned on all other elements. More specifically, if $u \in S_t$, then set $S_{t+1} = S_t - u$ with probability $\frac{1}{1+\lambda}$. Otherwise, if $u \notin S_t$ and $S_t + u \in \mathcal{I}(\mathcal{M}) \cap \mathcal{I}(\mathcal{N})$, then set $S_{t+1} = S_t + u$ with probability $\frac{\lambda}{1+\lambda}$. In all other cases, set $S_{t+1} = S_t$.

Let P_λ denote the corresponding transition probability matrix, and let $\mathcal{L}_\lambda := I - P_\lambda$. It is straightforward to verify that $P_\lambda, \mathcal{L}_\lambda$ are reversible with respect to μ_λ . For $\lambda > 0$, the chain is irreducible since every state can reach $\emptyset$ by deletions, and each such path can be reversed.

Theorem D.1. *For every $\mathcal{M}, \mathcal{N}$ with $n = |\mathcal{U}| \geq 1$ and every $\lambda \geq 0$, we have the spectral gap lower bound*

$$\gamma(\mathcal{L}_\lambda) \geq \frac{1}{n \cdot (1 + \lambda) \cdot (1 + \lambda \cdot (n - 1))}.$$

Following the strategy used above for common bases, we first establish a Poincaré inequality for a softened version of μ_λ using Theorem 1.14. We then deduce Theorem D.1 by a direct comparison of Dirichlet forms.

Lemma D.2. *For $0 \leq t \leq 1$, define the distribution*

$$\mu_\lambda^{(t)}(S) \propto \lambda^{|S|} \cdot t^{|S| - \text{rk}_{\mathcal{N}}(S)}, \quad \forall S \in \mathcal{I}(\mathcal{M}).$$

Then for every $f : \mathcal{I}(\mathcal{M}) \rightarrow \mathbb{R}$, we have the Poincaré Inequality

$$\text{Var}_{\mu_\lambda^{(t)}}[f] \leq \mathcal{E}_\lambda^{(t)}(f, f) := \lambda \cdot \sum_{S \in \mathcal{I}(\mathcal{M})} \sum_{\substack{u \notin S \\ S+u \in \mathcal{I}(\mathcal{M})}} \mu_\lambda^{(t)}(S) \cdot (f(S+u) - f(S))^2.$$

Remark D.3. As discussed in Remark 2.6, the “softening” of μ_λ into $\mu_\lambda^{(t)}$ can be viewed as a generalization of the “softening” used in the proof of Theorem 2.2.

Proof. For $\lambda > 0$ and $0 < t \leq 1$, define the weight functions

$$\pi(S) := \lambda^{|S|} \mathbf{1}[S \in \mathcal{I}(\mathcal{M})], \quad \nu(S) := t^{|S| - \text{rk}_{\mathcal{N}}(S)} \mathbf{1}[S \in \mathcal{I}(\mathcal{M})].$$

Their common support $\mathcal{I}(\mathcal{M})$ is downward closed and connected, and $\pi \odot \nu = \mu_\lambda^{(t)}$. We claim that π has upward 0-spectral curvature and ν^* has upward 1-spectral curvature. Assuming these claims, Theorem 1.14 with $\eta = 0$ and $\kappa = 1$ gives the desired inequality: the dynamics has rate $\pi(S+u)/\pi(S) = \lambda$ for every feasible addition, so its Dirichlet form is exactly $\mathcal{E}_\lambda^{(t)}(f, f)$ by reversibility. The case $t = 0$ follows by continuity of the finite sums, and $\lambda = 0$ gives a point mass at $\emptyset$, for which both sides vanish. It remains to verify the two claims.

First, for $S \in \mathcal{I}(\mathcal{M})$, the feasible additions are the nonloops of $\mathcal{M}/S$. Two distinct feasible additions u, v are called *parallel* in $\mathcal{M}/S$ if $S + u + v \notin \mathcal{I}(\mathcal{M})$. Together with equality, this is an equivalence relation on the nonloops of $\mathcal{M}/S$, partitioning them into equivalence classes. Thus

$A_S^{\pi,0}$ is a direct sum of blocks $\lambda^2 J$ on these classes and zero rows for infeasible additions, where J denotes an all-ones matrix. Thus $A_S^{\pi,0} \succeq 0$, proving the first claim.

Second, fix $S \in \mathcal{I}(\mathcal{M})$, so $\mathcal{U} \setminus S \in \text{supp } \nu^*$, and put $\mathcal{K} := (\mathcal{N}|_S)^*$. For every $T \subseteq S$, applying Eq. (29) to the matroid $\mathcal{N}|_S$ on ground set S gives

$$\text{rk}_{\mathcal{K}}(T) = |T| + \text{rk}_{\mathcal{N}|_S}(S \setminus T) - \text{rk}_{\mathcal{N}|_S}(S) = |T| + \text{rk}_{\mathcal{N}}(S \setminus T) - \text{rk}_{\mathcal{N}}(S).$$

Since $S \setminus T \in \mathcal{I}(\mathcal{M})$, the definitions yield

$$\frac{\nu^*((\mathcal{U} \setminus S) \cup T)}{\nu^*(\mathcal{U} \setminus S)} = \frac{\nu(S \setminus T)}{\nu(S)} = \frac{t^{|S \setminus T| - \text{rk}_{\mathcal{N}}(S \setminus T)}}{t^{|S| - \text{rk}_{\mathcal{N}}(S)}} = t^{-|T| + \text{rk}_{\mathcal{N}}(S) - \text{rk}_{\mathcal{N}}(S \setminus T)} = t^{-\text{rk}_{\mathcal{K}}(T)}.$$

Consequently, $A_{\mathcal{U} \setminus S}^{\nu^*,1}$ is a direct sum of blocks $\frac{1-t}{t^2} J$ on the equivalence classes under parallelism among the nonloops of $\mathcal{K}$, with zero rows at its loops and outside S . Since $0 < t \leq 1$, these blocks are positive semidefinite. $\square$

Proof of Theorem D.1. The case $\lambda = 0$ is trivial so assume $\lambda > 0$. Let $\Omega := \mathcal{I}(\mathcal{M}) \cap \mathcal{I}(\mathcal{N})$. For $S \in \mathcal{I}(\mathcal{M}) \setminus \Omega$, write $D_S := \{u \in S : S - u \in \Omega\}$ and $d_S := |D_S|$. Fix a function $f : \Omega \rightarrow \mathbb{R}$. We extend it to the function $f^\uparrow : \mathcal{I}(\mathcal{M}) \rightarrow \mathbb{R}$ as follow:

$$f^\uparrow(S) := \begin{cases} f(S), & S \in \Omega, \\ \frac{1}{d_S} \sum_{u \in D_S} f(S - u), & S \in \mathcal{I}(\mathcal{M}) \setminus \Omega, d_S > 0, \\ 0, & S \in \mathcal{I}(\mathcal{M}) \setminus \Omega, d_S = 0. \end{cases}$$

By Lemma D.2 at $t = 0$,

$$\text{Var}_{\mu_\lambda}[f] = \text{Var}_{\mu_\lambda^{(0)}}[f^\uparrow] \leq \mathcal{E}_\lambda^{(0)}(f^\uparrow, f^\uparrow).$$

We will finish the proof by showing that

$$\mathcal{E}_\lambda^{(0)}(f^\uparrow, f^\uparrow) \leq n(1 + \lambda)(1 + \lambda(n - 1)) \cdot \langle f, \mathcal{L}_\lambda f \rangle_{\mu_\lambda}.$$

For convenience, put

$$\mathcal{E}_\lambda(f, f) := n(1 + \lambda) \langle f, \mathcal{L}_\lambda f \rangle_{\mu_\lambda} = \lambda \sum_{\substack{S \in \Omega, u \notin S \\ S+u \in \Omega}} \mu_\lambda(S) (f(S+u) - f(S))^2.$$

By the definition of $f^\uparrow$,

$$\begin{aligned} \mathcal{E}_\lambda^{(0)}(f^\uparrow, f^\uparrow) &= \mathcal{E}_\lambda(f, f) + \lambda \sum_{S \in \mathcal{I}(\mathcal{M}) \setminus \Omega} \sum_{u \in D_S} \mu_\lambda(S - u) (f^\uparrow(S) - f^\uparrow(S - u))^2 \\ &= \mathcal{E}_\lambda(f, f) + \underbrace{\sum_{\substack{S \in \mathcal{I}(\mathcal{M}) \setminus \Omega \\ d_S \geq 2}} \frac{\lambda}{d_S} \sum_{\{u,v\} \subseteq D_S} \mu_\lambda(S - u) (f(S - u) - f(S - v))^2}_{(*)}. \end{aligned}$$

In the last equality, we used that for any probability measure ζ on a domain Ω and any test function $\varphi : \Omega \rightarrow \mathbb{R}$, we have $\text{Var}_{\omega \sim \zeta}[\varphi(\omega)] = \frac{1}{2} \cdot \mathbb{E}_{\omega, \omega' \sim \zeta}[(\varphi(\omega) - \varphi(\omega'))^2]$. The inner sum is over unordered pairs of distinct elements; its summand is symmetric because $\mu_\lambda(S - u) = \mu_\lambda(S - v)$.

We control $(*)$ by a multiple by $\mathcal{E}_\lambda(f, f)$. To do this, observe that for each $S \in \mathcal{I}(\mathcal{M}) \setminus \Omega$ with $d_S \geq 2$ and each pair $\{u, v\} \subseteq D_S$, its contribution to $(*)$ can be viewed as a “swap” move that transitions from $S - u$ to $S - u - v$ followed by $S - u - v \rightarrow S - v$. This swap can be simulated using two steps of Glauber dynamics P_λ , namely $S - u \rightarrow S - u - v$ followed by $S - u - v \rightarrow S - v$. With this intuition in mind, we invoke the numerical inequality $(a - b)^2 \leq 2((a - c)^2 + (b - c)^2)$ and $\mu_\lambda(S - u) = \lambda \mu_\lambda(S - u - v)$

to obtain

$$\begin{aligned}
(*) &\leq \sum_{\substack{S \in \mathcal{I}(\mathcal{M}) \setminus \Omega \\ d_S \geq 2}} \frac{2\lambda^2}{d_S} \sum_{\substack{u, v \in D_S \\ u \neq v}} \mu_\lambda(S - u - v) (f(S - u) - f(S - u - v))^2 \\
&\leq \lambda^2 \sum_{\substack{T \in \Omega, v \notin T \\ T+v \in \Omega}} (n - |T| - 1) \mu_\lambda(T) (f(T + v) - f(T))^2 \\
&\leq \lambda(n - 1) \mathcal{E}_\lambda(f, f).
\end{aligned}$$

(Setting $T = S - u - v$ and using $d_S \geq 2$)
 $(|T| \geq 0)$

This finishes the proof. □